

THE PYRAMID AS MEMORY HYPOTHESIS

The Hill That Rose from the Waters, Made into a Pyramid — Pyramid Texts, Utterance 600

Zep Tepi Series, The Destination — Paper I

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Abstract

“O Great Ennead, who are in Heliopolis... make this pyramid endure, and this his temple, for ever and ever.” — Pyramid Texts, Utterance 601 (Mercer 1952; cf. Faulkner 1969)

The Great Pyramid of Giza was called *Akhet-Khufu* — the Horizon of Khufu — by its builders. No mummy has been found in any of the three Giza pyramids. The chambers are bare. The sarcophagus in the King's Chamber was placed during construction, too wide for any passage it would have needed to traverse. The causeways point west. These are not anomalies requiring special pleading. They are features of a coherent design.

This paper proposes that the Giza complex is the permanent architectural encoding of the Zep Tepi cosmological system — the origin narrative of Egyptian civilisation, traced through a documented migration corridor from the Saharan pastoral tradition to the Nile Valley. At landscape scale, the complex instantiates a four-element relational system derived from the Richat origin landscape: central elevation, bounded sacred space, engineered transition water, and a western processional axis. The engineered harbour basin (Younes et al. 2024), the Ahramat Branch alongside 31 pyramids (Ghoneim et al. 2024), the concentric enclosure walls, and causeway orientations averaging 269.4° — within 0.6° of due west (Nell & Ruggles 2014) — are the cosmological argument made physical.

At dimensional scale, the Great Pyramid's six exterior measurements show correspondences through the Wedjat binary fraction system across seven knowledge domains: π , ϕ , $\sqrt{5}$ and $\sqrt{6}$; Earth's polar and equatorial radii (at a lower evidential tier); the golden angle; the anomalistic lunar month, 360-day schematic year, civil and sidereal years, and 70-day Sirius invisibility; and the synodic periods of Saturn, Jupiter, Venus and Mars. The Sothic cycle was tested and is reported below as one of the four documented misses. Table 3 sets out these ten — the same ten of fourteen reached in the Monte Carlo analysis; the mathematical constants and terrestrial radii lie outside that list and carry no Monte Carlo support. The stellar shafts encoded specific stars at the construction epoch; the exterior concavity marks the equinoxes; the Royal Cubit is self-referentially recoverable from the π correspondence. The Wedjat fraction system — documented in Old Kingdom grain measurement, the Rhind Mathematical Papyrus, and Egyptian temple orientations — is the proposed structural key.

Two specificity tests are reported. Comparing the sixty-fourths against neighbouring grids $k/48$, $k/60$ and $k/72$ at nine documented Old Kingdom pyramids, the Great Pyramid returns a margin of +5 where the mean across twenty thousand randomly dimensioned pyramids is -1.32 ($p = 0.0050$), holding across four of five tolerances. Holding the seked at $5\frac{1}{2}$ and varying the base across every whole cubit from 75 to 600, the built base of 440 ranks in the top 4.2 per cent of 526 candidates throughout the observational tolerance band and ranks third at its centre; against three hundred shadow target lists its median rank is 275 of 526, and it took the maximum in none of the three hundred trials ($p < 0.0033$). The standing of 440 is therefore specific to values Egyptian sources record as astronomically significant. Comparative analysis against nine major ritual landscapes confirms the full four-element relational integration at Giza is unparalleled before the 12th century CE. Nine falsifiable predictions are stated. The evidence draws from nine independent disciplines — hydrology, genetics, archaeozoology, textual analysis, archaeoastronomy, mathematical analysis, cultural memory theory, archaeometry, and comparative architecture. The convergence across independent lines is the argument.

Keywords: Great Pyramid of Giza; Khufu; Akhet-Khufu; Horus-eye fractions; Wedjat fraction system; royal cubit; seked; Rhind Mathematical Papyrus; Sothic cycle; Monte Carlo null models; Giza causeway orientations; Zep Tepi; Richat Structure; Shemsu Hor; Green Sahara; Nile paleohydrology; Old Kingdom Egypt; archaeoastronomy; cultural memory transmission; African Humid Period

Author's Note

A note on the subtitle: it alludes to Pyramid Texts Utterance 600, in which the creator Atum rises as the benben — the primordial mound that emerged from the waters of Nun at Heliopolis (Allen 2005), the mythic image from which the pyramid's own form is traditionally derived. The phrasing here is an evocation of that utterance, not a direct quotation.

Series Context and Key Terminology

This paper is part of the **Zep Tepi Series**. It builds directly on the findings established in the four preceding papers:

- **Paper I (Eye of the Sahara Hypothesis):** Identification of the Richat Structure as the surviving candidate origin landscape and documentation of the paleohydrological corridor.
- **Paper II (Zep Tepi Hypothesis):** Genomic continuity from Taforalt through Takarkori to the Nile Valley, the Wedjat sky-compass, and the cattle cult tradition.
- **Paper III (Shemsu Hor Hypothesis):** The Followers of Horus as the institutional carriers of this knowledge across the transition to the Dynastic period.
- **Phase Zero (Bridging Paper):** The Minor Step Pyramids as a coordinated territorial program encoding the Meskhetyu astronomical tradition, bridging the gap between the Shemsu Hor and Giza.

Two supplementary papers in the series bear directly on material discussed here and are referenced at the relevant points below:

- **The Sphinx Hypothesis (Supplementary to Corridor III):** A Predynastic origin framework for the Great Sphinx and the philological case for its original name Mehit, "she who restores the Wedjat" — relevant to the Sphinx-to-Second-Pyramid geometry and the Giza plateau layout discussed in this paper.
- **The Sekhmet-Heka Hypothesis (Supplementary to Corridor III):** The inversion of rationality in the Egyptian medical corpus, and the botanical and pharmacological knowledge carried along the Saharan corridor — relevant to the medical and botanical components of the knowledge architecture referenced in this paper.

The present paper asks the question the four principal papers leave open: if the Shemsu Hor carried the knowledge and Phase Zero expressed it at provincial scale, what was ultimately built when state capacity finally matched cosmological ambition?

This paper does not re-argue the findings of Papers I–III or Phase Zero. It assumes them and builds directly on them. For the full evidentiary chain — including climatic "flickering" mechanisms, isotopic and archaeozoological correlates, the Canopus orientation family across 350+ Egyptian temples, the genomic continuity from Taforalt to Takarkori to Nuwayrat, the institutional memory of the Shemsu Hor across fifteen centuries of king lists, and the architectural program of the Minor Step Pyramids — see the original papers.

Introduction

The ancient Egyptians did not call this site Giza. That name is medieval Arabic. They called it Imentet — The West — and Rostau (râ-stâw), the Entrance to the Passages of Osiris. The pyramid at its centre they called Akhet-Khufu: the Horizon of Khufu. Not the tomb. The horizon — the threshold where the sun transforms, where the mortal becomes divine, where the First Time is renewed. That

name survives in the Merer Papyrus, the eyewitness logbook of an official who transported its stones (Tallet 2017). It is not a later poetic attribution. It is the construction-era designation. Imentet: the direction the memory came from. Rostau: the transformation it was built to produce. This paper argues that the architecture says the same thing.

The standard interpretation of the Giza pyramids as royal tombs is internally coherent and accounts for much of the evidence. What it does not readily account for is the specific compound of features documented here: no mummy recovered from any of the three Giza pyramids, a sarcophagus placed during construction and never removed, interior chambers bare of any inscription or preparation for text (Petrie 1883, pp. 67–69), stellar shafts sealed at both ends with polished interior faces, and three successive causeways deviating in opposed senses about the east–west axis yet averaging within a degree of due west across three reigns (Nell & Ruggles 2014; Lehner 1985). Each feature has an individual explanation. The constraint is that any viable interpretation must account for all of them simultaneously — and must do so without treating the pyramid’s own documented name as incidental.

The preceding papers of this series have assembled the transmission chain that makes a different reading possible. A pastoralist population carrying the Zep Tepi cosmological system moved northeast along a documented Highland Corridor as the African Humid Period collapsed. Ancient genomes are now available from three geographically independent North African nodes (Salem et al. 2025; Lipson et al. 2025; Morez Jacobs et al. 2025), documenting a distinct North African lineage present across the region through the period in question. Salem et al. conclude that pastoralism entered this lineage by cultural diffusion rather than population replacement, and Lipson et al. document high continuity of local ancestry in the eastern Maghreb. The genetic record therefore supports the movement of practice and knowledge rather than large-scale replacement of populations — the mechanism this series proposes, and a weaker requirement than migration in the demographic sense. Their cosmological system encoded the Richat Structure as the landscape of the First Time. The Shemsu Hor are proposed here to have preserved and transmitted this knowledge across fifteen centuries of Egyptian institutional memory. The pyramid form itself was already encoded in stone at seven provincial cult centres before Khufu was born (Dreyer & Kaiser 1980; Belmonte, Shaltout & Fekri 2005). What was built at Giza in the 4th Dynasty is proposed here not to have been an invention but a culmination of a tradition whose origins are the subject of this series.

The mechanism by which cosmological knowledge survives across millennia is not conjectural — it is the subject of an established scholarly literature. Assmann (1995, 2001) distinguishes communicative memory (living within three to four generations) from cultural memory (preserved over centuries and millennia through canonical texts, ritual repetition, and institutional structures). Cultural memory of this kind operates through what Assmann calls “figures of memory” — narrative and symbolic patterns that anchor collective identity to a foundational past event. The Zep Tepi is precisely such a figure: a temporal anchor (“the First Time”) that Egyptian priestly institutions returned to consistently across fifteen centuries of dynastic record. The durability of such anchors is well-documented: Australian Aboriginal oral traditions have been demonstrated to preserve accurate coastal geography from events approximately 7,000 years ago (Nunn & Reid 2016). The transmission proposed here — approximately 1,500 years from the African Humid Period collapse to the 4th Dynasty — falls well within the documented range of canonical cultural memory in pre-literate and proto-literate institutional contexts. The Shemsu Hor records are proposed here as the institutional expression of that preserved memory, not as proof of literal genealogical continuity.

The following table defines the four distinct types of continuity claim made in this paper and the evidence type required for each. These categories are maintained as distinct throughout; claims should be evaluated at the level appropriate to their category.

Concept	Definition	Required Evidence Type	Examples in This Paper
Symbolic continuity	Recurrence of structurally equivalent motifs across	Iconographic and architectural parallels	Cattle cosmogram; mound-water motif; Wedjat

Concept	Definition	Required Evidence Type	Examples in This Paper
	time		geometry; Benben symbolism
Institutional continuity	Preservation of ritual systems within functioning institutions	Documentary or archaeological evidence of institutional succession	Pyramid priesthood records; Shemsu Hor title in Turin Canon and Tombos Stela
Cultural memory	Cosmological framework transmitted as shared social knowledge across generations	Canonical textual codification of earlier traditions	Edfu Building Texts encoding Zep Tepi; Pyramid Texts preserving stellar destiny theology
Historical continuity	Demonstrable biological, genealogical, or institutional lineage	Genetic, epigraphic, or archival evidence of actual succession	Salem et al. (2025) North African lineage documented across the corridor region; Van Neer et al. faunal continuity at HK6

The Subterranean Sanctuary: Rostau and the Design Template

The Westcar Papyrus (Berlin Papyrus 3033, c. 1650 BCE) records a primary source tradition that Khufu was not designing his pyramid without a template. The text states that Khufu spent time seeking the number of the secret chambers in the sanctuary of Thoth in order to make something similar for his own pyramid (Blackman 1988; Parkinson 1997). The magician Djedi knows where this number is recorded — in a flint chest in the room called the Inventory at Heliopolis — but cannot deliver it. The number is never recovered within the narrative. What the papyrus establishes without ambiguity is the deliberate intent: the Great Pyramid’s chamber design was to be modelled on a pre-existing sacred architectural template, and the numerical count of its hidden chambers was the design specification Khufu sought.

The Giza plateau was named *pr wsir nb rstaw* — the House of Osiris, Lord of the Underground Passages. That name is consistent with the underground complex having defined the site’s sacred character, on this reading preceding the monuments that rose above it. That complex — the multi-level subterranean structure now called the Osiris Shaft, located beneath Khafre’s causeway — is the only structure at Giza architecturally parallel to the Abydos Osireion, a parallel Hawass confirmed directly, calling it “the Osirion of Giza” (Hawass 2007). Surface luminescence dating of stone within the Osiris Shaft — samples OT1 and OT2, taken on the second underground level rather than from the artifact deposits that record subsequent ritual use across dynasties — yields ages of 2930 ± 600 BCE and 2870 ± 570 BCE, consistent with construction before or during Khufu’s reign (Liritzis and Vafiadou 2015, Table 1). The uncertainties are large, the sample count is two, and the authors themselves record no independent archaeological age estimate for these samples; the result is therefore indicative rather than decisive. No pharaoh inscribed his name on the structure as builder, inconsistent with a new royal project. Stadelmann concluded that Khafre’s causeway was built to conform to a pre-existing feature at its location (Stadelmann 1991). The engineering logic is straightforward: excavating a thirty-metre multi-level shaft through solid bedrock after a causeway of hundreds of thousands of tonnes was built above it is structurally reckless; building a causeway around a pre-existing underground feature is standard Egyptian practice. The Shetayet of Sokar — the god’s hidden tomb-shrine, located at Rosetau and understood in Egyptology as the Lower Egyptian counterpart of the Osiris tomb at Abydos (Edwards 1986) — anchors the plateau’s underworld geography, Sokar being ‘lord of Rosetau’. The structural argument advanced here is that the Osiris Shaft complex was the defining sacred feature of that Rosetau, not a later addition to it. This paper advances as a structured open question — and as Prediction 9 — that the Osiris Shaft complex is proposed here as the pre-

existing sanctuary that gave Rostau its name, and the most plausible candidate for the architectural template the Westcar Papyrus narrative records Khufu seeking.

This paper advances one central hypothesis: that the Giza pyramid complex constitutes a deliberate and permanent encoding of the Zep Tepi cosmological system — the origin landscape, stellar grammar, and cosmological memory of a Saharan pastoral tradition transmitted through the documented corridor established in the preceding papers of this series. All subsidiary arguments — the dimensional correspondences, the institutional transmission chain, the water-cosmological integration, and the stellar orientation system — are proposed as supporting evidence for this central claim. They are advanced in hierarchical order: the landscape-scale and hydrological evidence constitutes the strongest evidential tier; the mathematical encoding claims are the most novel and the most in need of independent verification. A reader who accepts only the landscape-scale tier and remains agnostic about the mathematical tier still has access to the paper's primary argument.

Methodologically, this paper operates as a cumulative probabilistic convergence model rather than a singular historical reconstruction. No single line of evidence presented here establishes the hypothesis independently. What the framework proposes is that the independent convergence of genomic migration data, paleohydrological reconstruction, zooarchaeological assemblage analysis, archaeoastronomical measurement, architectural correspondence, and textual transmission scholarship — across research programmes designed for entirely different purposes — on the same geographic corridor, temporal window, and cosmological grammar constitutes a probabilistically meaningful pattern. The hypothesis strengthens with each additional independent confirmation, and would weaken materially if independent streams were found to contradict rather than reinforce each other. This is the epistemological structure of consilience-based historical reconstruction, as formalised by Whewell (1840) and operationalised in contemporary landscape archaeology and archaeoastronomy.

This paper documents what that culmination encodes. At the landscape scale, the Giza complex instantiates the four-element relational system derived from the Richat origin landscape — central elevation, bounded sacred space, engineered transition water, and a western processional axis — confirmed across hydrology (Sheisha et al. 2022; Younes et al. 2024), archaeoastronomy (Belmonte & Lull 2023), and survey data (Nell & Ruggles 2014). At the dimensional scale, the Great Pyramid's exterior measurements show correspondences — proposed in this hypothesis to reflect deliberate encoding through the Wedjat binary fraction system — across seven independent knowledge domains, of which ten individual correspondences are set out with their complete arithmetic in Table 3 — the same ten of fourteen targets reached in the Monte Carlo analysis, the remaining four being documented misses. Their coverage of Old Kingdom astronomical knowledge — planetary, lunar, solar, and Sirius-calendar — is the core claim of this section. These two registers are one argument: a permanent architectural solution to the specific problem created when a cosmological system loses its geographical anchor. The paper establishes the evidence, tests it against nine comparative landscapes, and states nine falsifiable predictions.

A note on these two registers and their evidentiary standards: the landscape-scale claims — relational geometry, directional orientation, water-feature correspondence, and comparative architecture — are supported by independent satellite surveys, hydrological reconstruction, archaeoastronomical measurement, and cross-disciplinary confirmation. They constitute the primary evidential core of this hypothesis. The dimensional-scale claims — mathematical correspondences through the Wedjat fraction system — are supported by arithmetic calculation, a fraction-system control test, and the Monte Carlo analysis reported below; they nonetheless remain the more novel register and the one most in need of independent replication. Throughout this paper, these two registers are maintained as distinct: landscape-scale observations are described in terms of correspondence and confirmation; dimensional-scale observations are explicitly framed as proposed and pending independent verification. A shift in register is always explicitly marked in the text; claims should be evaluated at the evidentiary level appropriate to the register in which they appear.

The argument does not end at Giza. The same cosmological grammar — central island, encircling water, bounded sacred space, east-west axis — was reinstated in megalithic stone at Abydos, at the primary Shemsu Hor institutional site, approximately 580 years after Giza — the interval given by surface luminescence dating of the Osireion's granite to 1980 ± 160 BCE (Liritzis and Vafiadou 2015). That structure, the Osireion, stands as the final architectural expression of the Nun cosmological tradition before that capacity was permanently extinguished; it is not examined in detail here.

A note on evidentiary categories: the evidentiary categories used in this paper are hierarchical rather than equivalent. Peer-reviewed archaeological and archaeometric findings constitute the paper's foundational evidentiary layer. Established interpretive scholarship is treated as supporting analysis. Later textual traditions with acknowledged temporal distance are used as contextual or corroborative material only. Original observations introduced in this paper are explicitly identified as hypothesis-generating rather than established findings.

The Anomaly: What the Evidence Demands We Explain

A note to readers: This paper is a continuation of the Zep Tepi Series. It does not re-establish the core findings of The Corridor Series: that the Richat Structure is the sole surviving candidate for the Saharan origin landscape under defined constraints (Corridor I); that the Zep Tepi cosmological system originated in the Green Sahara and was transmitted via the Wedjat sky-compass and cattle cosmogram (Corridor II); and that the Shemsu Hor preserved this knowledge across fifteen centuries of Egyptian memory (Corridor III). This paper is the first of The Destination Series. Readers unfamiliar with these foundations should consult those papers first. The present paper asks: given all that, what was built at Giza?

No Mummies Have Been Found in the Giza Pyramids

Mainstream Egyptology has long interpreted the Giza pyramids as royal tombs, pointing to the presence of sarcophagi, mortuary temples, and later textual traditions. However, the absence of primary burial equipment — mummies, canopic jars, grave goods, or identifying inscriptions — in the original chambers of all three Giza pyramids is striking. This paper does not dispute that the pyramids served a funerary function. It argues that the design priorities — particularly the engineered water, western axis, bare walls, and stellar alignments — are better explained by a cosmological purpose within which the king's transformation was nested. The two readings are complementary at different interpretive scales.

No mummy has ever been found inside any of the three great Giza pyramids. The King's Chamber in the Great Pyramid of Khufu contains a lidless granite sarcophagus — too wide to have been brought through the ascending passage, meaning it was placed during construction — but it is empty. No mummy, no canopic jars, no burial goods, no identifying inscriptions were present when Caliph al-Ma'mun breached the pyramid in approximately 820 CE.

Fragmentary bone material of uncertain date has been found in the burial chambers of the other two Giza pyramids. When Giovanni Belzoni entered the burial chamber of Khafre's pyramid in 1818, he found the granite sarcophagus containing bone fragments and rags, later identified as those of a bull and now linked to a Coptic-period intrusive deposit (Belzoni 1820; Lehner 1997). In 1837, Richard William Howard Vyse cleared the lower burial chamber of Menkaure's pyramid and discovered a large uninscribed basalt sarcophagus (empty, no inscriptions or burial goods), which was subsequently lost at sea (Vyse 1840). A separate wooden coffin with later-period human remains was also found. No intact burial, no identifiable royal remains, and no primary funerary equipment have been recovered from any of the three Giza pyramids.

The Amenpanufer confession (Papyrus Mayer B, Group VI) documents systematic stripping of royal mummies in the 20th Dynasty — approximately 1,200 years after Khufu's death. This accounts for the

absence of reusable valuables (gold, jewellery, canopic equipment) from later Valley of the Kings burials. It does not account for the total absence of any such traces in the Giza primary chambers: the contrast with later pyramids such as Unas and Teti, where Pyramid Texts and fragmentary physical remains are both present, remains stark and unexplained by robbery alone.

The sarcophagus in Khufu's King's Chamber is intact, lidless, and empty. It shows no evidence of burning or ash deposits. Had robbers destroyed the body inside the sarcophagus, residue would remain; none does. This rules out in-situ destruction, but not the more common practice of removing a body intact to strip it elsewhere, which would leave no trace. The undamaged, residue-free sarcophagus is therefore consistent with robbery as much as with non-burial, and is offered as one observation among several rather than as decisive on its own. However, the confession does not explain:

Hawass himself has publicly stated his belief that "Khufu's real burial chamber is still hidden inside the pyramid" (Piers Morgan Uncensored, 15 May 2026) — the field's own leading authority acknowledging the pyramid's primary purpose remains unresolved. This is corroborated by the peer-reviewed scanning record itself: the discovery of the Big Void (Morishima et al. 2017, *Nature*), the North Face Corridor (Procureur et al. 2023, *Nature Communications*), and the Menkaure voids (Helal et al. 2025, NDT & E International) confirm that the pyramid's interior geometry is not fully mapped. Hawass's November 2025 announcement of a 30-metre sealed corridor (Egypt Independent, 9 November 2025), pending peer-reviewed publication, adds further weight to the observation that undiscovered internal features may yet substantially alter the interpretive landscape.

1. **The sarcophagus:** The granite sarcophagus in Khufu's King's Chamber is too wide for the passages, meaning it was placed during construction and never extracted. It is empty.
2. **The structural design:** The ascending passage, Grand Gallery, and relieving chambers show no evidence of ever having been sealed and reopened — consistent with intentional non-interment design.
3. **The absence of Pyramid Texts:** Later pyramids (Unas, Teti) preserve fragments; Giza preserves nothing — even from secondary disturbances.
4. **The bare limestone:** The walls of the King's Chamber and Queen's Chamber show no cutting marks, no smoothing for inscriptions, no preparation for text — this is construction-finished surface, not stripped surface (Petrie 1883, pp. 67–69).

Robbery explains missing mummies. It does not explain the architectural design logic.

What the Egyptians Themselves Said: The Herdsmen and Shepherd Traditions

Herodotus (c. 450 BCE) recorded that his Egyptian priestly sources consistently placed Khufu's body elsewhere: "**in a place under the earth, on an island surrounded by water drawn from the Nile**" (Herodotus, *Histories* II.124). He also reported that the Egyptians of his time refused to speak the names of Khufu and Khafre — instead calling the pyramids after "the shepherd Philitis, who then pastured his flocks in this place" (II.128). Diodorus Siculus (c. 60 BCE), drawing on separate Egyptian sources, recorded a second tradition: "some say that [the pyramids] were built by the aforesaid kings, others say by herdsmen who had at one time held power" (Diodorus, *Bibliotheca Historica* I.64). Two independent Greek authors, four centuries apart, both transmit the same embedded memory: Egyptian oral and priestly tradition associated the construction of the pyramids with a pastoral people. As documented in Corridor III of this series, the Shemsu Hor were the cultural memory of Saharan pastoralists who arrived in Egypt from the west during the predynastic period (Corridor III, Levy 2026c).

A note on the Herodotus 'island surrounded by water' passage: mainstream Egyptology reads this as describing the pyramid's position — elevated above the Nile floodplain, approached by water via the

Khufu Branch canal (Sheisha et al. 2022). That reading encounters a difficulty the text itself poses: the passage specifies a place **“under the earth”**. The Great Pyramid is not under the earth. The Osiris Shaft is. Level 3 of the Osiris Shaft — approximately thirty metres below the plateau surface — features a central island permanently surrounded by a water channel fed by the Nile water table: precisely **“under the earth, on an island surrounded by water drawn from the Nile.”** Hawass identified this chamber as the symbolic Osirian burial at Giza — the Osirion of Giza — and it is the structure this paper identifies as the pre-existing Rostau sanctuary beneath the plateau. The most literal reading of Herodotus is that his Egyptian priestly sources were directing him to the underground sanctuary, not to the pyramid above it: in Egyptian tradition, Khufu’s sacred identity resided in his transformation as Osiris in the subterranean chamber below, not in the stone structure above. The mainstream reading and the present reading are not mutually exclusive — but the Osiris Shaft satisfies the passage’s literal description with a precision the pyramid’s above-ground position cannot match. A further philological note strengthens this reading: the Godley Loeb translation of Herodotus II.127 (the immediately adjacent passage, describing Khafre’s pyramid) reads: “the river comes in through a built passage and encircles an island, in which, they say, Cheops himself lies” — a description that, read carefully, locates Khufu’s burial within the water-circled island described under Khafre’s causeway, not in the pyramid above. The two water-feature traditions Herodotus records — the underground chambers accessed by Nile canal (II.124–126, Great Pyramid) and the canal-encircled island (II.127, Khafre complex) — are not contradictory: they describe different cosmological registers of the same sacred site, consistent with the reading advanced here that the Osiris Shaft’s subterranean Nun-island and the surface hydraulic infrastructure constitute a single integrated cosmological landscape.

A fourth witness to this tradition comes from within Egyptian literary culture itself. The Westcar Papyrus (P. Berlin 3033, c. 1650 BCE), set at Khufu’s court, portrays the king not as a tomb-builder but as a seeker of hidden knowledge — specifically, the secret chambers of Thoth. In a key passage, the sage Djedi tells Khufu about the number of hidden rooms in the sanctuary of Thoth, and the text connects this esoteric knowledge to Khufu’s building ambitions, with some scholars reading the language as carrying resonances of the king’s desire to create something comparable to the horizon (Blackman 1988; noting that the text’s primary register is literary entertainment and this reading is not universal). The resonance, where present, is striking: an Egyptian literary text, composed roughly a millennium after Khufu’s reign, associates the king with esoteric knowledge and cosmic aspiration rather than burial — echoing the monument’s own attested name *Akhet-Khufu* in the contemporary Merer Papyrus (Tallet 2017). The Westcar Papyrus was composed roughly a millennium after Khufu’s reign, yet it preserves a cultural memory in which the pyramid is inseparable from esoteric knowledge and cosmic aspiration — not from burial. Taken together, the Merer Papyrus, Herodotus, Diodorus Siculus, and the Westcar Papyrus represent four ancient sources of distinct type, provenance, and evidentiary weight — from contemporary administrative record to Greek historiography to Egyptian literary tradition, composed across a span of roughly a millennium — none of which primarily frames the Great Pyramid as a tomb. This convergence is noted as corroborative context, not as proof of the cosmological hypothesis; the sources differ too greatly in reliability and intent to be treated as a unified evidential body.

No Hieroglyphs Inside the 4th Dynasty Pyramids

The Egyptians were obsessive documenters. Yet the three Great Pyramids at Giza have almost completely bare interiors. Not a single dedicatory inscription, spell, or caption. The only text found inside the Great Pyramid is construction graffiti in the relieving chambers — work crew names never intended to be seen.

This absence is anomalous for a culture that inscribed its tombs and temples exhaustively, yet it is what a mnemonic reading predicts. In the comparative framework of Kelly (2015), a monument built to encode knowledge does so structurally — in form, proportion, and alignment — rather than in text: the information is carried by the architecture itself rather than written upon it. On that reading the bare

interiors are not an absence to be explained away but the expected signature of a structure whose content is encoded in its dimensions.

The argument advanced here is not that the Giza pyramids had no funerary role. The king's burial and apotheosis were real and ritually significant. The claim is that the primary design logic — the bare walls, the engineered water, the western axis, the stellar alignments, the absence of canopic equipment — is better explained as a cosmological monument than as a tomb that happens to have anomalous features. The two purposes are nested, not opposed. **A more precise formulation is that the pyramids' design is not optimized for burial** — the absence of canopic equipment, identifying inscriptions, and burial goods suggests that preservation of the body was not the primary architectural driver.

The content of the 5th Dynasty inscriptions confirms this reading. Utterance 213, inscribed on the south wall of the Unas pyramid's burial chamber (c. 2345 BCE), opens the entire inscription program with a formula that defines its purpose: 'O Unas, you have not gone dead, you have gone alive to sit on the throne of Osiris.' Not burial — transformation. The pyramid texts do not describe a dead king lying in a tomb. They describe a living king moving through a cosmological sequence. The 4th Dynasty builders required no written caption for this sequence because, as the texts themselves demonstrate, the sequence was already understood in its architectural form. The Pyramid Texts may be the verbal transcription of an architectural reality that preceded them by two centuries.

One structural observation is worth registering here for future investigation: the Great Pyramid contains three main chambers arranged on a strict vertical axis (Subterranean, Queen's, King's). This ascending vertical arrangement — with chambers rising progressively from bedrock through the pyramid's mass to near the apex — has no close parallel in any other Old Kingdom pyramid in the standard comparative record. Djoser's Step Pyramid presents a vertical arrangement of subterranean chambers below the pyramid but does not extend this progression into the pyramid mass above bedrock level; the ascending arrangement of internal chambers within the body of the structure itself is specific to the Great Pyramid. By contrast, standard Old Kingdom pyramid design places the burial chamber below the pyramid in the bedrock (Lehner 1997, pp. 108–113). The Bent Pyramid of Sneferu contains an upper chamber within the pyramid's masonry body (entered via the west-face passage ~33 m above ground), which might appear to parallel the Great Pyramid's ascending arrangement. However, archaeological evidence establishes that this upper chamber was a structural response to cracking discovered during construction, not a planned cosmological sequence: the tunnel connecting the two chambers was cut after initial construction as a remedial shoring measure, and the upper chamber was partially filled with rough masonry (Nuzzolo 2015). The ascending three-chamber progression in the Great Pyramid — Subterranean (bedrock), Queen's (lower body), King's (upper body), spanning the full vertical axis from geological base to near-apex — has no parallel in which all three tiers were an original, unified design intention rather than a structural accident or engineering correction. Belmonte and Lull (2023, pp. 396–398) confirm this by reading Sneferu's two Dahshur pyramids as a paired external cosmological statement — the pyramids together forming a symbolic dual whole, not an attempt at internal vertical ascent architecture — which situates Khufu's three-tier interior sequence as the specific innovation the 4th Dynasty achieved and that no predecessor had attempted. The ascending three-chamber arrangement is an anomaly that requires explanation within any model of the pyramid's function.

The Bare Walls: Why the Great Pyramid Has No Hieroglyphs

The standard Egyptological explanation for the bare interior is temporal: the tradition of decorating pyramid interiors with Pyramid Texts began with Unas (c. 2350 BCE), 200 years after Khufu. This is accurate but does not explain why the tradition began when it did.

This paper proposes a complementary explanation: the Pyramid Texts appear in the 5th Dynasty because the living understanding of the cosmological system was beginning to fade. The Shemsu Hor

period ended at approximately 3100 BCE. The Great Pyramid was completed approximately 540 years later — 20 generations. At that point, the cosmological tradition was still institutionally alive. The form of the pyramid was readable. No caption was necessary.

By the 5th Dynasty (c. 2494–2345 BCE), another 200 years had elapsed. The direct embodied understanding that made architectural form self-explanatory may have been attenuating. The Pyramid Texts are the verbal caption for an image that had once been self-evident.

The linguistic and architectural evidence supporting this temporal layering is well established in the Egyptological literature. James P. Allen, in the introduction to his standard translation of the Pyramid Texts, documents two convergent lines of evidence for the texts' antiquity relative to their 5th Dynasty inscription. First, "the language of the texts... is not the speech of the time they were first inscribed (ca. 2350 BC), but an older form that had been preserved in liturgical use, perhaps for as much as half a century or more." The texts were not composed fresh for Unas; they were copied from older sources preserving an already-archaic linguistic register. Second, "the architecture of the chambers in which they are inscribed also shows a similar lag, representing a form that had been established more than a century before Unas" (Allen 2005, p. 4). Morales (2016) documents the broader transmission mechanism: the Pyramid Texts passed through oral ritual performance, scribal manuscript culture, and finally wall inscription across successive media — each stage preserving material already archaic at the moment of transcription (Morales 2016; cf. Allen 2005, p. 4). The ritual and spatial logic encoded in the texts belongs to the pyramid-building tradition of Khufu's era, not to the moment of their first inscription.

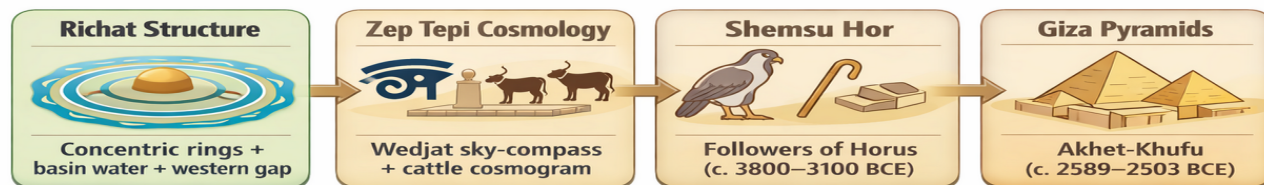
The Abusir Papyri (24th century BCE, Fifth Dynasty) document that pyramid complexes were functioning cult centers with priests and offering rosters (Posener-Kriéger 1976). The Fifth Dynasty wrote extensive administrative texts about pyramid cults. The Fourth Dynasty did not write inside pyramids because the living tradition of embodied cosmological knowledge may have made text unnecessary — in this reading, the stone itself was the statement.

A further line of evidence deepens this temporal picture. Manetho, the Egyptian priest-historian who compiled the *Aegyptiaca* in the early 3rd century BCE from native temple records, places pyramid construction in the First Dynasty under King Vavenephis: "Vavenephis, for 42 years. In his time famine seized the land. He reared pyramids near the town of Cho" (Waddell 1940, Loeb Classical Library, p. 33). The site identified as Cho — Sakkara — was, as the Loeb edition's own note records, confirmed by W.B. Emery's excavations of 1935–1938, which uncovered First Dynasty tombs containing stepped-brick pyramid constructions (Waddell 1940, p. 33 n. 3). This places the origins of the pyramid form approximately four centuries before Khufu, around 3000 BCE. Direct archaeological corroboration comes from the seven Minor Step Pyramids of the late 3rd and early 4th Dynasty at provincial centers across Upper Egypt — Elephantine, Naga el-Ghoneimiya (Edfu South), el-Kula at Hierakonpolis, Naqada, Sinki at Abydos, Zawiet el-Meitin, and Seila — identified as a coherent group by Dreyer and Kaiser (1980) and astronomically analyzed by Belmonte (2005). These structures are oriented to Meskhetyu (the Bull's Foreleg) at quarter-cardinal axes that cannot be explained by Nile flow or local topography (Levy 2026d, Phase Zero). The pyramid form was therefore already a century old, with stellar orientation procedures already established at provincial cult centers, by the time Khufu's architects laid out Akhet-Khufu. The 4th Dynasty monuments at Giza were not an invention but the culminating monumental expression of a tradition already ancient when Khufu was born. If the pyramid form was already centuries old, neither was the cosmological grammar it expressed a 4th Dynasty novelty. The bare walls of Khufu's chambers require no text not because the tradition had not yet developed, but because it was still fully embodied. This is precisely the pattern James Henry Breasted identified in 1912: the solar religion of the 5th Dynasty absorbed and transcribed an older stellar tradition whose living expression had been architectural, not textual (Breasted 1912). Breasted's "absorption" model was a description of a fading, not an origin — and Allen's linguistic evidence confirms that the Pyramid Texts were transcribed from material already archaic when they

were first inscribed (Allen 2005, p. 4). On this hypothesis, the stone spoke while the tradition lived. When it began to fade, the scribes began to write.

The Predynastic Grammar (c. 3,700–3,100 BCE)

The Shemsu Hor — the Followers of Horus — are the Egyptian cultural memory of the Saharan migrants who arrived in the Nile Valley during the 3800–3100 BCE window. **This paper assumes the reader is familiar with the full evidentiary chain established in Corridor Series (Corridor III, Levy 2026c).** In summary: the Shemsu Hor occupy a non-optional structural slot in Egyptian king-list architecture across fifteen centuries. Later pharaohs cite their written annals. Ptolemaic builders cite a goatskin scroll containing architectural plans.



African Humid Period

Arid Sahara

Transmission chain from the **Green Sahara** to Giza. Richat Structure (c. 14,800–5000 BP) → Zep Tepi cosmological system (Wedjat, cattle **cosmogram**) → Shemsu Hor transmission vector → 4th Dynasty Giza pyramids. Bottom gradient: green (humid) to tan (arid) showing climate change driving *along the Tamanrasset–Kufra paleoniver corridor* over ~12,000 years.

Phase Zero: The Bull in the Sky (Levy 2026d) establishes the institutional transmission chain across the 540-year window between the end of the Shemsu Hor period (c. 3100 BCE) and the completion of the Great Pyramid (c. 2560 BCE), through the Minor Step Pyramid tradition, the Djoser complex, and the archaeozoological evidence for Shemsu Hor continuity documented at Hierakonpolis. That argument is not reproduced in this paper; this paper assumes its findings as established.

The External Architecture of Giza (c. 2,589–2,560 BCE)

The Akhet: Not a Tomb but a Horizon

The original Egyptian name for the Great Pyramid was *Akhet-Khufu* — 'Horizon of Khufu.' The Merer Papyrus — the eyewitness logbook of an inspector who transported stone blocks during Khufu's reign — repeatedly refers to the structure by this name, most explicitly in the dated entries of Logbook A (Tallet 2017; Tallet & Lehner 2021). The *Akhet* is the liminal zone where the sun rises and sets, where the king transforms from human to divine. It is a threshold, not a storage place for a body.

The Pyramid Texts confirm what the name encodes. Inscribed on the antechamber walls of the Unas pyramid (c. 2345 BCE), Utterance 260 states: '*Unas comes out on this day in the true form of a living Akh-spirit.*' Not a dead king lying in a tomb — a **living** Akh-spirit emerging. Utterance 257 (antechamber north wall) then announces the completion: '*Unas takes heaven, he splits its brass... Unas goes to peace, being alive in the West... Jubilate for Unas, he has taken the Horizon!*' The pyramid's name — Akhet-Khufu, the place where Khufu takes the Horizon — is the event these inscriptions describe. The Pyramid Texts are the verbal transcript of what the 4th Dynasty architecture was built to produce.

The Benben and the Primordial Mound

Egyptian creation mythology consistently describes the origin of the world as a pyramid-shaped mound — the Benben — rising from the primordial waters of Nun. Pyramid Texts Utterance 600 addresses Atum directly: "O Atum-Khepri, when you became high as the mound, you rose up as the benben in the mansion of the bird Benu in Heliopolis" (Allen 2005, pp. 252–253). The pyramid form is this cosmology made architectural.

That the pyramid was understood as the physical vehicle of this ascent — not merely its symbol — is confirmed by the inscriptions on the passage leading to the burial chamber of the Unas pyramid (c. 2345 BCE). Utterance 267, inscribed on the passage south wall at the spatial threshold between antechamber and burial chamber, states directly: *'The earth is beaten into steps for him towards heaven, that he may mount on it towards heaven, and he rises on the smoke of the great fumigation. He flies away, this Unas, as a goose, he alights as a scarab on the empty throne which is in your boat, O Re.'* The physical structure is the steps. The architecture does not represent the ascent — it enacts it, and the inscriptions confirm this at the precise spatial location where the ascent is entered.

 **CRITICAL CAVEAT ON TEXTUAL SOURCES:** The textual sources cited here (Pyramid Texts, Coffin Texts, Edfu Building Texts, Dendera inscriptions) postdate the 4th Dynasty by centuries to millennia. They are **not** treated as direct historical records of Old Kingdom belief. Rather, they are treated as later codifications of cosmological concepts that the Egyptians themselves attributed to earlier periods — a standard method in Egyptology (Assmann 2001; Hornung 1996). This temporal distance is acknowledged as a limitation.

The following table makes explicit the chronological tiers of textual evidence used in this paper and the evidentiary weight assigned to each. Old Kingdom sources are given primary evidential status; Ptolemaic and later sources are used interpretively, with the temporal distance explicitly noted as a caveat throughout.

Source Layer	Period	Example Sources	Evidentiary Weight	Use in This Paper
Old Kingdom	c. 2686–2181 BCE	Pyramid Texts; Merer Papyrus	Primary — contemporaneous	Stellar theology; construction context; royal name
Middle Kingdom	c. 2055–1650 BCE	Coffin Texts; Westcar Papyrus	Secondary — c. 500 years after	Design template tradition; Khufu seeking plans
New Kingdom	c. 1550–1070 BCE	Book of the Dead; Ramesside sources	Secondary — c. 1000 years after	Cosmological continuity evidence
Late/Ptolemaic	664–30 BCE	Edfu Building Texts; Dendera	Interpretive — c. 2000 years after	Zep Tepi codification (explicitly caveated)
Graeco-Roman	c. 450 BCE–4th CE	Herodotus; Diodorus; Manetho	Late interpretive — memory residue	Corroborative; no independent architectural value
Medieval Arabic	9th–14th CE	Al-Maqrizi; Surid traditions	Memory residue only	Cultural echo; explicitly lowest evidentiary weight

The Zep Tepi Cosmology and the Green Sahara

This paper proposes that the Zep Tepi cosmological system originated in the Green Sahara and was carried to the Nile Valley by pastoralists displaced by the end of the African Humid Period. **This paper assumes the reader is familiar with the full evidentiary chain established in Corridor I and II**

(Levy 2026a, 2026b). In summary: Corridor I documents the Highland Corridor of elevated massifs and plateaus (Tassili n'Ajjer, Tadrart Acacus/Takarkori, Messak Settafet, Gilf Kebir) — supplemented by the Tamanrasset and Kufra paleorivers as eastern paleohydrological segments — as the primary migration route, identifies the Richat Structure as the probable landscape encoded as the primordial mound, and provides a 25-element testing protocol. Corridor II establishes the genomic continuity of the Saharan pastoralist lineage and documents the Wedjat sky-compass.

The identification of Richat as the specific origin landscape rests on a conjunction of three relatively rare characteristics: (1) concentric ring geometry — nested circular boundaries around a central elevation, uncommon in natural landscapes and preserved in the Zep Tepi cosmology as concentric sacred space; (2) basin-style water surrounding the mound — not linear or adjacent, but encircling the entire formation, matching the ‘island surrounded by water’ description in Herodotus and the Nun cosmology; and (3) documented human presence at the structure itself during the Green Sahara window — Acheulean and Neolithic tool assemblages within the same Holocene sediments that record the paleolakes. The elimination analysis is set out in full in Corridor I and extended in Corridor II (Levy 2026a, Section 3; Levy 2026b). Twenty-eight Saharan formations have now been evaluated against four criteria fixed in advance: sufficient physical scale to sustain a population across generations; independently documented Holocene water at the structure itself rather than merely in the surrounding region; documented physical evidence of human presence at the structure during that window; and a central elevation rising from that water. Twenty-seven fail at least one. Ten of those assessed are impact craters predating the Green Sahara, several by tens of millions of years. The massifs and plateaus — Tassili n'Ajjer, the Tadrart Acacus, the Gilf Kebir — have documented regional occupation but are ridges and tablelands rather than a unified mound within a water-retaining basin, and Nabta Playa is a basin, the geometric inverse of a mound. The most instructive failure is Semsiyat Dome: it shares the Richat’s exact geological origin and lies only 45 kilometres away, yet at roughly a tenth of the diameter it could never have sustained a lasting population. Scale, not ring morphology alone, does real work in the elimination. Only the Richat passes all four, and it alone advances to the seven-criteria textual filter applied in Corridor II, which it also passes. The hypothesis does not depend on the Richat specifically, but on a western Saharan source fitting the descriptors; the Richat is the only currently known candidate that fits all of them.

The Relational System: A Four-Element Framework

The hypothesis proposes that Giza instantiates a specific four-element relational system derived from the Richat landscape. Each element is defined below with measurable criteria.

Element	Definition	Richat Expression	Giza Expression	Testable Criterion
Central Elevation	A raised focal point at the geometric center	Central uplift (~80m above inner ring)	Khufu pyramid apex (146.6m)	Central structure dominant; largest single monument at site
Bounded Sacred Space	Enclosing circular/rectangular boundary	5+ concentric cuestas; enclosure walls	Mortuary temple + valley temple + enclosure wall	Boundary walls documented archaeologically
Transition Water	Water surrounding/approaching the mound	Paleolakes in annular depressions during Green Sahara	Ahramat Nile Branch; artificial harbor channels (Ghoneim et al. 2024)	Hydrological reconstruction confirms water proximity at construction epoch
Processional Axis	Directional path from water to	Wadi Bamouéré draining west from	Causeway + valley temple approach from	Causeway axis oriented to Nile; western alignment

Element	Definition	Richat Expression	Giza Expression	Testable Criterion
	monument	the outer ring; Wadi Akerdil east (Sao et al. 2008)	Nile	documented

This operationalization distinguishes the hypothesis from generic "sacred landscape" claims. Each element is independently verifiable at both the origin (Richat) and destination (Giza).

Every major physical element of the Giza pyramid complex corresponds to an element of the Zep Tepi cosmological system.

Table 1: Architectural Homology with Zep Tepi Cosmological Template

Zep Tepi Element	Physical Expression at Giza	Evidence / Source
Primordial mound (Benben) rising from waters	The pyramid itself — a mound rising to a pyramidal apex	Pyramid Texts Utterances 600, 587
Primordial waters of Nun	Khufu Branch canal (Sheisha et al. 2022) and Ahramat Branch (Ghoneim et al. 2024); engineered harbour basin (Younes et al. 2024)	Sheisha et al. 2022 PNAS; Ghoneim et al. 2024; Younes et al. 2024
Concentric rings	Enclosure walls (peribolos) surrounding the sacred precinct	Physical remains, Giza Plateau Mapping Project
Western origin — the homeland	Causeway directed westward toward the desert	Egyptian cosmological association of west with origin
Pyramidal form as 'first solid land'	Giza plateau — a natural limestone spur chosen because it was already elevated	Geological survey; Lehner 1997
Solar axis — where first sun touched the mound	Alignment to true north achieved via meridian transit of Meskhetyu (Bull's Foreleg / Ursa Major); within ~3.5 arc minutes	Belmonte 2001 (Meskhetyu pairing); Dash 2017; Spence 2000 (Kochab–Mizar pairing)

Causeway Orientations: A Consistent Western Pattern

The western orientation of the Giza causeways is not merely a qualitative impression. A Total Station survey of the Giza complex by Nell and Ruggles (2014) measured Khafre's causeway at azimuth 103° 26' from eight surveyed points, and Menkaure's at 89° 10' from six — that is, 283.4° and 269.2° respectively when expressed in the westward sense, toward the pyramid. Khufu's causeway is largely destroyed and was not surveyed by them; earlier measurements give a deviation from cardinality of -14° to -15°, an azimuth near 255° (Lauer 1947; Goyon 1969; Lehner 1985). The three do not cluster tightly. What they do is deviate in opposed senses about the east–west axis: Khufu roughly 14.5° south of west, Khafre roughly 13.4° north of west, and Menkaure within a degree of cardinal. Nell and Ruggles themselves describe the approximate symmetry of the Khufu and Khafre causeways about due east–west as noteworthy, while cautioning that it need not have been an end in itself for the

builders. The mean of the three azimuths is 269.4° , within 0.6° of due west. On the reading advanced here, the axis rather than any single causeway carries the orientation: three reigns produced three different deviations that resolve, taken together, to the cardinal westward line. This is a weaker pattern than uniform clustering would be, and it is stated as such.

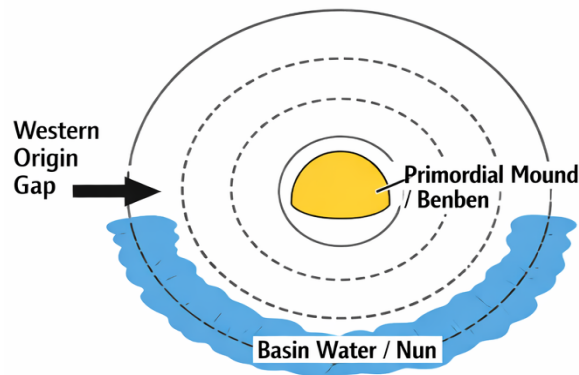
The broader survey confirms that east-west orientation in relation to sunrise and sunset was "a, or even the, key factor" (Nell & Ruggles 2014) in the layout of many associated structures. Khufu's mortuary temple reflects the orientation of the Great Pyramid itself, with a deviation from cardinality of about one arc minute; Khafre's valley temple and the adjacent Sphinx temple are aligned very close to the meridian, and Menkaure's valley temple gives an east–west alignment of $90.2^\circ/270.2^\circ$. Khufu's own valley temple is destroyed and could not be surveyed. One further finding of that survey bears directly on the western axis. The mastabas of Khufu's western cemetery deviate from cardinality by about -2.3° , where every other surveyed structure in his complex falls between about -0.6° and $+1.0^\circ$ — the mortuary temple at 0.0° , the surviving queen's pyramid at -0.3° , the southern cemetery at -0.6° , and the four rows of the eastern cemetery running from -0.5° to $+1.0^\circ$. Nell and Ruggles observe that the western cemetery is the one group whose eastern horizon the Great Pyramid itself obscures, and propose that these tombs were therefore oriented on sunset on the western horizon rather than on sunrise in the east — the same day, the opposite direction. They add, as their own suggestion, that the western orientations may instead reflect an older tradition in which Osiris, who had assumed the title Foremost amongst the Westerners, governed the direction in which the sun enters the netherworld. Either reading places a deliberate westward orientation at Giza, surveyed and published, independent of the causeways. The causeways thus participate in a directional grammar in which the western horizon—associated in Egyptian cosmology with the setting sun, the entrance to the netherworld, and the land of the dead—appears privileged. Khafre's northward deviation (283.4°) has an established alternative explanation: Nell and Ruggles, following Lehner (1997), suggest it arose from aligning the axis of the Sphinx temple, the south side of the Sphinx, and the south side of Khafre's pyramid. Magli (2009) additionally reads the Giza causeways within a solar framework. Neither explanation is excluded by the reading offered here, and both should be weighed against it. The two orientations — western origin and eastern theological culmination — operate within a single architectural complex, unified rather than contradictory. The organisation of all three causeways about the east–west axis, however, remains a feature of the Giza complex that invites explanation beyond purely utilitarian models.

The Giza Diagonal: Alignment to the Primeval Mound

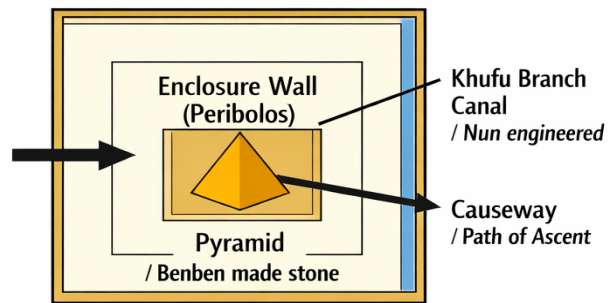
The directional grammar of Giza extends beyond the causeways to the placement of the pyramids themselves. Mark Lehner, director of Ancient Egypt Research Associates, documented that "a line extended through the southeast corners of the three pyramids... points directly toward the temple of Heliopolis, the center of the solar cult... This 'Giza Diagonal' suggests that the placement of each subsequent pyramid was not merely a matter of finding suitable bedrock, but was constrained by a predetermined sacred orientation toward the site of the Primeval Mound" (Lehner 1985b: 136). While the alignment is approximate rather than survey-grade precise, and visibility from Giza to Heliopolis remains debated, the diagonal pattern across three successive reigns is difficult to explain as coincidence. Magli (2009, 2013) has argued that the alignment represents a deliberate symbolic orientation toward the Benben cult center.

Heliopolis was the cult center of the Benben — the primordial mound of Egyptian creation theology. The alignment of Giza's pyramids toward Heliopolis is not merely a visual convenience. It is a theological statement: the pyramids, as architectural instantiations of the Benben, orient toward the cult center where that theology was most fully articulated. The Giza Diagonal encodes a relationship between the architectural mound — the pyramid — and the theological mound, the Benben at Heliopolis. Within the framework of this series, Heliopolis is the formalised theological expression of the same cosmological grammar whose origin landscape is argued in Corridor I to be the Richat Structure. Giza points not toward the origin but toward where the origin was remembered.

Richat Structure — Natural Cosmological Template



Giza Pyramid Complex — Architectural Expression



Same relational structure — different geometric expression

Not visual mimicry — conceptual translation

On the visual relationship between Richat and Giza: The correspondence proposed here is not visual or geometric. The Richat Structure and the Giza complex differ in scale, symmetry, and spatial organization. Richat is a large circular geological formation with concentric rings; Giza consists of angular, rectilinear architectural complexes. Richat's water system was basin-like and surrounding; the Nile branches at Giza were linear and localized. The argument advanced here is therefore not one of visual imitation. Rather, it concerns the instantiation of a cosmological grammar: a central elevated form, spatial separation between a sacred core and the surrounding world, water associated with the transition zone, and a directional orientation toward the west.

The western causeway at Giza exemplifies this directional grammar. The three causeway orientations documented by Nell and Ruggles (2014) and by earlier surveys deviate in opposed senses about the east–west axis and average to within a degree of due west, a configuration consistent with a cosmological rather than a purely topographic determinant, though not by itself decisive between them. These elements are argued to have been re-expressed within the constraints and conventions of Old Kingdom Egyptian architecture. Such transformations from natural or symbolic forms into rectilinear architectural expression are consistent with broader patterns in Egyptian sacred design (e.g., the circular heb-sed court becoming the square temple plan). The homology proposed is structural and conceptual, not visual.

The Pyramid Texts anchor this western orientation explicitly in the theology of transformation. Utterance 257 (antechamber north wall of the Unas pyramid, c. 2345 BCE) announces: '*Unas goes to peace, being alive in the West.*' The king does not merely travel toward the west — he **lives** in it. And Utterance 214 (sarcophagus chamber south wall) identifies the western direction as a title of the fully transformed king: '*so that you become the One-at-the-Head-of-the-Westerners.*' 'Foremost of the Westerners' is among the highest epithets of the transformed king in Egyptian theology. The Giza causeway does not simply face west; it faces the direction the Pyramid Texts themselves name as the dwelling place of the glorified dead.

The Bull's Foreleg: Giza's Stellar Anchor

The most precise architectural feature of the Great Pyramid — its alignment to true north within ~3.5 arc minutes — was achieved, on the most extensively developed peer-reviewed model, by observation of the constellation Meskhetyu (the Bull's Foreleg, equivalent to the modern Plough or

Ursa Major). Belmonte (2001) demonstrates that the simultaneous meridian transit of Megrez and Phecda — two stars of the Bull's Foreleg — provided the reference for accurate north-south orientation, with the alignment achieving its greatest precision at approximately 2562 BCE. In their 2023 synthesis of Egyptian astronomy, Belmonte and Lull reaffirm this hypothesis, noting that "this working hypothesis posed no apparent chronological problems" and that "Khufu's pyramid could have been aligned at dates close to 2556 BC, with Meskhetyu in lower culmination" (Belmonte & Lull 2023, p. 15). The same procedure has been independently identified at the predynastic structure HK29A at Hierakonpolis (Kom el-Ahmar) and at the Osireion of Seti I at Abydos (Belmonte 2001, Fig. 8) — spanning more than two millennia of Egyptian sacred architecture.

The alignment procedure itself belonged to a ritual tradition of far greater antiquity: the "stretching of the cord" ceremony, whose surviving Ptolemaic temple texts "repeatedly mention Meskhetyu... as the appropriate celestial target" toward which the pharaoh looked when fixing a temple's orientation (Belmonte & Lull 2023, p. 15). At Dendera and Edfu, "the northern, imperishable, ancient Egyptian constellation of Meskhetyu is advocated as the target the Pharaoh was looking at when fixing the orientation of the temples" (Belmonte & Lull 2023, p. 378). The ceremony depicted on these temple walls preserves a ritual grammar whose architectural expression at Giza predates its surviving textual form by more than two millennia.

Meskhetyu was, in Belmonte's phrasing, "one of the most important asterisms of Egyptian religion since at least the Old Kingdom, if not earlier" (Lull & Belmonte 2006). It appears in the Pyramid Texts as the "imperishable star" per excellence — the destination of the king's ascent. Belmonte and Lull (2023, p. 252) state its identification unequivocally: "Without the slightest doubt, this asterism is none other than the Plough or Big Dipper, the most characteristic element of the constellation of Ursa Major." The constellation's dual identity — adze and bull's foreleg — is documented in the Egyptian textual record itself: "The earliest surviving mentions of Meskhetyu... refer to this constellation with the adze determinative... Later, Meskhetyu is written with the bull's leg determinant, thus recognizing the way in which this constellation was also observed" (Belmonte & Lull 2023, p. 252). This iconographic evolution is confirmed on Middle Kingdom coffin lids, where "Meskhetyu is represented in the shape of a bull's leg, and the seven stars indicated in it lend it a shape that reminds us of the Wagon" (Belmonte & Lull 2023, p. 252). The same constellation is used in the Opening of the Mouth ritual, where the bull's foreleg is the implement by which the senses of the deceased are restored. The province of Letopolis, whose capital sat near Heliopolis, bore the name "Bull's Foreleg" (Belmonte 2010).

This is structurally significant. The cattle cosmogram established in Corridor II (Levy 2026b) and the predynastic cattle cult at Hierakonpolis HK6 documented in Shemsu Hor (Corridor III, Levy 2026c) — including the deliberate horn modification of livestock first attested anywhere in the world (Van Neer et al. 2024) — share a single unifying symbol with the constellation that anchored the Great Pyramid's most precise architectural alignment. The iconographic evolution from adze to bull's foreleg traces the same symbolic chain — implement, animal, constellation — that connects the pastoralist bull cult at HK6 to the astronomical orientation at Giza. The bull on the ground at Hierakonpolis and the bull in the sky over Giza are not separate phenomena. They are the same symbol expressed at different scales: ritual, anatomical, celestial, and architectural. The Bull's Foreleg connects the cattle pastoralism of the Saharan corridor to the stone of the Old Kingdom by the most direct route imaginable — the orientation procedure used to set the cardinal axis of the monument itself.

Phase Zero: The Pyramid Tradition Predates Khufu

A standard objection to any cosmological reading of Giza is that the pyramid form itself emerged in the 4th Dynasty as a royal innovation. The archaeological record contradicts this. Seven small step pyramids — collectively known as the Minor Step Pyramids (MSPs) — were built in the late 3rd and early 4th Dynasty at provincial centers across Upper and Middle Egypt, predating Khufu by approximately a century. Dreyer and Kaiser (1980) first identified them as a coherent group; Ćwiek

(1998) systematized their dating and function; Belmonte, Shaltout and Fekri (2005) measured their orientations and demonstrated that they were astronomically motivated, not topographically; and Marouard and Papazian (2012) have re-investigated the Edfu MSP in the field, confirming the late 3rd / early 4th Dynasty horizon. The MSPs are located at or near the most important provincial capitals of the Old Kingdom: Abu (Elephantine), Behedet (Edfu), Nekhen (el-Kula at Hierakonpolis), Nubt (Naqada), and This-Abydos (Sinki).

The fullest pre-Khufu expression of this tradition at royal scale is Djoser's Step Pyramid at Saqqara (c. 2,650 BCE), designed by Imhotep. The Step Pyramid is the first full-scale stone pyramid in the Egyptian record and the first monument to integrate the pyramid form with a comprehensive royal cosmological program: stellar north orientation, enclosure walls mimicking the palace facade (per-ineb), and a Heb-Sed court designed specifically to provide a location for the king's ka to perform regenerative rituals after death. Djoser's complex is architecturally pivotal because it demonstrates that the institutional infrastructure — a dedicated priestly class, a stellar-oriented pyramid, a ka-ritual tradition — was fully operational at royal scale some decades before the 4th Dynasty. The MSPs and Djoser's Saqqara complex together establish that Khufu's architects inherited a tradition already mature in both its cosmological grammar and its monumental expression.

The el-Kula pyramid is the case that closes the loop. Located approximately six kilometers from the predynastic cattle cult center at Hierakonpolis HK6, el-Kula is oriented at a quarter-cardinal axis $\sim 44^\circ$ west of north, with a secondary axis declination of $66\frac{3}{4}^\circ$ placing the orientation on Meskhetyu at lower culmination in the Old Kingdom epoch (Belmonte, Shaltout & Fekri 2005, Table 1). Its sister structure at El Ghoneimiya marks "the last point of Egypt where Meskhetyu is circumpolar." In their 2023 synthesis, Belmonte and Lull reaffirm that el-Kula, Sinki, and El Ghoneimiya form a "quarter-cardinal family" whose orientations "could be explained if they were orientated towards the lower culmination of Meskhetyu, the Bull's Foreleg, the most characteristic Egyptian circumpolar constellation" (Belmonte & Lull 2023, p. 390). Phase Zero (Levy 2026d) develops this point at length: el-Kula is the architectural-astronomical expression of the same Bull constellation whose physical analogue — the cattle herd — defined the material culture at the same site.

The implication for the present paper is direct. The cosmological grammar of the Giza complex did not appear ex nihilo in the 4th Dynasty. By the time Khufu's architects laid out Akhet-Khufu, the pyramid form, the stellar orientation procedure, and the bull-symbol cosmology had already been operating in stone for at least a century at provincial centers across Upper Egypt. The 4th Dynasty did not invent the pyramid tradition. It monumentalized it. The MSPs are the missing intermediate stage between the predynastic cattle cult at HK6 (c. 3700 BCE) and the Giza complex (c. 2560 BCE) — an architectural bridge that materially confirms the institutional continuity required by the Shemsu Hor transmission model (Corridor III).

The Zep Tepi cosmology describes the primordial mound emerging from the waters of Nun — a landscape of water and elevation, of boundaries and directional axes. The preceding sections have established that the Giza pyramids encode this relational system in their external form. But that encoding depends on a physical premise: the water must actually have been there. This section reviews the hydrological evidence confirming that it was.

The Khufu Branch Canal (2022 PNAS Study)

For centuries, a fundamental logistics problem remained unsolved: the Nile today runs several kilometres from the Giza plateau. How did multi-tonne stone blocks get from the river to the construction site? In 2022, Hader Sheisha and colleagues published findings in *PNAS* confirming the existence of the 'Khufu Branch' — a now-buried Nile channel that ran directly to the base of the Giza plateau during the Old Kingdom. Using pollen-derived vegetation patterns from sediment cores, they reconstructed 8,000 years of fluvial variation on the Giza floodplain and found that the Khufu branch remained at roughly 40% of its Holocene maximum water level through the reigns of Khufu, Khafre and Menkaure — a level they conclude facilitated the transportation of construction materials to the

site. Merer's diary confirms the practical use, as the inspector describes the 'Lake of Khufu' as a delivery point for limestone blocks (Tallet 2017; Tallet & Lehner 2021).

The Ahramat Branch (2024 Study) and the Engineered Harbour

In 2024, Eman Ghoneim and colleagues published findings in *Communications Earth & Environment* identifying a larger extinct Nile branch — the Ahramat ('Pyramids') Branch — running alongside 31 Egyptian pyramids, using radar satellite imagery, geophysical survey and sediment cores. Many of the causeways associated with those pyramids run perpendicular to the reconstructed branch and terminate at valley temples, which Ghoneim and colleagues interpret as river harbours.

The engineering detail matters. A 2024 study led by Younes, Marriner, Kaniewski, and Sheisha extracted sediment core Giza-3 from the proposed harbour basin itself. Their palaeoenvironmental reconstruction confirms that the harbour was not a natural embayment — it was a deliberately cut and maintained basin, active across the reigns of Khufu, Khafre, and Menkaure (Younes et al. 2024). The Egyptians did not simply build at the water's edge. They engineered the water's edge to be there.

The Cosmological Significance of the Water

The canal and the river branch were integral to the design. The pyramid was meant to rise from water. The Zep Tepi describes the primordial mound emerging from the waters of Nun. The architects engineered the water to be there. When viewed from the Nile, the pyramid would have appeared to rise directly from the water — exactly as the Benben rises from Nun.

A note on functional versus symbolic interpretation: The proposed framework does not deny that the Khufu Branch canal served logistical purposes (stone transport) or that the causeway facilitated construction. Rather, it argues that functional systems were organized in accordance with a pre-existing cosmological structure. Dual-use is not a weakness of the hypothesis but an expectation: sacred architecture routinely co-opts practical infrastructure. The Egyptians did not have to choose between building a canal for transport and building a canal that symbolically enacted Nun. They did both simultaneously.

The valley temples are positioned at the water's edge, with causeways running up to the mortuary temples. The causeway is the cosmological axis of ascent — the path from water to mound, from Nun to Benben, from mortal to divine. Every procession along the causeway enacted the Zep Tepi.

This cosmological logic was not confined to the plateau's surface. The Osiris Shaft — a multi-tiered subterranean structure descending 30 metres into the Giza bedrock, located beneath the causeway connecting the Sphinx to the Second Pyramid (for the Sphinx's own dating and proposed original name, see Levy 2026e) — provides independent confirmation that water's cosmological role at Giza was enacted at every level of the landscape. At its lowest tier, the builders descended to the actual Nile water table and constructed a central island surrounded by a groundwater-fed moat. The excavator Zahi Hawass reads the design as an enactment of Nun cosmology. The trench formed by the emplacement, he writes, “takes the shape of a house plan, the hieroglyph pr”, and “is generally filled with water”, so that “the burial chamber is thus similar to an island with the sarcophagus in the middle surrounded by water in the shape of the pr sign”. He links that water to Osiris's role as god of vegetation and resurrection and to his identification as creator god, “appearing on the primeval mound surrounded by the Nun”, and states his conclusion as a contention: “It is my contention that this shaft complex is a dummy tomb for the god Osiris” (Hawass, Z.A. 2007. “The Discovery of the Osiris Shaft at Giza.” In Hawass and Richards (eds.), *The Archaeology and Art of Ancient Egypt: Essays in Honor of David B. O'Connor*, Vol. I. ASAE Cahier 36. Cairo: Supreme Council of Antiquities, pp. 379–397). The ancient Egyptians named the entire Giza plateau pr wsir nb rstaw — “the House of Osiris, Lord of Rosetau” — with Rosetau referring specifically to the network of underground passageways beneath it, and the hieroglyphic word pr (“house”) was inscribed in the floor of the lowest chamber itself

(Hawass 2007; for the Shetayet of Sokar at Rosetau, Edwards 1986). The plateau's own ancient name encodes its underground cosmological function: the entire Giza complex, from its engineered harbour at the surface to the water table 30 metres below, was understood as a single cosmological landscape. The Osireion of Seti I at Abydos replicates the same island-in-water design, identified by Egyptologists as a physical enactment of Nun with the Primordial Mound at its centre (Frankfort, de Buck & Gunn 1933). That two major sacred sites — Giza and Abydos — built to the same subterranean Nun cosmology template establishes an architectural typology: the island-in-water Nun chamber was a documented method of physically instantiating the creation moment at Egyptian sacred sites. The causeway carried the cosmological grammar above ground; the Osiris Shaft carried it below.

The Osiris Shaft and the Pyramid: A Designed Cosmological Pair

The tradition of paired cosmological structures at a single royal site is documented in Egyptian architecture before Khufu. Magli (2016, *Time and Mind* 9(1): 57–74) argues that duality in the funerary cult is a defining characteristic of the pyramid tradition from its origins: Djoser's Step Pyramid complex at Saqqara contains two almost identical underground tombs — one beneath the pyramid and one at the south boundary — and Snefru's Dahshur project pairs two pyramids with complementary cosmological functions. Magli's central argument in that same *Time and Mind* paper is that the Giza plateau itself was conceived as a unified "written landscape" in which the Khufu and Khafre complexes form a single double project — a cosmological statement neither complex can make alone. The Osiris Shaft sits beneath the Khafre causeway, inside this documented landscape. Academic Egyptology further recognises that the Duat was conceptualised simultaneously as subterranean and celestial — not one or the other (Hornung 1999, *The Ancient Egyptian Books of the Afterlife*, Cornell University Press; Zago, S. 2022. *A Journey through the Beyond*. Lockwood Press — documenting that within the Pyramid Texts "the Duat's location and features [were] conceived as flexible and variable"). The pyramid's ascending chambers enact the celestial Duat; the Osiris Shaft enacts the subterranean Duat. Both are the Duat. The shaft enacts the same three cosmological registers as the pyramid — the living world, the realm of the dead, the primordial waters — but stacked top-to-bottom as descent, inverted against the pyramid's bottom-to-top ascent. On the hypothesis that Giza encodes the Zep Tepi cosmological system at landscape scale, the pyramid and the shaft may together constitute its complete expression: descent to the primordial waters of Nun and ascent to the Imperishable Stars, the full arc of the cosmological circuit encoded permanently across 270 metres of plateau.

The Edfu Building Texts as Corroboration

The Edfu Building Texts describe the Seven Sages — ancestors who survived the destruction of the 'Homeland of the Primeval Ones' in the Far West — as the builders of the original sacred mound in Egypt. The Egyptologist E.A.E. Reymond, in her analysis of the Edfu Building Texts, describes the Homeland of the Primeval Ones as featuring a sacred island or mound set within primeval waters, with associated enclosures and water channels that echo concentric-ring geometry (Reymond 1969, pp. 75 ff.). Their activities included initiating construction at 'the Great Primeval Mound' and establishing the architectural template for all subsequent Egyptian sacred buildings.

The institutional context for this transmission is materially documented at Edfu itself. The temple's "house of the book" (pr mDAṯ) has preserved the titles of at least 31 works inscribed on its walls, including several dedicated to astronomy (Belmonte & Lull 2023, pp. 106–107). Among these, one scroll is entitled "Knowledge of the periodic returns of the sun and the moon" and another bears the title "The government of the periodic returns of the stars" — texts that, in the assessment of Belmonte and Lull, "had to include long lists of observations that would allow the calculation of the risings, settings, or transits of all these celestial bodies" (Belmonte & Lull 2023, p. 107). The presence of a dedicated astronomical archive at the very temple whose Building Texts invoke the architectural knowledge of the Shemsu Hor is consistent with the model of institutional transmission proposed here: an active scribal tradition preserving and applying celestial observations across millennia, from the time of the Followers of Horus to the Ptolemaic period.

The medieval Islamic tradition regarding the pyramids represents a sophisticated intellectual effort to integrate pre-Islamic monuments into a sacred, universal history. As Colavito (2021) puts it in a general-readership survey of the tradition, Islamic scholars "recreated the pyramids as evidence of a vanished antediluvian world of scientific wonders whose glories anticipated those of Islam"; the scholarly treatment of the same material is Fodor (1970) and Smith (2007).

The Surid Legend: Antediluvian Knowledge in Stone

The most detailed Arabic tradition attributes the pyramids to Surid ibn Salhouk (also known as Saurid), a legendary king said to have ruled Egypt 300 years before the biblical Great Flood. According to al-Mas'udi's 10th-century *Akhbar al-Zaman*, Surid dreamed of a cataclysm — "the fixed stars had come down" and "had grabbed the people and had hurled them between two mountains" — which he interpreted as divine warning of an impending deluge. In response, he commanded the construction of the pyramids as repositories for "the sciences of arithmetic, geometry, astronomy, and medicine," along with the mummified bodies of his royal ancestors and their treasures.

Scholarly analysis traces this legend to Coptic Christian adaptations of older Gnostic or Sabian lore around the 4th–5th centuries CE, with Surid representing a conflation of the historical pharaoh Khufu (Suphis in Manetho) with prophetic archetypes such as the biblical Enoch, the Islamic Idris, and the Greco-Egyptian Hermes Trismegistus (Fodor 1970). The name "Surid" itself derives from "Suphis" — the Greek rendering of Khufu's name. As Smith (2007) notes, "The story of Surid and his antediluvian construction of the pyramids assigns to them a place in sacred history and establishes a neutral narrative ground upon which Muslims and Christians could agree" (Smith 2007).

The legend is not treated here as literal history. But the methodological literature on traditional transmission cautions equally against the opposite error of dismissing such material as pure invention. Vansina (1985) established that traditions may preserve recoverable information across long transmission chains when assessed critically, and Barber and Barber (2004) document cases in which myth demonstrably encodes real events — the Klamath account of the formation of Crater Lake, for instance, corresponds to the geological record of the eruption and was transmitted for roughly eight thousand years. The relevant question for the Surid tradition is therefore not whether it is history, but whether its specific and unusual claim — that the pyramids were built as repositories of arithmetic, geometry, and astronomy — preserves a kernel of the encoding logic proposed here, however distorted by its later Coptic and Hermetic transmission.

Al-Ma'mun and the Opening of the Great Pyramid

The Abbasid Caliph al-Ma'mun (r. 813–833 CE) is credited with the first documented entry into the Great Pyramid during the Islamic period. According to Arabic sources, he was "seized with the desire to open them" and employed fire, acid, iron tools, and catapults to breach the structure (Al-Maqrizi, *Al-Khitat*). While the caliph's brief stay in Egypt makes his personal involvement unlikely, scholars note that "it may be presumed that it was about that time that further progress into the interior of the Pyramid of Cheops was made for the first time in the Muslim period," motivated by the widespread belief that "rich treasures were concealed there" (*Encyclopaedia of Islam*, Brill).

The Flood Narrative: Knowledge Preserved Before Catastrophe

The Islamic tradition's emphasis on the pyramids as antediluvian repositories of knowledge has thematic resonance with the Zep Tepi framework. This resonance is noted as corroborative folklore, not as historical evidence. The Surid legend is a post-pharaonic Arabic construction with no documented continuity from Old Kingdom sources; it cannot serve as independent confirmation of any claim made in this paper and is presented here solely to establish the depth and persistence of the knowledge-preservation reading of the pyramids across cultural traditions. A 17th-century translation of an Arabic manuscript records that Abd al-Hokim reported: "The greatest part of chronologers

affirmed that he who built the pyramids was Surid ibn Salhouk, the king of Egypt, who was before the Flood 300 years" (Greaves 1646, citing *al-Morat Alzeman*).

Parallel to the Zep Tepi Hypothesis: This tradition — of knowledge encoded in stone before a catastrophic deluge — mirrors the Green Sahara's termination by the African Humid Period's "flickering" dry events (Trauth et al. 2024), which this series has argued created the migration pressure that carried the Zep Tepi cosmology to the Nile Valley. The Islamic scholars who walked into the empty chambers and sensed that knowledge was present were not wrong. They simply could not identify the form in which it was encoded. The knowledge was not inside the pyramid in scrolls or tablets. It was the pyramid's form itself.

Early Attempts at Hieroglyphic Decipherment

Significantly, Arabic scholars made early attempts to decipher hieroglyphs. In the 10th century, the Iraqi scholar Ibn Wahshiyya published *Kitab Shawq al-Mustaham (Ancient Alphabets and Hieroglyphic Characters Explained)*, in which he recognized that Egyptian symbols represented words — an insight that anticipated Champollion by 800 years. The 1806 English translation notes that Ibn Wahshiyya "claimed to have been able to decipher many ancient Egyptian hieroglyphic signs, identifying their phonetic values and meanings". While some scholars are skeptical of his success, others credit him with anticipating Champollion and Young by recognizing that hieroglyphic signs had phonetic as well as symbolic values. This intellectual heritage places Islamic scholarship at the origins of Egyptology.

Modern non-destructive scanning technologies have opened a new chapter. The ScanPyramids project (Cairo University and France's HIP Institute, launched 2015) has used muon tomography, ground-penetrating radar, and electrical resistivity to detect previously unknown interior spaces.

The Big Void (2017)

In November 2017, the ScanPyramids team published in *Nature* the discovery of a large unknown void above the Grand Gallery (Morishima et al. 2017). Confirmed by three independent detection methods, the void is at least 30 metres long. Its purpose remains unestablished. Two peer-reviewed construction-infrastructure interpretations have since been proposed. Rosell Roig (2026), using a coupled parametric-geometry / discrete-event-logistics / finite-element framework, suggests that several muography anomalies — the North Face Corridor, together with reported cavities and notches, though explicitly not the Big Void, which his model does not predict — may be partially related to backfilled construction infrastructure: temporary edge-integrated ramps later closed top-down. Scheuring (2025) independently argues that the Grand Gallery and Ascending Passage functioned as sliding-ramps for a counter-weight pulley system, with the Big Void and North Face Corridor representing upper sliding-ramp infrastructure. Both authors are explicit that these are hypothesis-generating correspondences rather than confirmed void geometries, and that current scanning resolution cannot distinguish construction infrastructure from other heterogeneities. These interpretations are mentioned here for completeness; they are not in tension with the cosmological reading developed in this paper, since construction technique and architectural symbolism operate on independent dimensions.

The North Face Corridor (2023)

In March 2023, the ScanPyramids team published in *Nature Communications* the confirmation of a hidden corridor on the pyramid's north face (Procureur et al. 2023). The corridor is approximately 9 metres long, located behind the chevron blocks above the main entrance. The endoscope captured bare limestone walls and a vaulted ceiling — no artifacts. The corridor was subsequently confirmed by two independent multimodal studies: Schumacher et al. (2025) combined ground-penetrating radar, ultrasonic testing, and electrical resistivity tomography in a discrete-wavelet-transform image fusion, and Pugacheva et al. (2025) applied ERT alone to provide volumetric data, reporting average internal dimensions of approximately 2.5 m by 2.5 m, beginning at a depth of about 1 m and extending at least

2 m into the pyramid. The corridor's existence is therefore established by four independent geophysical methods (muon tomography, GPR, UST, ERT) across three peer-reviewed publications. Rosell Roig (2026) reports that the four-ramp configuration of the IER predicts an upper haulage lane crossing the north face at approximately 39.4 m elevation with a horizontal setback of about 13.4 m — a position the author identifies as a quantitative geometric coincidence with the NFC depth envelope reconstructed from Pugacheva et al. (2025). The author explicitly characterizes this as geometric correspondence rather than functional attribution, and the present paper retains the cosmological reading of the chevron blocks (their visual resemblance to the akhet hieroglyph) as a separate, complementary line of analysis.

The V-shaped chevron blocks above the main entrance bear a visual resemblance to the akhet hieroglyph — the horizon. The pyramid's own name, Akhet-Khufu, means 'Horizon of Khufu.'

Menkaure Voids (2025) and the Hawass Sealed Corridor (2025)

In late 2025, researchers published findings in *NDT & E International* identifying two previously unknown air-filled voids behind the eastern granite facade of the Menkaure pyramid (Helal et al. 2025). The first void is located at a depth of approximately 1.4 m, with dimensions of roughly 1.0 m by 1.5 m; the second is at a depth of approximately 1.13 m and measures approximately 0.9 m by 0.7 m. Both lie behind a section of unusually polished granite blocks (about 4 m by 6 m) on the eastern face whose finish resembles the polished surround of the known northern entrance. Helal and colleagues interpret the voids as supporting the hypothesis — previously raised on architectural grounds — that the eastern face once concealed a second, unopened entrance.

In November 2025, Egyptologist Zahi Hawass announced at a public book fair the discovery of a 30-metre corridor inside the Great Pyramid ending at a sealed door (Egypt Independent, November 9, 2025; Khaleej Times, November 9, 2025). **At the time of writing, this finding remains a news announcement only; no peer-reviewed publication has appeared. Pending peer-reviewed publication, this paper does not rely on this unverified discovery as evidence.** The present analysis therefore restricts itself to the peer-reviewed discoveries (Big Void, North Face Corridor, Menkaure voids).

Together, the peer-reviewed discoveries (Big Void, North Face Corridor, Menkaure voids) confirm that the Giza pyramids are not fully mapped. These structures, built 4,500 years ago, are still revealing their geometry. Ongoing systematic scanning has now extended to the Pyramid of Khafre: the SciDEP (Scintillator Imaging Detector for the Egyptian Pyramids) Collaboration is constructing muon telescopes to scan Khafre from multiple viewpoints, both inside the king's burial chamber and from the exterior (Gamal et al. 2025). No interior scan data has been published from Khafre at the time of writing; but the instrumentation development and simulation framework are complete, and the project represents the systematic extension of pyramid scanning beyond Khufu.

The preceding sections have established the landscape-scale argument: that the Giza complex instantiates, at architectural scale, the four-element relational system that originated in the Richat landscape and was carried northeast through three thousand years of documented institutional memory. The following section examines the second register of the hypothesis. Where the landscape-scale argument is primarily spatial and relational, the dimensional-scale argument is mathematical and astronomical: it proposes that the pyramid's exterior measurements preserve, through the Wedjat binary fraction system, a structured body of knowledge whose scope extends from fundamental constants of geometry to multi-generational stellar cycles. The two registers are independent lines of evidence. Their convergence on the same structure is the core of the hypothesis.

The Encoded Memory: What the Pyramid May Preserve

This section presents a structured investigation into mathematical and astronomical correspondences between the Great Pyramid's surveyed exterior dimensions and a range of constants, cycles, and quantities spanning multiple domains of knowledge. All calculations were performed directly against Petrie's (1883) and Dash's (2017) surveyed dimensions using the exact Wedjat binary fractions. Results were independently verified by arithmetic computation (see AI Use Disclosure). The mathematical relationships documented here are arithmetically verifiable by any reader. The interpretation — that they represent deliberate, structured knowledge preservation by the tradition described in The Corridor Series — is this hypothesis's central claim. It is not proof. Readers are invited to evaluate whether the pattern of correspondences, their precision, their domain coherence, and their structural consistency are more consistent with deliberate encoding or with coincidental numerical proximity.

The section is tiered. Findings whose underlying geometric and metrological relationships are documented in peer-reviewed literature are identified as established, with citations. Mathematical correspondences proposed as novel observations specific to this hypothesis are explicitly noted as such. The distinction is maintained throughout. No claim is presented as more certain than the evidence supports.

The peer-reviewed foundation for the mathematical claims in this section is threefold. At the metrological level, Zapassky, Gadot, Finkelstein and Benenson (2012, PLoS ONE) demonstrate through volumetric analysis of hundreds of ancient Egyptian and Phoenician ceramic vessels that the royal cubit was not an arbitrary anthropometric measure but the fundamental unit of a sphere-based geometric measurement system: a sphere whose circumference equals one royal cubit has a volume of exactly $\frac{1}{2}$ hekat. The authors note that royal cubit rods kept in Egyptian temples carried inscriptions affirming their spiritual significance — “The cubit is life, prosperity, and health, the repeller of the rebel” — consistent with this paper's argument that Egyptian metrology was cosmologically embedded. At the construction geometry level, Rossi (2004, Cambridge University Press) documents that the seked of 5.5 palms — confirmed for the Great Pyramid across survey data from over 80 pyramids — produces π -proportions and ϕ -proportions as verifiable geometric consequences of the construction method; the canonical dimensions (base 440 cubits, height 280 cubits) yield accurate approximations of π and ϕ through this seked to within 1% (Rossi 2004; Herz-Fischler 2000). This paper's additional claim is that the seked of $5\frac{1}{2}$ was selected from the Wedjat fraction system precisely because it produces these relationships, not merely as a practical construction convenience. That claim requires a direct answer to the standard objection, which is set out and addressed below. At the astronomical level, the Sothic cycle target value used below (1,461 Egyptian civil years, the identical interval to the 1,460 Julian years in which the cycle is conventionally quoted) was documented in Egyptological scholarship by Poole (1851, *Horae Aegyptiacae*) — one hundred and seventy-five years before this paper — as a universally acknowledged fact of Egyptian astronomical knowledge. It is not a value invented by this hypothesis.

Relation to Prior Scholarship

Nothing in the arithmetic below is original to this paper, and the debts should be stated plainly. That the Egyptians employed a closed series of binary submultiples — $\frac{1}{2}$, $\frac{1}{4}$, $\frac{1}{8}$, $\frac{1}{16}$, $\frac{1}{32}$ and $\frac{1}{64}$ — in the division of the hekat is established through the work of Gillings (1972) and Clagett (1999) on the Rhind Mathematical Papyrus and the associated corpus, and the internal arithmetic of that series has been analysed at length in studies of the Akhmim Wooden Tablet, the Egyptian Mathematical Leather Roll and the Rhind $2/n$ table. Pommerening (2003) demonstrated that the same series governs pharmaceutical measurement in the Ebers Papyrus, an attestation independent of grain administration. The identification of the series with the anatomical components of the wedjat eye originates with Möller (1911) and entered the standard reference literature through Gardiner (1927). The geometric consequences of the seked, including the π - and ϕ -proportions discussed above, were

established by Rossi (2004) and Herz-Fischler (2000). The dimensions themselves derive from Petrie (1883) and the re-survey of Dash (2017). This paper contributes none of these findings and depends on all of them.

The derivation of the series from the wedjat has since been challenged. Ritter (2002) argued that the hieratic capacity-measure submultiples, in their origin and their written form, did not visually derive from the pictographic components of the eye, and that the association was a later interpretive overlay rather than the source of the notation. That argument is, in the present author's view, well founded, and it is accepted here. Its scope should be stated precisely, since three separable questions are easily conflated. The first is whether the hieratic fraction signs derive visually from eye-parts; this is Ritter's subject, and his conclusion is accepted. The second is whether the individual senses were associated with the Eye of Horus formula in Old Kingdom ritual; this is independently attested in the Unas offering liturgy (Pyramid Texts, Utterances 22–171) and is not addressed by Ritter. The third is whether the six senses correspond to the six fraction values in a specific ordered sequence; that correspondence is an inference rather than a directly attested Old Kingdom statement, and it is treated throughout this series at a lower evidential tier than the attestations that support it. Ritter's finding bears on the first question only. It does not reach the second or third, and it does not touch the existence, composition or use of the binary series itself. The series is attested independently of the symbol in grain administration (Gillings 1972; Clagett 1999) and in pharmacy (Pommerening 2003), two domains that would not share a notation by convention alone. Ritter's own result is in one respect load-bearing for the present analysis rather than against it: he establishes that $1/64$ is a hard lower bound, below which scribes abandoned the series entirely for no notation — a structural boundary the analysis below relies upon. The present paper employs the series and its combinations only. No step in the arithmetic that follows depends on the sense correspondence or on the derivation of the sign, and the results would stand unchanged if both were set aside. The term 'Wedjat fractions' is retained as the conventional designation for the series, in the sense in which the literature uses it, and not as an assertion about its origin.

What this paper proposes as new is neither the fractions nor the constants but the method. The series is here treated as a constrained computational grammar and tested as such: a fixed set of six exterior dimensions, the six fractions and their sums, the two operations attested in the Rhind Papyrus, a stated tolerance, and a target list fixed in advance — with the resulting correspondence count compared against rival historical fraction systems downsampled to equal cardinality, against randomly generated systems of matched size and span, against evenly spaced systems of matched spacing, against the complete grids of neighbouring denominators, and against randomly generated pyramid proportions. A search of the published literature has not identified a prior application of the Egyptian binary series to monument dimensions under declared operational constraints with statistical controls of this kind. Existing work on the series concerns its internal arithmetic, its attestation, or its iconographic history; existing work on pyramid metrology proceeds in modern units and without a fraction-system null. The combination is, to the present author's knowledge, untried, and is offered on that basis rather than as a claim of priority over any particular finding.

The Seked Objection

The strongest standing objection to any claim that the Great Pyramid encodes π or ϕ is that both follow automatically from its slope, and that the slope was chosen for practical reasons. The objection deserves to be stated at full strength. Egyptian builders expressed inclination as a seked — a number of horizontal palms and digits per royal cubit of rise — and the royal cubit divides into seven palms of four digits, twenty-eight digits in all. A seked of $5 \frac{1}{2}$ palms is therefore 22 digits horizontal to 28 vertical. The reciprocal, $28/22$, equals 1.272727. It happens that $4/\pi$ equals 1.273240, a difference of 0.040%, and that $\sqrt{\phi}$ equals 1.272020, a difference of 0.056%. Both are well-documented coincidences arising from the number seven. On this reading, a builder who selects a whole-and-half-palm seked for ease of construction obtains π -proportions and ϕ -proportions in the finished monument without possessing, or intending, either constant.

Two things follow from this, and they should be separated. The first is granted without reservation: because the ratio alone determines both relationships, π and ϕ are not two independent correspondences but a single fact expressed twice, and every pyramid built to a seked of $5 \frac{1}{2}$ possesses both regardless of its size. They are accordingly listed in this section as established findings, already documented as consequences of the construction method (Rossi 2004; Herz-Fischler 2000), rather than as novel claims. The second does not follow. The objection assumes that a seked of $5 \frac{1}{2}$ was adopted for convenience, and offers no evidence that it was. Convenience does not select it. A builder working in whole and quarter palms had at least eight equally practical values available across the range used for large pyramids, and their perimeter-to-height ratios are 5.714 at seked 5, 6.000 at $5 \frac{1}{4}$, 6.286 at $5 \frac{1}{2}$, 6.571 at $5 \frac{3}{4}$, 6.857 at 6, 7.143 at $6 \frac{1}{4}$, 7.429 at $6 \frac{1}{2}$ and 8.000 at 7. Against $2\pi = 6.28319$, the errors are 9.05%, 4.51%, 0.040%, 4.59%, 9.14%, 13.7%, 18.2% and 27.3% respectively. One value out of eight lands on 2π , and it lands two orders of magnitude closer than its nearest neighbour. Khafre's pyramid, built on the same plateau within the same dynasty by the same institutional apparatus, uses a seked of $5 \frac{1}{4}$ and misses 2π by 4.5%. Practical convenience therefore cannot account for the selection, since equally convenient alternatives were used and produce nothing. The objection explains π and ϕ given the seked; it does not explain the seked. That question remains open, and this paper's proposal — that $5 \frac{1}{2}$ was selected from within the Wedjat system because of what it produces — remains a live candidate answer rather than one the objection has displaced.

A related objection holds that the Egyptians cannot have encoded π or ϕ because they possessed no such constants: there is no Egyptian word for either, and no surviving text treats them as abstract quantities. The premise is correct and the inference does not follow. Nothing in this section requires that the builders held an abstract constant, named it, or reasoned about it as the Greeks later did. What it requires is that their own arithmetic produced the values, and that they used the arithmetic that does. A ratio does not need a name to be operative. A seked of $5 \frac{1}{2}$ palms yields a perimeter-to-height relation of $22/7$ whether or not anyone present has a term for what $22/7$ approximates, and the Wedjat fractions yield the correspondences documented above whether or not anyone present conceives of them as a numerical system. The claim advanced here concerns method, not vocabulary.

A point of evidence tells against reading the seked value backwards out of Egyptian circle mathematics. The Rhind Mathematical Papyrus gives an explicit rule for the area of a circle in Problem 50 — take eight-ninths of the diameter and square it — which corresponds to a π of $256/81$, or 3.1605, in error by 0.602%. The value implied by a seked of $5 \frac{1}{2}$ is $22/7$, or 3.14286, in error by 0.040%. The seked value is some fifteen times the more accurate. Whatever produced it, it was not derived from the circle rule the Egyptians actually wrote down, and the two are independent. This is consistent with the reading offered here: the accuracy resides in the fraction system and the proportions built from it rather than in any stated theory of the circle, which is what encoding through method rather than through doctrine would look like.

The objection does not, however, extend to the astronomical and calendrical correspondences, and the reason is visible within the comparative sample analysed above. Three of the nine documented pyramids share a seked of exactly $5 \frac{1}{2}$ palms: the Great Pyramid (base 440, height 280), Meidum (base 275, height 175), and Unas (base 110, height 70). All three therefore possess identical π -proportions and identical ϕ -proportions; whatever the seked confers, it confers equally on each. Their scores against the fourteen frozen targets are 10 of 14, 7 of 14 and 8 of 14 respectively. A two-to-three-target spread separates monuments that the seked cannot distinguish at all.

The seked therefore explains π and ϕ , and explains nothing else. What varies across the three is the absolute dimension — 440 cubits against 275 and 110 — and it is the absolute dimension, operated on by the Wedjat fractions, that the astronomical correspondences track. This is also why the second Monte Carlo test varies base and seked jointly rather than slope alone: the discriminating variable is scale, not inclination. A reader who accepts the seked objection in full may treat π and ϕ as a single relationship rather than two, and may set them aside entirely, without touching the result on which this paper rests.

Table 2 below documents each primary dimension used in the mathematical encoding analysis, its measured value, measurement source, uncertainty range, and whether the claimed correspondence survives under error propagation. All correspondences survive their respective tolerance ranges.

Dimension	Measured Value	Source	Uncertainty	Claimed Correspondence	Survives Error?
Base length	440 rc	Petrie (1883); Dash (2017)	± 0.10 rc (0.02%)	π via perimeter÷height	Yes — at $\pm 0.5\%$
Height (original)	280 rc	Petrie (1883); Lehner (1997)	± 0.19 rc (0.07%)	π , ϕ via seked geometry	Yes — at $\pm 0.5\%$
Slope angle (seked)	5.5 palms/cubit = $51^\circ 50'$	Rhind Papyrus; Rossi (2004)	$\pm 2'$	Seked-derived π and ϕ	Yes — Herz-Fischler (2000)
Royal Cubit	0.5236 m	Petrie (1883); Zapassky et al. (2012)	± 0.001 m	$\pi/6$ m (arithmetic)	Yes — arithmetic identity
Orientation	0.067° W of true N	Dash (2017); Nell & Ruggles (2014)	$\pm 3.5'$ (0.058°)	Western processional axis	Yes — directional claim
KC north shaft	$32^\circ 28' 16''$	Belmonte & Lull (2023)	$\pm 0.1^\circ$	α Draconis at epoch	Yes — Trimble (1964)
KC south shaft	$45^\circ 00' 00''$	Belmonte & Lull (2023)	$\pm 0.1^\circ$	Orion's Belt at epoch	Yes — Trimble (1964)

Declared Methodological Constraints

A note on evidence independence: the nine disciplines cited in this paper are not uniformly independent of one another. Mathematical analysis and archaeoastronomical analysis both operate on the same dimensional and survey data; their conclusions are correlated rather than independent confirmations. The genuinely independent evidence streams — those that could in principle falsify the hypothesis without reference to any other stream — are the genetic migration data, the paleohydrological reconstruction, the zooarchaeological assemblage analysis at Hierakonpolis, and the epigraphic transmission record. These four streams converge on the same geographic corridor and temporal window from entirely separate research programmes, without reference to pyramid dimensions or astronomical alignments. The convergence argument is strongest and most structurally independent in these four streams. The mathematical and archaeoastronomical streams constitute a separate, internally coherent convergence argument that shares a data set with each other and should be evaluated on that basis.

To address the concern that a large search space makes pattern emergence statistically inevitable, the following constraints governed this investigation and were fixed before results were examined. They are stated here so any reader can replicate or challenge the search. Dimensions: six exterior measurements only — base (440 cubits), height (280 cubits), half-base (220 cubits), apothem (356.09 cubits), base diagonal (622.25 cubits), and perimeter (1,760 cubits) — all derived from Petrie (1883) and Dash (2017). No internal chamber dimensions are used in this section. Fractions: the exact six Wedjat fractions ($\frac{1}{2}$, $\frac{1}{4}$, $\frac{1}{8}$, $\frac{1}{16}$, $\frac{1}{32}$, $\frac{1}{64}$) and their sums only. No arbitrary fractions. No non-Wedjat denominators. That this is the historically attested set is not an inference of this paper: Clagett (1999,

p. 58) states that the first six terms of the Egyptian unit-fraction series were commonly used for fractions of a heqat and were written with special Horus-eye symbols, identifying them as $\frac{1}{2}$, $\frac{1}{4}$, $\frac{1}{8}$, $\frac{1}{16}$, $\frac{1}{32}$ and $\frac{1}{64}$ as they occur in the Rhind Mathematical Papyrus. Operations: multiplication and division only in the frozen confirmatory analysis. Cross-dimension addition and subtraction — likewise attested in the Rhind Mathematical Papyrus — produce additional correspondences but are excluded from the statistical test to avoid inflating the expression space. This exclusion is conservative rather than neutral. Clagett (1999, pp. 57–58) describes Egyptian multiplication as proceeding by doubling and halving, with the final product formed by adding partial products, so addition is intrinsic to the arithmetic rather than foreign to it. Barring it narrows the achievable set below what Egyptian practice would permit. No higher-order transformations, no logarithms, no trigonometric functions. Threshold: results within 0.2% of a target value were recorded. All results — including non-matches — were reported. The post-hoc nature of this search is a genuine limitation, acknowledged throughout this section: the correspondences were identified before any protocol existed. The confirmatory analysis reported below is governed by the frozen analysis specification set out in this section and implemented in the deposited code, which fixes the target list, dimensions, fraction system, operations, tolerance, null models and decision rule. That specification converts the reported statistics into the application of a fixed, publicly deposited procedure that any reader can inspect and re-run. The domain coherence of the results — that the constrained search recovers correspondences spanning all major Old Kingdom astronomical knowledge domains simultaneously — is the observation the hypothesis advances, not a statistical proof of encoding. The structural tests proposed in Predictions 6, 7, and 8 define the methodological path toward confirmation or falsification of the encoding claim.

A note on targets removed during methodological tightening. An earlier form of this analysis included four further targets. Three — the Saros eclipse cycle (6,585.32 days), the Metonic cycle (6,939.69 days) and the precessional rate — were reachable from the declared dimensions, the Saros by $\text{apothem} \times 13 \div (45/64)$ and the Metonic by $\text{diagonal} \times 17 \times (21/32)$, and all three were removed. They were removed because those routes require integer multipliers of 13 and 17, which lie outside the two operations permitted above. The exclusion is deliberately stricter than Egyptian practice would license: Clagett (1999, pp. 57–58) shows that any whole-number multiplier could be assembled by selectively adding terms of the doubling series, so 13 ($1 + 4 + 8$) and 17 ($1 + 16$) were both trivially constructible by ordinary Egyptian method. These were therefore reachable hits, discarded as successes obtained by means this analysis forbids rather than as failures; the constraint was enforced against the reported result rather than in its favour. The Saros and Metonic are in any case Babylonian constructions with no Old Kingdom Egyptian attestation, so their exclusion is doubly warranted. The fourth, the tropical year, was removed for a different reason: it is not distinguished from the sidereal year in Old Kingdom sources, and the same formula reaches both within the tolerance, so counting it separately would have double-counted a single correspondence. Each removal lowered the reported count.

Design specification rather than as-built measurement. The dimensions analysed throughout this section are the design values — a base of 440 royal cubits and a seked of $5 \frac{1}{2}$ palms — not the as-built measurements of the standing monument. This distinction matters and is stated explicitly. No Egyptian document records the figure 440; it is a reconstruction from survey. Petrie's (1883) mean base of 230.33 ± 0.05 m converts, at a royal cubit of 0.5236 m, to 439.90 ± 0.095 cubits. The reconstruction is nonetheless secure on three independent grounds. First, 440 lies 1.08 standard deviations from the surveyed mean, while the neighbouring integers 439 and 441 lie 9.4 and 11.6 standard deviations away; 440 is the only whole-cubit value compatible with the survey. Second, the result is stable across the full range of attested cubit lengths: values from 0.5232 to 0.5240 m yield 439.56 to 440.23 cubits, rounding to 440 throughout. Third, and independently of the base, a base of 440 with a height of 280 yields a seked of exactly $5 \frac{1}{2}$ palms — a whole-palm design value of the kind documented in Rhind Mathematical Papyrus Problems 56–58 — producing a slope of 51.843° , within 0.002° of Petrie's measured $51^\circ 50' 40''$. Two design parameters therefore resolve independently to simple Egyptian values. Because an encoding hypothesis is a claim about intent, and intent resides in the specification rather than in the executed stone, the design values are the appropriate object of

analysis. The 0.023% divergence between the reconstructed specification and the surveyed monument is a measure of construction accuracy, not evidence against the specification. Readers should note that correspondence counts computed on the surveyed value rather than the design value are lower, and that the analysis reported here is of the plan.

Fraction-system control. Whether the Wedjat system is specifically effective for these targets, or whether any comparable fraction system would do as well, is not settled by inspection; it is settled by the head-to-head and equal-cardinality tests reported in Part 3 of the deposited analysis. Those tests score four real historical fraction systems — Egyptian unit fractions, Babylonian sexagesimal, small integer ratios and decimal — against the same six dimensions, the same fourteen targets, the same two operations and the same tolerance, and then downsample each rival to the same sixty-three fractions the Wedjat system has, so that the comparison does not simply reward whichever system contains the most fractions. The Wedjat set reaches ten of fourteen with sixty-three fractions, a rate of 15.87 correspondences per hundred; Egyptian unit fractions reach eleven but require 1,923 fractions to do it, a rate of 0.57. Downsampled to equal size, no rival reaches the Wedjat score in twenty thousand draws — every rival returns $p = 0.0000$ — and across the tolerance band from $\pm 0.10\%$ to $\pm 0.30\%$ the highest rival probability is 0.0032. That is the control, and it is computed in the deposited script rather than asserted here.

Even-spacing control. One structural feature of the Wedjat system requires its own control. The six Wedjat signs combine additively — the notation in which Egyptian scribes wrote hekat capacities — and their sixty-three sums are exactly the sixty-fourths, $k/64$ for $k = 1$ to 63. That set is a perfectly uniform grid, whereas fraction sets drawn at random cluster and leave gaps, and a uniform grid covers a numerical range more efficiently than a clustered one. A null model built from randomly drawn fractions therefore sets a slightly lower bar than the Wedjat set faces. The test was accordingly repeated against fraction sets of identical cardinality and span but with the values evenly spaced and randomly offset, holding spacing constant so that only the denominator varies. Under this more conservative null the Great Pyramid still exceeds the comparison systems, though at reduced significance: 10 of 14 against a mean of 6.00 ($p = 0.055$, not significant), and 8 of 11 against 4.25 ($p = 0.031$); on the maximally-independent nine-target core the result also fails to reach significance (6 of 9 against 3.39, $p = 0.100$). Roughly an order of magnitude of the significance reported under the randomly-drawn null is therefore attributable to even spacing rather than to the Wedjat system specifically. This is reported rather than omitted.

Denominator specificity. The even-spacing control raises the sharper question of whether the denominator 64 matters at all, or whether any sufficiently even grid would serve. This is testable deterministically by scoring the complete grids k/M against the same fourteen targets at the same threshold. Grids of comparable density and identical construction score four to five: $k/48$ gives 4, $k/60$ gives 4, $k/72$ gives 5. The sixty-fourths give 10. The neighbouring denominators are not intermediate between these values; the transition is a discontinuity rather than a gradient, and it cannot be attributed to grid density, spacing, or cardinality, since those are held constant across the comparison. The denser grid $k/128$ scores 11 of 14, but with 127 fractions it is markedly less efficient per fraction (8.66 correspondences per hundred fractions against Wedjat's 15.87), and $1/128$ is not an attested Horus-eye sign — the series terminates at $1/64$ — so $k/128$ represents a modern extension of the notation rather than a historical rival system. On this evidence the specificity claimed here attaches to the denominator itself, not to even spacing in general.

Null Model and Statistical Framework

The following null hypothesis governs the statistical evaluation of this section.

H_0 (Null Hypothesis): The exterior dimensions of the Great Pyramid were determined by practical architectural, engineering, and cosmological considerations independent of systematic astronomical encoding through the Wedjat fraction system. Any mathematical or astronomical correspondences

identifiable through Wedjat fraction operations are the product of coincidental numerical proximity within the search space.

H_1 (This Hypothesis): The exterior dimensions were chosen to preserve a structured body of mathematical and astronomical knowledge through the Wedjat fraction system. Under H_1 , Wedjat fraction operations on the six primary exterior dimensions should produce correspondence rates with real astronomical and mathematical targets that significantly exceed chance, should exhibit structural patterns consistent with systematic domain-organised encoding, and should show selective absences consistent with the observational limits of the tradition described in The Corridor Series.

Predictions under H_0 : Wedjat fraction operations on pyramid dimensions should produce correspondence rates with real astronomical constants statistically indistinguishable from randomly generated values in the same numerical ranges. Hits should be distributed randomly across domains without structural clustering by formula type or astronomical category.

Predictions under H_1 : Correspondence rate with real targets significantly exceeds chance. Structurally related astronomical bodies share formula structures (e.g., same pyramid dimension and operation class). Hits cluster in domains with strong Egyptian observational tradition. Constants lacking that tradition (Mercury; modern physics constants) are absent or underrepresented.

Test and result: The constrained search (six declared dimensions $\times$ Wedjat binary fractions $\times$ declared operations, applied to fourteen astronomical and mathematical targets at $\leq 0.2\%$ precision threshold) produces the correspondences documented throughout this section. Each correspondence is arithmetically verified. Taken individually, any single correspondence could be coincidental. Taken collectively, they span every major domain of Old Kingdom astronomical knowledge: all four of the visible planets tracked here, the anomalistic lunar month, multiple year types, and the Sirius-Osirian calendar cluster reaching from the 70-day period to an unsuccessful approach to the Sothic cycle — all from the same six dimensions using the same Egyptian fraction system. The domain coherence of this pattern is the observation the hypothesis advances.

Structural evidence bearing on H_1 : Saturn and Jupiter are both yielded by base $\times$ (Wedjat fraction) — the same pyramid dimension and the same operation, with different fractions for different bodies. Venus and Mars are yielded by apothem and height respectively, each divided by a Wedjat fraction — a consistent operation class for a different pair of bodies. Six mathematical constants are found, all Egyptian-constructible. Earth's two principal radii are recoverable through purely Egyptian-arithmetic formulae, at the lower evidential tier defined in the Earth's Geometry section. Mercury is not found, consistent with its observational difficulty. Euler's number, $\sqrt{3}$, and the speed of light are not found, consistent with their absence from any Egyptian observational tradition.

Interpretation and limitations: The search was conducted post-hoc, and the target list was compiled from astronomically meaningful values rather than randomly selected. These constraints prevent this section from constituting statistical proof of deliberate encoding, and that claim is not made here. What is presented is an observation: a constrained Wedjat-fraction operation on Giza's six declared exterior dimensions produces arithmetically verified correspondences spanning all major domains of Old Kingdom astronomical practice — planetary synodic cycles, lunar periodicity, solar and lunar calendrics, and the Sirius-Osirian ritual and long-period calendar system. The domain-complete pattern is the claim. Whether it reflects deliberate encoding is the hypothesis. The structural tests for that claim are stated as Predictions 6 and 7.

A note on the target list as constrained prior: the fourteen targets used in both Monte Carlo tests are not a random sample of astronomical quantities. They constitute a constrained prior — specifically, the periods and cycles that appear in the canonical Egyptological record of Old Kingdom astronomical knowledge and practice: Faulkner (1966) identifies Venus in the Pyramid Texts as the Morning Star and, on his reading, also as the Lone Star, placing a named planet in the Old Kingdom's earliest religious corpus; Clagett (1995) documents the decans and the stellar rising and setting cycles; Parker

(1950) documents lunar and Sothic calendrics — noting that his calendar model, while widely accepted, is not settled: Clagett (1995) records that Parker's views "are anything but certain," and Krauss (1998) notes challenges from Helck, Luft and Leitz; the Rhind Mathematical Papyrus (c. 1650 BCE) documents the civil year and schematic year; the 70-day Sirius invisibility period is attested across multiple Old Kingdom and later sources. The Monte Carlo tests therefore ask a precise question: does the Wedjat fraction system specifically recover the documented Egyptian astronomical knowledge base? This is the correct question under H_1 , because the hypothesis is that specifically Egyptian astronomical knowledge was encoded. A test using arbitrarily selected astronomical constants — including modern quantities unknown to the Egyptians — would be methodologically weaker, not stronger, because it would dilute the signal with noise. The fourteen targets are the signal. The fraction-system control reported in Part 3 of the deposited analysis is the primary methodological control because it holds this target list constant and asks only whether the Wedjat system is doing the work — answering the target-selection objection directly.

On the attestation of the planetary targets. The four planetary synodic periods are the least directly documented members of the list, and the difficulty should be stated rather than passed over: no Egyptian source at any period records a planetary period as a number, and the surviving numerical planetary tables are Greco-Roman and calculated rather than observed. Two considerations bear on their inclusion. First, the planets were watched and named within the tradition from an early date: Faulkner (1966) identifies the Morning Star of the Pyramid Texts as Venus at dawn, and argues that the Lone Star is Venus after sunset, making Venus the one planet securely attested in the Old Kingdom's earliest religious corpus — and the alternation between those two apparitions is precisely the observation whose period is the synodic cycle. Second, the absence of a written period is what this hypothesis predicts rather than an anomaly it must absorb: the argument advanced throughout is that such quantities were held in the architectural register while the tradition was intact, and committed to papyrus only as it faded. That Clagett (1995), in a source book of 575 pages on Egyptian calendars, clocks and astronomy, includes no chapter and no document on the planets is consistent with both readings and decides between neither. The planetary targets are accordingly advanced on a weaker documentary footing than the calendrical ones, and are so marked.

On late attestation and older content. Several targets in this list are documented in sources later than the Fourth Dynasty, and a reader may reasonably ask why late documentation is treated as evidence of earlier knowledge. The principle is not special pleading but standard Egyptological method. Clagett (1995, Preface) states it directly: the carrier of a document frequently dates much later than the document itself. He instances the decanal transit tables in the cenotaph of Seti I (c. 1306–1290 BCE), whose content dates from at least as early as the reign of Sesostri III in the Twelfth Dynasty, and the Ramesside Star Clock, whose earliest surviving copies are in the tomb of Ramesses VI (c. 1151–1143 BCE) but whose internal evidence places it between roughly 1500 and 1470 BCE. Late attestation therefore establishes a *terminus ante quem* for a tradition, not its date of origin. This does not license unlimited backdating, and it is not offered as proof that any specific quantity was known in the Fourth Dynasty; it establishes only that the inference is a normal one in the field rather than an accommodation invented here. The exclusions applied above follow the same rule in the opposite direction: the thirty-six decans, first attested on coffin lids around 2150 BCE, are omitted precisely because no claim of Old Kingdom currency is made for them.

On the anomalistic month. Of the fourteen targets this is the least directly attested, and the position is better stated than assumed. No Egyptian source records an anomalistic period; Parker (1950, §7) gives the sidereal month at approximately $27\frac{1}{3}$ days, a value the base does not reach within the tolerance used here. The case for the anomalistic cycle is physical rather than textual. On Parker's reconstruction the Egyptian lunar month began on the morning the old crescent of the waning moon could no longer be seen before sunrise (Parker 1950, quoted at length in Clagett 1995, ch. 3), and Parker (1950, §§11–12) identifies the first of the three factors governing that visibility as the anomaly of the moon — its varying distance from the earth. The anomalistic cycle is thus the period of the variable that determines the observable event on which the Egyptian lunar calendar was regulated,

and a tradition watching that event across centuries samples the anomaly whether or not it named a period for it. This is an inference and is offered as one. A reader who declines it should know exactly what that costs, and the leave-one-out analysis states it plainly: removing the anomalistic month and re-running the second Monte Carlo test on the remaining thirteen returns $p = 0.064$, which does not reach the conventional threshold. It is the only deletion in the series that does not. This target is therefore load-bearing in a way the others are not, and a reader who rejects the argument above should treat the correspondence-count result as unsupported rather than merely weakened. The designated primary test, which does not depend on the target count, is unaffected.

A tested domain that does not encode. The target list contains periods and one geometric constant; it contains no stellar positions. That boundary was tested rather than assumed, and it was tested with the same control used for the base sweep, so that the two domains can be compared directly. Declinations, rising azimuths and culmination altitudes were computed for eleven stars of documented significance in the Egyptian and Saharan record — Thuban, Sirius, Canopus, Arcturus, Alnitak, Alnilam, Kochab, Mizar, Alkaid, Dubhe and Alpha Centauri — precessed to 2560 BCE with proper motion applied, together with the obliquity of the ecliptic at that epoch, Giza's latitude and the circumpolar limit. Arcturus is included because Malville et al. (2007) revised the northern Nabta Playa alignments from Dubhe to Arcturus, dating them to three epochs between 4530 and 3630 BCE; both stars are retained here. Five of the eleven — Thuban, Kochab, Mizar, Alkaid and Dubhe — are circumpolar at Giza in 2560 BCE and never cross the horizon, so they contribute a declination and a culmination altitude but no rising azimuth. That leaves thirty-one angular quantities, which were scored against the same six dimensions, sixty-three Wedjat fractions, two operations and $\pm 0.2\%$ tolerance used throughout. The computation and the control below are performed by `stellar_control.py`, deposited with this paper. Five reached tolerance: Alnitak's culmination altitude at 0.060%, Mizar's declination at 0.138%, Canopus's rising azimuth at 0.157%, Sirius's culmination altitude at 0.163% and Canopus's declination at 0.198%. Five of thirty-one is 16.1%. The control determines what that figure means. Each angular target was replaced by a random value drawn from the same category range — a declination stays a declination, an azimuth an azimuth — and the pyramid was re-scored against the resulting shadow list ($N = 200,000$, seed 42). The expected number of hits by chance is 4.83 of thirty-one, or 15.6%, and the probability of reaching five or more is 0.5431. The observed stellar result is therefore not merely close to chance but statistically indistinguishable from it. Set against the calendrical suite scored by the identical procedure, the contrast is the point. The calendrical targets return ten of fourteen against a chance expectation of 6.25, at $p = 0.023$; the stellar angles return five of thirty-one against a chance expectation of 4.83, at $p = 0.543$. The same monument, the same six dimensions, the same fraction system, the same operations and the same tolerance produce a result above chance in one domain and at chance in the other. The difference lies in the target set rather than in any general flexibility of the pyramid's arithmetic, and this is the most direct available answer to the objection that a monument with many measurements will reach any list of numbers. One feature of the stellar result deserves emphasis because it runs against the interpretation offered here and is reported for that reason. The stars that reach tolerance are not obscure ones. They are Alnitak, the belt star at which the King's Chamber southern shaft is directed; Sirius, whose heliacal rising governed the Egyptian year; and Canopus, which accounts for two of the five hits. These are among the most significant stars in the tradition, and a reader is free to pursue any of them. The finding is that even when the hits fall on the stars that matter most, the total remains what chance predicts, so no stellar encoding is claimed here. Conversely, several quantities central to the monument fail: Thuban, which the northern shaft addresses, misses on both available measures; the obliquity of the ecliptic misses at 0.327%; and Giza's own latitude misses at 2.15%. The stellar domain is accordingly reported as tested and not encoded. The division this suggests is consistent with the rest of the paper: the dimensions carry periods, while the architecture — shaft angles, cardinal orientation, the causeway bearings — carries directions. It also marks a limit on the method that should be stated rather than left implicit. The candidate space for stellar targets is large and weakly bounded; adding stars adds hits at roughly the chance rate, as the expansion of this test from nine stars to eleven demonstrated. A search across enough stars, quantities and epochs will always

produce correspondences. The calendrical result is defensible not because it is larger but because its candidate set is small, externally documented, and fixed before scoring.

The correspondences documented in this section were identified through a constrained post-hoc search. Their strength lies in their structural coherence and cultural alignment rather than in the manner of their discovery. The methodological constraints — six declared dimensions, six Wedjat fractions, two operations, and a 0.2% precision threshold — are designed to limit the search space and reduce the risk of spurious pattern-matching. The fraction-system control in Part 3 of the deposited analysis and the unit-system specificity test (Prediction 8) are the primary methodological controls for this section. Readers should weigh each correspondence against those constraints, and against the probability that a domain-coherent, culturally aligned pattern of this density could arise from an unconstrained search of arbitrary numbers.

A further control addresses the concern that the fourteen targets — drawn from the documented Egyptian record, but assembled after the correspondences were observed — might simply be values the framework can reach. In a target-value null, each target was replaced by a random value drawn from the plausible physical range of its own quantity type. The frozen ranges are lunar periods 25–35 days, planetary synodic periods 100–800 days, the geometric angle 90–180°, Sirius invisibility 50–120 days, year lengths 300–400 days, and the long cycle 1,000–2,000 years, as implemented in the deposited code. The Great Pyramid was then scored against the resulting shadow list (N = 100,000, seed 42). At the baseline $\pm 0.2\%$ tolerance the pyramid reaches the real constants markedly more often than these random plausible values — 10 of 14 against a null mean of 6.25 ($p = 0.023$), 8 of 11 against 4.44 on the independent set ($p = 0.017$), and 6 of 9 against 3.53 on the maximally-independent set ($p = 0.064$, not significant) — indicating that the specific constants are targeted, not merely that the ranges are hittable. As with the other tests, the significance of the correspondence count is tolerance-dependent. The deposited code performs one tolerance sweep, on the matched-cardinality fraction null: it is significant at $\pm 0.20\%$ and $\pm 0.25\%$ and at the tightest $\pm 0.05\%$ step, and not significant at $\pm 0.10\%$, $\pm 0.15\%$, $\pm 0.30\%$ or looser. This is precisely why the correspondence count is not the primary test. The designated primary criterion — the equal-cardinality system-specificity comparison reported below — is tolerance-independent by construction, and it holds across the entire band from $\pm 0.10\%$ to $\pm 0.30\%$, with a maximum rival probability of 0.0032. The count is reported here with its profile; the argument does not rest on it. The target-value null reported here was run at the frozen $\pm 0.2\%$ tolerance only; no tolerance profile is claimed for it. The chance baseline itself (approximately 6 of 14) reflects the density of the achievable-value set; the observed excess is real but modest, and is reported here alongside its tolerance profile rather than at a single favourable threshold.

The Mathematical Vocabulary: Correspondences with Fundamental Constants

The pyramid's slope is governed by a seked of 5.5 palms per cubit of rise — the ratio 14:11 in palm units. This is the single design decision from which all slope-derived correspondences follow. It is established in peer-reviewed Egyptology that the resulting proportions (height 280 cubits, base 440 cubits) yield a perimeter-to-height ratio of 22/7, an approximation of π with 0.040% error (Rossi 2004; Verner 2001; Herz-Fischler 2000). Herz-Fischler (2000) additionally notes that these approximations can arise naturally from the seked construction without requiring explicit knowledge of the transcendental numbers themselves — a position this hypothesis acknowledges: the seked choice produces the correspondences; whether that choice was made with awareness of its mathematical implications is the interpretive claim. It is similarly documented that the apothem — the slant distance from base midpoint to apex, derived by Pythagoras as $\sqrt{(220^2 + 280^2)} = 356.09$ cubits — divided by the half-base yields 1.61859, an approximation of ϕ with 0.034% error; this is noted in architectural mathematics literature (Rossi 2004). These two correspondences arise inevitably from the single choice of 280:220 and require no supplementary interpretation.

The following additional correspondences are presented as original observations within this hypothesis; no peer-reviewed source is known to have proposed them previously. Direct calculation shows that the exterior surface area ratios yield ϕ , $\phi-1$, and ϕ^2 simultaneously from the same proportions — three area relationships converging on the golden ratio (algebraically interrelated, since $\phi-1 = 1/\phi$ and $\phi^2 = \phi + 1$, and therefore not independent of one another) (errors 0.034%, 0.090%, 0.056% respectively). The base diagonal ($440\sqrt{2} = 622.25$ cubits) multiplied by the Wedjat fraction combination $23/64$ and divided by 100 yields 2.23623, within 0.007% of $\sqrt{5}$. The height multiplied by the combined first three Wedjat fractions ($7/8$) and divided by 100 yields 2.45000, within 0.021% of $\sqrt{6}$. The base multiplied by the Wedjat combination $5/16$ yields 137.500, within 0.006% of the golden angle ($360^\circ/\phi^2 = 137.508^\circ$) — the angle at which successive elements in plant growth are positioned to achieve maximum exposure and minimum overlap. It governs the spiral packing of seeds in sunflower heads, scales in pine cones, and the unfurling of leaves along a stem across the plant kingdom. Any pastoralist community observing nature across generations would encounter this pattern daily. It is the most precise single result in this investigation (error 0.006%), and it is produced by one of the simplest Wedjat combinations: the second and fourth fractions summed, applied to the base. This hypothesis proposes its presence is consistent with a tradition that understood the mathematical grammar underlying natural growth — knowledge accumulated, as this series argues, by a people who lived within and studied the natural world of the Green Sahara over millennia. The golden angle's mathematical necessity in plant growth is documented in peer-reviewed literature: Douady & Couder (1992, *Physical Review Letters*, 68:2098) demonstrated it as a self-organized physical process; Okabe (2015, *Scientific Reports*, 5:15358) demonstrated its biophysical optimality across vascular plants. A sustained pastoralist tradition across the Green Sahara would have encountered this pattern in the vegetation of their environment across millennia. This hypothesis proposes that these correspondences, taken together with the established π and ϕ relationships, suggest a mathematical vocabulary encoded in the pyramid's exterior geometry — the foundational constants from which measurement, proportion, and natural pattern can be reconstructed.

A note on Egyptian context: no surviving Egyptian text names the Golden Angle or the constant ϕ explicitly. The first formal definition of the 'extreme and mean ratio' appears in Euclid's *Elements* (c. 300 BCE), approximately 2,300 years after the pyramid was built. What the Egyptian mathematical record does document is the *seked* system: Rhind Mathematical Papyrus Problems 56–58 (c. 1650 BCE, copied from a 12th Dynasty original) show scribes computing pyramid slopes in palms per cubit. The Great Pyramid's *seked* of 5.5 palms per cubit produces a face angle of approximately 51.84° , which is geometrically equivalent to a ϕ -proportioned structure; the Golden Angle is a mathematical consequence of those same proportions. The Horus Eye fractions — the operational fraction set preserved in the papyri — are the same fractions this paper identifies as the pyramid's mathematical vocabulary (Clagett 1999; Robins & Shute 1987; Imhausen 2006). Whether the Golden Angle was a deliberate encoding or an emergent property of *seked*-based construction cannot be resolved from surviving sources. Its presence in the dataset is reported as a geometric observation, registered with the lowest error of any result (0.006%).

The Unit of Measure

It is documented that the Royal Cubit, as measured from surviving cubit rods and from Petrie's (1883) survey, is approximately 0.5236 m — within 0.0002% of $\pi/6$ (Petrie 1883; Rossi 2004). This is a measured property of the unit, not a hypothesis. This hypothesis further proposes a consequence: the primary unit used to build the pyramid can be derived from the pyramid's own π correspondence. A future civilisation that measures the structure, identifies the perimeter-to-height relationship, and then divides π by 6 recovers the original construction unit. The standard is self-referential. This hypothesis proposes that this self-referential property was intentional — that the architects embedded the primary construction unit within the monument's own dimensional grammar, recoverable by any civilisation capable of measuring the structure and recognising the π relationship.

Cardinal Orientation and the Wedjat Fraction of the Circle

It is established that the Great Pyramid's four exterior faces are oriented to the cardinal directions to within approximately 3.5 arc minutes, one of the most precise alignments known from the ancient world (Dash 2017; Spence 2000). Expressed as fractions of a full circle: each face-to-face angle is 90° = one quarter of 360° ; the face diagonals bisect at 45° = one eighth of 360° ; the north-south axis spans 180° = one half of 360° . These are precisely the second, third, and first Wedjat fractions ($\frac{1}{4}$, $\frac{1}{8}$, and $\frac{1}{2}$ respectively). Separately, Belmonte (2001) has documented in peer-reviewed archaeoastronomical literature that Egyptian temple orientations systematically employed rotations of $\frac{1}{8}$, $\frac{1}{4}$, and $\frac{1}{2}$ of a full circle — the same three fractions — as a hierarchical orientation system across multiple construction periods. This hypothesis proposes that the pyramid's precisely achieved cardinal alignment is consistent with the same Wedjat fraction system governing orientation as well as dimension — that the structure's spatial position in the landscape was oriented by the same mathematical framework that this hypothesis proposes governs its dimensional encoding. The Belmonte pattern provides independent peer-reviewed evidence that these specific angular fractions were architecturally functional in Egyptian construction practice.

Correspondences with Earth's Geometry

Direct calculation confirms the following mathematical relationships, presented here without claiming their intentionality is established. All comparisons here are computed in royal cubits — the unit in which the pyramid was designed and built — with metric values given in parentheses only for the reader's reference, since the metre is a modern unit the builders did not use. At the metrological value of 0.5236 m per royal cubit (Petrie 1883; Zapassky et al. 2012), Earth's polar radius is 12,140,474 cubits (6,356.752 km, WGS84) and its equatorial radius is 12,181,316 cubits (6,378.137 km). Both principal radii are recoverable using only components attested in Egyptian practice. The sum of the base and its diagonal, multiplied by 10,000 (an attested decimal power with its own hieroglyph) and divided by the Wedjat sum $\frac{7}{8}$ ($= \frac{1}{2} + \frac{1}{4} + \frac{1}{8}$), yields 12,140,045 cubits (6,356.528 km), within 0.0035% of Earth's polar radius. The sum of the height and the diagonal, multiplied by the 360-day schematic year, by 100, and by the Wedjat sum $\frac{3}{8}$ ($= \frac{1}{4} + \frac{1}{8}$), yields 12,180,429 cubits (6,377.672 km), within 0.0073% of Earth's equatorial radius. No modern constant enters either formula. Earth's two principal radii differ by approximately 40,842 cubits (21 km) because Earth is an oblate spheroid, not a perfect sphere. Two limitations designate this observation as a lower evidential tier than the frozen fourteen-target suite. First, these formulae expand the operation class: pairwise sums of dimensions and calendrical multipliers do not appear in the fourteen-target Monte Carlo analysis, whose statistics this subsection therefore cannot inherit. Second, under the denser combinatorial grid this expansion creates, closeness of this order is not rare: random targets of the same magnitude are matched at comparable precision (median best-error $\approx 0.009\%$; Monte Carlo control, $N = 50,000$, seed 42). The correspondence is accordingly presented as a demonstration of possibility within Egyptian arithmetic — the terrestrial radii are expressible in the builders' own numerical system to high precision — not as a statistically distinctive encoding. Whether a more constrained Egyptian route exists that would make the terrestrial match as rare as the planetary ones remains an open question.

Correspondences with Lunar and Solar Cycles

The following correspondences are original observations within this hypothesis. Direct calculation shows that the base multiplied by the fourth Wedjat fraction ($\frac{1}{16}$) yields 27.5 days, within 0.196% of the anomalistic lunar month (27.554 days). The apothem added to the fifth Wedjat fraction of the height ($\frac{1}{32} \times 280 = 8.75$) yields 364.840 days, within 0.044% of the Egyptian civil year of 365 days; this formula was independently verified by arithmetic computation (see AI Use Disclosure). The base minus the height multiplied by the Wedjat combination $\frac{17}{64}$ ($= \frac{1}{4} + \frac{1}{64}$) yields 365.625 days, within 0.101% of the sidereal year (365.256 days). The fraction $\frac{1}{64}$ is Thoth's restoring portion in the Wedjat mythology; Thoth is specifically the Egyptian deity of time measurement and the lunar calendar

(Wilkinson 2001). This hypothesis notes this thematic convergence while acknowledging it cannot substitute for evidence of intent.

Two additional correspondences connect the exterior dimensions to the Sirius-centred timekeeping system. First, the 360-day schematic year: the base of the Egyptian civil calendar was twelve months of thirty days each — 360 days before the five epagomenal days were added — and this administrative foundation appears throughout the Rhind Mathematical Papyrus. The half-base (220 cubits) added to the height multiplied by the primary Wedjat fraction ($\frac{1}{2} \times 280 = 140$) yields $220 + 140 = 360.000$ days exactly — the simplest addition formula in the dataset, using two dimensions and the largest Wedjat fraction. Second, the 70-day period of Sirius invisibility: Sirius was absent from the night sky for approximately 70 days each year as seen from the Giza latitude, a period the Egyptians encoded as the canonical duration of mummification and directly associated with the death of Osiris and the mourning of Isis (Parker 1950, §21). The height of the pyramid (280 cubits) multiplied by the second Wedjat fraction ($\frac{1}{4}$) yields 70 days exactly. This is the simplest possible correspondence in the entire dataset — a single dimension and a single fraction — yet it connects the pyramid's height to the star whose alignment the Queen's Chamber southern shaft encodes, and whose long cycle the perimeter approaches without reaching, as set out below.

Proposed Correspondences with Planetary Synodic Periods

The correspondences presented in this subsection are original mathematical observations within this hypothesis. No peer-reviewed source is known to have previously proposed that the Great Pyramid's exterior dimensions encode planetary synodic periods through Wedjat fraction operations. Readers are asked to evaluate these as candidate correspondences, not established encodings. It is documented that the ancient Egyptians observed and named the planets: Faulkner (1966) identifies Venus in the Pyramid Texts, the earliest Egyptian religious corpus, as the Morning Star and, on his reading, also as the Lone Star. No Egyptian source records a planetary period as a number, and that limitation is stated in full in the note on planetary attestation above. This hypothesis proposes that synodic periods accumulated through that observational tradition are preserved in the pyramid's exterior dimensions through the Wedjat system. Direct calculation confirms the following. Saturn (synodic period 378.09 days): $\text{base} \times (55/64) = 378.125$ days, within 0.009%. Jupiter (398.88 days): $\text{base} \times (29/32) = 398.750$ days, within 0.033%. Venus (583.92 days): $\text{apothem} \div (39/64) = 584.353$ days, within 0.074%. Mars (779.94 days): $\text{height} \div (23/64) = 779.130$ days, within 0.104%. All four fractions are confirmed exact sums of distinct Wedjat parts. All four results were independently verified by arithmetic computation (see AI Use Disclosure).

This hypothesis proposes that the structural pattern of these correspondences — Saturn and Jupiter both yielded by $\text{base} \times (\text{Wedjat fraction})$, sharing a formula structure across related bodies; Venus and Mars yielded by apothem and height respectively through division — is more consistent with systematic encoding than with chance. A systematic search across all formula combinations at the same precision threshold was conducted; Mercury was not found within acceptable error bounds. The structural patterning — Saturn and Jupiter sharing a formula family, Venus and Mars sharing the division operation — is noted as consistent with systematic encoding. Mercury was not found within the 0.2% threshold; this absence is reported as a genuine null result.

Proposed Stellar Cycle Correspondence: The Sothic Period

The Sothic cycle is the longest practical astronomical period the Egyptians are documented to have tracked. Because the civil year of 365 days is shorter than the true solar year by approximately 0.25 days, the heliacal rising of Sirius (the calendar anchor of Egyptian civilisation) drifted through the civil calendar, completing a full 365-day drift and returning to its original calendar date after 1,461 civil years. This is the Sothic cycle — a period spanning five to six human generations, directly observable through sustained astronomical record-keeping, and documented as culturally significant in Egyptian inscriptions across multiple periods. The connection between the Sothic cycle and the civil year's 0.25-day annual drift makes it, in effect, the long-term correction cycle for the 365-day calendar: the civil

year encodes the daily timekeeping; the Sothic cycle would encode the multi-generational recalibration, and is the layer tested here without success. A methodological note on evidentiary framing: the period is the arithmetic consequence of the civil calendar's own structure — $365 \text{ days} \times 4 \text{ years per day of drift} = 1,460 \text{ solar years}$, which is 1,461 years of the Egyptian 365-day civil calendar — derivable by any tradition that tracked the Sothic rising against the civil calendar and noted the annual drift rate. Both the Kahun Papyrus (c. 1800 BCE) and the Ebers Papyrus document Egyptian observation of Sothic risings against the civil calendar (Parker 1950); whether Old Kingdom astronomers named the period explicitly or possessed the annual observations from which it is arithmetically calculable, the correspondence proposed here encodes the mathematical foundation of the calendar system itself. This framing makes the claim more defensible: it does not require a named Old Kingdom Sothic doctrine, only the annual record-keeping already documented in the papyri. The perimeter of the Great Pyramid, multiplied by the Wedjat combination $\frac{1}{2} + \frac{1}{4} + \frac{1}{16} + \frac{1}{64} (= \frac{53}{64})$, yields 1,457.5. Against the Sothic cycle counted in Egyptian civil years — 1,461 — that is an error of 0.240%, which falls outside the $\pm 0.2\%$ tolerance, and the target is accordingly recorded as one of the four documented misses. It is reported here rather than omitted because the near-approach is a genuine one and because the unit decision that produces the miss is itself a finding: against the same cycle counted in Julian years (1,460) the error is 0.171% and the target would be a hit. The Julian year is a Roman unit the Egyptians did not possess, and this paper declines it on the same grounds that it declines the metre. The result is a target that fails on the honest convention and succeeds on the convenient one; it is scored as a failure. This would otherwise have been the only correspondence in the dataset linking an exterior dimension directly to a known long-period stellar cycle, and it connects the dimensional encoding system to the Queen's Chamber shaft alignment with Sirius and the 70-day invisibility period documented above — three independent encoding layers converging on the same astronomical object.

Calendar and Sirius Cluster: Summary

The three relations above form a coherent Sirius-centred cluster, two of which reach tolerance and one of which does not. The 360-day schematic year is the administrative and mathematical foundation of the civil calendar — the base structure before the five epagomenal days are added — encoded exactly as $\text{half-base} + \text{height} \times \frac{1}{2} = 360.000$ (an addition-based correspondence, excluded from the frozen $\times / \div$ statistical test, which reaches the 360-day year separately). The 70-day invisibility period is the annual ritual anchor of the Osirian mythology, the canonical duration of mummification. The Sothic cycle is the multi-generational correction period that closes the drift between the 365-day civil year and the stellar year anchored by Sirius; it is the layer the perimeter approaches but does not reach. Together with the existing civil and sidereal year correspondences already documented in this section, they demonstrate that the pyramid's exterior dimensions preserve the full layered structure of Egyptian calendrical astronomy — from administrative practicality (360-day base), to annual ritual theology (70-day Sirius absence), to long-period stellar recalibration (Sothic cycle). This is the only cluster in the dataset where independent correspondences converge on a single astronomical object through different dimensional mechanisms — two of them reaching tolerance and the third, the Sothic cycle, falling outside it: half-base and height combined (360-day), height alone (70-day), and perimeter (Sothic cycle). The cluster is powerful precisely because it is not imposed: each of the three correspondences was identified independently, uses a different pyramid dimension, and encodes a different time-scale of the same stellar phenomenon — daily structure, annual ritual, and multi-generational recalibration — spanning from 70 days to the Sothic period, the last of these a near-approach rather than a correspondence.

Calendar Coherence: The Wedjat Sum as Lunar Reconciliation Key

A second-order observation distinct from the direct dimensional correspondences above emerges from the 360-day schematic year already encoded in the pyramid's proportions. The complete Wedjat sum ($\frac{63}{64}$) applied to the 360-day schematic year yields 354.375 days — within 0.002% of the lunar year (12 synodic months of 29.531 days = 354.367 days). This is not a dimensional encoding: no

single pyramid dimension multiplied by a Wedjat fraction produces the lunar year within the 0.2% threshold of the Monte Carlo analysis. It is instead a property of the Wedjat system applied to an already-encoded constant, demonstrating that the fraction system is internally coherent with Egyptian calendar reconciliation practice. Parker (1950) documents that the Egyptians maintained a lunar religious calendar alongside the administrative calendar continuously from the Old Kingdom onward; the Kahun Papyrus (c. 1800 BCE) and the Ebers Papyrus calendar correlate both systems in a single document.

The operation is equivalently expressed as the 360-day year minus Thoth's restoring portion: $360 - (360 \times 1/64) = 354.375$ days. The $1/64$ fraction — 5.625 days annually — approximates the actual drift between the schematic and lunar calendars (5.633 days per year, error 0.14%). In the Wedjat mythology, 63/64 is the damaged eye: what remains after Set's fragmentation. The $1/64$ is Thoth's restoring portion: the knowledge required to complete what was broken. In the calendar, the lunar year is the damaged state — 63/64 of the administrative whole — and every year Thoth's $1/64$ marks the drift that must be tracked to keep the religious and administrative calendars in alignment. The pyramid encodes not only the 360-day administrative year in its dimensions but the lunar religious calendar as a consequence of that encoding, mediated by the same fraction system used to measure grain and calculate construction slopes.

The 64-year cycle that follows requires no precision threshold: $64 \times (360/64) = 360$ exactly. After 64 years, the cumulative drift between the schematic and lunar calendars equals precisely one full 360-day year. The Wedjat denominator (64) is the length of the complete calendar cycle. The system is self-referential: the fraction with the largest denominator ($1/64$) produces the annual drift, and the denominator itself is the number of years before the drift closes and the cycle returns to its origin. This relationship is proposed here as a hypothesis consistent with documented Egyptian calendar practice; it connects the administrative, religious, and mathematical domains of the pyramid's encoding through a single arithmetic property of the Wedjat fraction system.

Correspondence	Dimension & Formula	Result	Error	Egyptian Significance
360-day schematic year	Half-base + Height $\times \frac{1}{2}$ ($220 + 280 \times \frac{1}{2}$)	360.000 days	0.000% (exact)	Administrative and mathematical foundation (Rhind Papyrus); 12×30 base structure
70-day Sirius invisibility	Height $\times \frac{1}{4}$	70.0 days	0.000% (exact)	Mummification canon; Osiris/Isis ritual cycle; canonical Sirius absence period

The Stellar Timestamps: Established Shaft Alignments

It is established in peer-reviewed literature that the Great Pyramid's internal shafts, which exit on the exterior faces at precise angles, were aligned to specific stars at the construction epoch. Trimble (1964) identified the astronomical function of the King's Chamber shafts. The King's Chamber southern shaft (approximately 45°) targeted Orion's Belt as a constellation-level astronomical object, with Anitak (ζ Orionis) as the theologically primary member — the brightest Belt star and the one most directly associated with Osiris-Sah in the Pyramid Texts corpus (Belmonte & Lull 2023, p. 15). The shaft angle of $\sim 45^\circ 13'$ (Petrie 1883) falls within the culmination range of Orion's Belt at the construction epoch; the correspondence is to the constellation of Osiris as a whole, consistent with the Pyramid Texts' language throughout. A further line of support for the Canopus-proxy argument emerges from classical sources: Plutarch (De Iside et Osiride, Moralia 359E) documents that the Egyptians identified the constellation Argo Navis — whose brightest star is Canopus — as the Boat of Osiris. Osiris was therefore associated with two stellar forms simultaneously: Sah/Orion as his constellation-body accessible from northern latitudes, and the Boat of Osiris/Canopus-Argo as his

southern stellar identity at the limit of visibility. The King's Chamber south shaft encodes the accessible Osirian constellation at 45°; Canopus, the Boat of Osiris, fell below the range that an enclosed shaft through the pyramid's mass could carry at 30°N (demonstrated in the Wedjat as Portable Celestial Map section below). The shaft therefore encodes not the absence of Canopus but the accessible component of a dual Osirian identity whose other component was permanently fixed at the architectural horizon — a constraint the monument's latitude itself imposed. The northern shaft (approximately 32°) aligned to Alpha Draconis, the then-pole star (Trimble 1964). The Queen's Chamber southern shaft aligned to Sirius, whose heliacal rising anchored the Egyptian civil calendar (Belmonte & Lull 2023). Spence (2000, *Nature* 408:320–324) dated the start of construction to c. 2,480 BCE by a different route — modelling the precession of the simultaneous meridian transit of Kochab (β Ursae Minoris) and Mizar (ζ Ursae Majoris) against the cardinal orientation errors of successive Old Kingdom pyramids, not from the shaft angles. Her result has been contested on methodological grounds (Rawlins & Pickering 2001; Belmonte 2001) and is cited here as one dating model among several. The shaft angles independently encode real stellar positions at a specific epoch. This hypothesis notes a further implication: the shaft alignments, combined with a known precessional rate, allow any future observer to calculate backwards to the construction date. Within the framework of this hypothesis, this is interpreted as a self-dating feature — the monument records not only knowledge but the epoch in which it was deposited.

The Concavity: A Documented Equinox Instrument

It is documented that the Great Pyramid is the only known pyramid with concave faces — an inward hollow running from base to apex along the centreline of each face (Petrie 1883; Edwards 1975; Nell & Ruggles 2014). The feature is invisible from ground level under normal conditions. It is documented that at the spring and autumn equinoxes the Sun's angle is precisely sufficient to cast a shadow revealing the inward hollow, briefly exposing the eight-sided geometry (Edwards 1975). This hypothesis proposes that the concavity functions as a passive solar calendar instrument, marking the equinoxes annually without additional apparatus. A claim in popular circulation — that the concavity depth encodes Earth's curvature — was directly tested and refuted: Earth's surface curvature across 230 metres produces a sag of approximately 1 mm; the concavity depth is 940 mm, a factor of 903 larger. This claim is not supported by calculation and is not advanced here. A second caveat is required, and it is not one this paper can settle. Monnier (2022) argues that the hollowing is not a design feature at all but an artifact of the monument's later history, modelling the impacts of casing stones falling on the exposed faces together with the dismantling of the summit — nearly 270 m³ of blocks brought down onto faces already stripped — and obtaining indentations accentuated at the centres of the faces that closely match the deviation map produced by the Scanning the Pyramids project. Laser scanning and photogrammetry also show the profiles to be far from uniform, with some courses barely recessed and maximum recession exceeding two metres in places; and Petrie was measuring the exposed core rather than the finished casing, which he himself suggested may have thickened along the midline to compensate. The equinox argument advanced here depends only on the hollow existing and catching low-angle light, not on its depth or its intentionality, and it survives Monnier's reading of the cause. But no single concavity value can be treated as a measurement, and for that reason the concavity is not used as a dimension anywhere in the statistical analysis.

The Encoding System: The Wedjat as the Proposed Key

The Wedjat fraction system — $\frac{1}{2}$, $\frac{1}{4}$, $\frac{1}{8}$, $\frac{1}{16}$, $\frac{1}{32}$, $\frac{1}{64}$, summing to $\frac{63}{64}$ with Thoth's $\frac{1}{64}$ restoring the whole — is documented in peer-reviewed Egyptology as the Egyptian system of binary measurement, appearing in the Rhind Mathematical Papyrus and in medical and administrative contexts from the Fifth Dynasty onwards (Clagett 1999; Pommerening 2003). It is further documented by Belmonte (2001) that this fraction system governed temple spatial orientations. This hypothesis proposes — and cannot at this stage prove — that the Wedjat also served as the decoding key for the correspondences documented in this section. Without the Wedjat, these correspondences are invisible

in the pyramid's dimensions. With it, a structured pattern emerges across multiple astronomical and mathematical domains. This hypothesis interprets that emergence as evidence of design.

Summary Table of Correspondences

Table 3 presents every documented correspondence. The “Status” column distinguishes established findings (supported by independent peer-reviewed literature), verified mathematical relationships (arithmetically confirmed, no prior peer-reviewed precedent for deliberate encoding), and original proposals within this hypothesis. Operation class is marked where it differs across entries. Formulae tagged $\times\div$ fall inside the frozen confirmatory analysis; those tagged addition or subtraction use the cross-dimensional operations attested in the Rhind Mathematical Papyrus but deliberately excluded from the Monte Carlo test, as declared in the Declared Methodological Constraints. Where a target is reachable by both operation classes the two routes differ, and the statistical test reports the $\times\div$ route even where it carries the larger error — the civil year at 0.171% by $\text{base}\times(53/64)$ rather than 0.044% by addition, and the sidereal year at 0.126% by $\text{height}\div(49/64)$ rather than 0.101% by subtraction. The statistics therefore rest on the more conservative figures, and a reader comparing this table with the Statistical Appendix should expect the two to differ for those entries.

Established (peer-reviewed) | Constructibility: Egyptian mathematics or direct observation — π via seked: 3.14286, error 0.040% (Rossi 2004; Verner 2001) | ϕ via apothem: 1.61859, error 0.034% (Rossi 2004) | Royal Cubit = $\pi/6$ m (Petrie 1883) | Cardinal orientation $\pm 3.5'$ (Dash 2017; Spence 2000) | Shaft alignments: Orion, Sirius, Alpha Draconis, Kochab (Trimble 1964; Spence 2000) | Concavity equinox effect (Edwards 1975; Petrie 1883).

Verified mathematical correspondences, no prior peer-reviewed precedent for deliberate encoding — [Egyptian constructible: derivable from seked, Wedjat, and Pythagorean methods] ϕ area triple (ϕ , $\phi-1$, ϕ^2 from exterior areas, errors 0.034%/0.090%/0.056%) | $\sqrt{5}$ via $\text{diagonal}\times(23/64)/100$, error 0.007% | $\sqrt{6}$ via $\text{height}\times(7/8)/100$, error 0.021% | Golden angle via $\text{base}\times(5/16)$, error 0.006% | [Requires carried knowledge: only meaningful if Earth's dimensions were known] Earth polar radius via $(\text{base}+\text{diagonal})\times 10,000\div(7/8)$, error 0.0035% | Earth equatorial radius via $(\text{height}+\text{diagonal})\times 360\times 100\times(3/8)$, error 0.0073% [lower tier: Egyptian-arithmetic recoverability; see Correspondences with Earth's Geometry] | [Observationally accessible: derivable from sustained sky-watching] Anomalistic month via $\text{base}\times(1/16)$, error 0.196% [$\times\div$ — in the statistical test] | Civil year via $\text{apothem}+\text{height}\times(1/32)$, error 0.044% [addition — outside the statistical test] | Sidereal year via $\text{base}-\text{height}\times(17/64)$, error 0.101% [subtraction — outside the statistical test].

Original proposals within this hypothesis [Requires carried knowledge: only meaningful if these cycles were observationally recorded] — Saturn (378.09d): $\text{base}\times(55/64) = 378.125\text{d}$, error 0.009% | Jupiter (398.88d): $\text{base}\times(29/32) = 398.750\text{d}$, error 0.033% | Venus (583.92d): $\text{apothem}\div(39/64) = 584.353\text{d}$, error 0.074% | Mars (779.94d): $\text{height}\div(23/64) = 779.130\text{d}$, error 0.104% | Mercury: not found.

What Was Explicitly Tested and Not Found

In the interest of transparency, the following were explicitly tested and not found, or were found and refuted. Speed of light: refuted. The claim is that the Great Pyramid stands at latitude 29.9792458°N and that the speed of light in vacuum is 299,792,458 m/s. The correspondence fails on four independent grounds. First, and decisively, the pyramid is not a point: its 230.33 m base spans 0.00208° of latitude — 7.48 arcseconds — so the monument occupies every latitude from 29.97821°N to 29.98028°N. Roughly twenty thousand distinct seven-decimal latitude values intersect the structure, and the digits beyond the third decimal place are therefore not a property of the monument but a choice of which point on it is measured. Second, the quoted line does not pass through the apex, which lies at approximately 29.9791750°N, some eight metres to the south. Third, the figure 299,792,458 is not a measured quantity but a definition: since the 17th General Conference on Weights and Measures (1983) the metre has been defined as the distance light travels in 1/299,792,458 of a second, the digits having been selected to preserve the length of the metre

established by the French meridian survey of the 1790s. The number is an artefact of modern metrological convention. Fourth, the comparison equates a value in degrees with a value in metres per second, which requires disregarding both the decimal point and the units. The finite speed of light was itself not established until Rømer (1676). This correspondence is accordingly recorded here as refuted, not merely unconfirmed. Euler's number e : systematic search of all primary dimension ratios found nothing within 0.5% of $e = 2.71828$; a formula in popular circulation claimed to yield e but in fact yields ϕ , a misidentification. Sidereal year in the base: the base measures 440 Royal Cubits; 365 cubits $\times$ 0.5236 m falls approximately 39 metres short of the actual base in every standard Egyptian unit system. Earth's curvature encoded in the concavity: calculation shows the concavity is 903 times too deep for Earth's surface curvature across 230 m. Mercury's synodic period: not found within acceptable error bounds after systematic search. $\sqrt{3}$, $\sqrt{7}$, the Euler-Mascheroni constant, and $\ln 2$: not found within 0.3% after systematic search.

Independent Arithmetic Verification

All findings in this section were independently verified by arithmetic computation, beginning from a sceptical position. The initial objection that ϕ had a larger error than π was resolved after direct calculation confirmed it. The planetary correspondences, the year-from-apothem formula, and the Earth radii relationships were each independently computed and confirmed. Two conclusions emerged from verification: that mathematics alone cannot establish intent, and that a fully controlled search, specified in advance and applied to a comparison dataset of comparable monuments, would strengthen the statistical argument. Both are reflected in the framing throughout this section. The AI tools used in this verification are disclosed in the AI Use Disclosure. The mathematical relationships documented here are verified; the interpretation is hypothesis.

The Proposed Scope of the External Memory

If the correspondences documented in this section are accepted as deliberate — and this hypothesis proposes that they are, while acknowledging that this is a claim requiring ongoing scrutiny and eventual peer review — then what may be preserved in the pyramid's exterior spans seven independent knowledge domains, comprising a mathematical vocabulary (π , ϕ , $\sqrt{5}$, $\sqrt{6}$, the golden angle, and the Royal Cubit as their practical embodiment); the physical scale of the Earth in both its principal dimensions (at a designated lower evidential tier); the Moon's anomalistic cycle; the Sun's civil and sidereal year, together with the 360-day schematic base and an equinox instrument built into the exterior geometry; the synodic periods of four of the five planets visible to the naked eye; an attempted but unreached approach to the Sothic cycle that would have connected the 365-day civil year to its long-term stellar recalibration; and the 70-day period of Sirius invisibility linking the pyramid's height to the embalming canon and Osiris mythology. The shaft angles additionally encode the sky's stellar configuration at the construction epoch, enabling precession-based dating by any future observer who knows the precessional rate. Of these, the fourteen astronomical and calendrical targets — and only those — were submitted to the Monte Carlo analysis, of which eleven are reached within the frozen tolerance (Test 1, $p = 0.01497$). The mathematical constants and the terrestrial radii were not among the frozen targets and carry no Monte Carlo support.

This hypothesis does not claim these correspondences constitute proof of deliberate encoding. It proposes that their precision, domain coherence, structural consistency, verified presence of absences (Mercury, e , $\sqrt{3}$), and alignment with the documented domains of the Wedjat fraction system — a system independently attested in Egyptian mathematical, medical, architectural, and calendrical practice — make deliberate encoding the more parsimonious explanation. The Corridor Series establish the who, where, and why of the transmission. This section documents the what: the domains of knowledge that the exterior of the pyramid may have been built to carry. The internal structure of the pyramid, with its own proposed encoding, lies outside the scope of this paper.

Table 3. Verified Wedjat Correspondences: Complete Arithmetic

Operation class: the golden angle, anomalistic month, Saturn, Jupiter, Venus and Mars entries use multiplication or division only and fall inside the frozen confirmatory analysis. The Sothic cycle is not listed here: it was tested and falls outside tolerance, and is reported among the misses. The civil year, sidereal year and 360-day year entries use cross-dimensional addition or subtraction, attested in the Rhind Mathematical Papyrus but excluded from the Monte Carlo test; where a target is reachable by both classes the statistical analysis reports the multiplication or division route, which for those three targets carries the larger error. The 70-day Sirius entry uses multiplication and is inside the test.

Target	Formula (Royal Cubits)	Result	Target value	Error
Golden Angle (137.508°)	$440 \times (5/16) = 440 \times 0.3125$	137.500	137.508°	0.0058%
Anomalistic month (27.554 d)	$440 \times (1/16) = 440 \times 0.0625$	27.500	27.554 d	0.1960%
Civil year (365.000 d)	$356.09 + 280 \times (1/32)$	364.840	365.000 d	0.0439%
Sidereal year (365.256 d)	$440 - 280 \times (17/64)$	365.625	365.256 d	0.1010%
360-day year (360.000 d)	$220 + 280 \times \frac{1}{2}$	360.000	360.000 d	0.0000%
70-day Sirius (70.000 d)	$280 \times \frac{1}{4}$	70.000	70.000 d	0.0000%
Saturn (378.09 d)	$440 \times (55/64)$	378.125	378.09 d	0.0093%
Jupiter (398.88 d)	$440 \times (29/32)$	398.750	398.88 d	0.0326%
Venus (583.92 d)	$356.09 \div (39/64)$	584.353	583.92 d	0.0741%
Mars (779.94 d)	$280 \div (23/64)$	779.130	779.94 d	0.1038%

What Was Ultimately Built

The Wedjat as a Portable Celestial Map and the Pyramid as a Fixed Anchor

The Wedjat sky-compass as a portable celestial map is fully documented in Corridor II (Levy 2026b, Section 5). Canopus — whose Egyptian identification remains open, with Shaltout, Belmonte and Fekri (2007) proposing, as a possibility they explicitly flag for further study, that it is the Great Star (sb3 '3) of Pyramid Texts Utterance 882 that traverses the sky with Sah, and with classical sources identifying its constellation Argo Navis as the Boat of Osiris (Plutarch, *De Iside et Osiride*, *Moralia* 359E; discussed below); for Canopus as an orientation target in Egyptian sacred architecture see Belmonte, Pérez Die and Díaz-Iglesias Llanos (2016) — sank progressively lower as ancestral populations migrated northeast. At the Richat Structure (21.1°N), Canopus was a consistent visible feature. At Giza (30.0°N), during the pyramid-building era (~2500 BCE), Canopus culminated at

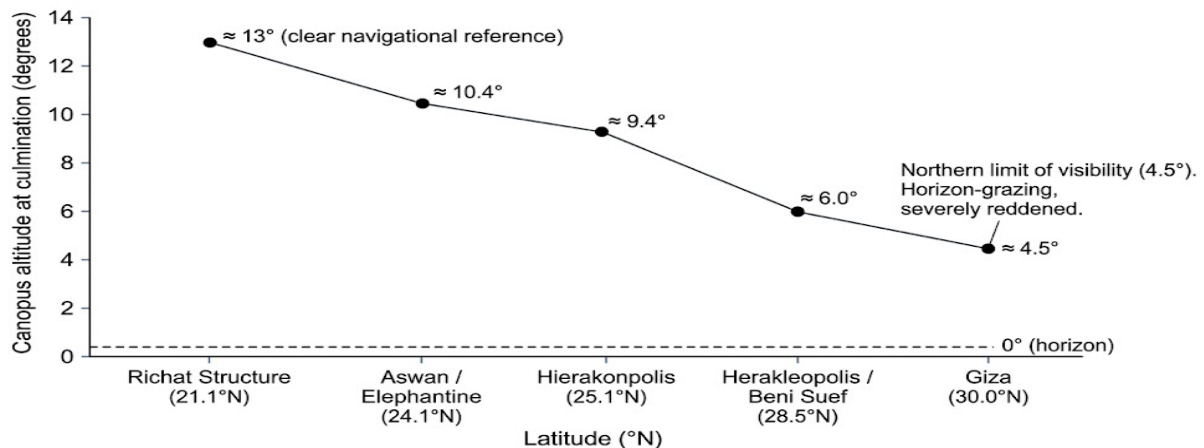
approximately 4.6° — horizon-grazing. The visibility limit for Canopus at approximately 5500 BCE was 27.7°N ; precession lifted this limit gradually northward, reaching $\sim 34.6^\circ\text{N}$ by pyramid-building time. Canopus was below the horizon at the Giza plateau's latitude of 30°N during the Green Sahara peak and lay near the northern edge of the star's practical visibility zone by the Old Kingdom.[^][Altitudes calculated using the precession model documented in Corridor II (Levy 2026b, Appendix A).]

A foundational study by Alexander Badawy and astronomer Virginia Trimble first proved that the so-called air-shafts in the King's Chamber were orientated to the stars of Orion's Belt (Osiris) in the south and to the circumpolar stars (Alpha Draconis) in the north (Badawy 1964; Trimble 1964). This paper proposes the pyramid encodes the stars of the Saharan ancestral tradition — but at Giza latitudes, Canopus could not be targeted by any interior shaft within the pyramid's mass. The reason is not ritual hesitation or architectural convention: it is geometry. At 30°N and ~ 2500 BCE, Canopus's declination of approximately -55.4° gives it a maximum altitude above the southern horizon of approximately 4.6° . All Canopus altitudes quoted in this paper derive from a single computation rather than being calculated separately at each point of use. The method is stated here in full: J2000.0 catalogue position, IAU 1976 rigorous precession, altitude at upper culmination as $90^\circ - \text{latitude} + \text{declination}$, without refraction or extinction. The script self-tests against the figures published in Corridor I and reproduces them to within 0.06° . Two quantities should be kept distinct, since they are easily conflated: Canopus's declination at a given epoch, computed here, and the declination of the Canopus temple-orientation peak at -53.5° reported by Shaltout and Belmonte (2005: 283), which is a property of their survey of temple azimuths and not of the star at any date. The latter must not be substituted into an altitude calculation. One further point of direction: Canopus's declination becomes less negative through the periods this series covers, so its altitude at a fixed latitude rises with time. Earlier epochs give lower altitudes, not higher — the northward migration of the visibility limit recorded in Corridor I is the same fact seen from the other side. The obstacle this creates is structural rather than strictly geometric, and the distinction should be stated precisely. A shaft leaving the King's Chamber floor (~ 43 m above the base) at 4.6° would meet the pyramid's sloping south face after roughly 77 metres of horizontal run, against approximately 46 metres for the actual 45° shaft and 54 metres for the 32.5° northern shaft. Such a channel is not impossible to lay out. It is, however, nearly horizontal across some seventy-seven metres of masonry, rising barely six metres over that distance — a configuration without parallel in Egyptian construction, structurally compromised through the pyramid's core, and useless as a sighting line to a star so severely reddened and dimmed by atmospheric extinction at that altitude that it could not serve as a reliable observational target in any case. The four shafts that were built span 32.5° to 45° , the range within which an inclined channel through a pyramidal mass remains structurally and observationally viable. Canopus at 4.6° falls outside it. The builders did not target Canopus in stone because the pyramid's latitude had placed the star below the range that the medium could carry. The pyramid therefore encodes not the star itself but the threshold — the fixed monumental anchor at the northern end of the Canopus sky road, the permanent 'You Are Here' marker placed exactly where direct stellar encoding ended. The Wedjat functioned as the portable complement: by observing the altitude of Canopus, any observer could determine latitude along this same sky road — and the monument at 30°N marked the point where the portable instrument reached its architectural limit.

A structural parallel to the Wedjat system itself emerges here. The binary fraction system documented by Ritter (2002) as operational grain-measurement mathematics has a hard lower bound: the $\frac{1}{64}$ is the smallest fraction expressible within it. Below $\frac{1}{64}$, Egyptian scribes used an entirely different notation — *ro*, an open mouth hieroglyph — for smaller quantities. The $\frac{1}{64}$ is therefore not merely the smallest step in the series; it is the threshold at which the existing system requires replacement by something else. Within Corridor II's framework, in which the Wedjat encodes the Canopus visibility gradient, the parallel is exact. At Giza — 30°N , pyramid-building era — Canopus culminates at approximately 4.6° . It is present above the horizon; it is not invisible. But as set out above, it falls below the range an enclosed shaft through the pyramid's mass can carry: the required channel would run nearly horizontal across some seventy-seven metres of masonry, and a star at that altitude is too reddened and dimmed by atmospheric extinction to serve as an observational target. The known

shafts in the Great Pyramid range from 32.5° to 45° (Badawy 1964; Trimble 1964).

The absence of a Canopus shaft is therefore not a ritual avoidance but a consequence of the monument's latitude acting on the limits of the medium: at 4.6° the star had fallen below the range an inclined channel through a pyramidal mass can usefully carry. Two thresholds at the same latitude: the smallest step the Wedjat system can express, and the lowest altitude at which Canopus can be targeted directly in stone. Both require what lies beyond the existing system. The Wedjat's missing 1/64 was supplied by Thoth's externalised memory (Ritter 2002; Gillings 1972). This paper proposes the pyramid encodes the threshold itself in stone — not the star, which had reached the geometric limit of direct architectural expression, but the latitude at which it crossed that limit (cf. Corridor II, Levy 2026b, Section 5). No prior peer-reviewed source has drawn this specific connection; it is presented as a synthesis within the framework of this series.



Canopus altitude at culmination decreasing from ~13° at Richat (21.1°N) to ~4.5° at Giza (30.0°N) in 5500 BCE. The Giza plateau lies at the northern limit of Canopus visibility. Data from precession-corrected coordinates (Paper II model).

Primary Purpose: Cosmological Encoding

A multi-generational migration of the kind documented in The Corridor Series does not merely displace a population geographically. It gradually erodes the fixed standards on which a cosmological system depends. Over centuries of movement, the cubit drifts as it passes through successive artisanal traditions without a master standard; the calendar loses its agricultural anchor as seasonal markers shift with latitude; Canopus — the navigational star that oriented the Saharan homeland — sinks progressively below the horizon with each generation born further northeast; and the landscape that gave the Zep Tepi its relational grammar — the primordial mound, the concentric waters, the break in the ring — recedes first into living memory and eventually into mythologised legend. By the 4th Dynasty, the Green Sahara was a dry wasteland. The people who carried its cosmology no longer had a living landscape to point to as their zero point. They had institutional memory, the Shemsu Hor transmission chain, and the mathematical encoding of the Wedjat — but no fixed physical anchor. The pyramid is the solution to exactly this problem. Positioned at 30°N, oriented to Meskhetyu within ~3.5 arc minutes, clad at its apex in Aswan granite from the southern threshold, aligned in its causeway toward the western origin, and engineered to outlast any other material made by human hands, the Great Pyramid is the zero point of the Zep Tepi cosmological system fixed in stone: an indestructible calibration standard permanently encoding every stellar reference, every cosmological relationship, and the memory of the landscape that gave rise to them — at a single location the desert could never take away.

The mythological dimension of this knowledge-preservation function is explicit in the Pyramid Texts. The Cannibal Hymn, inscribed on the antechamber east gable of the Unas pyramid (Utterances 273–274, c. 2345 BCE), describes the king's apotheosis as the absorption of divine knowledge: *'Unas has*

swallowed the wisdom (sjA) of every god... Unas' span of life is Eternity, his limit is Everlastingness.' The sjA — the divine faculty of perception and knowledge — is precisely what the king incorporates in his transformation. This is the ritual framework within which the pyramid's mathematical and astronomical encoding operates. The architecture that preserves knowledge externally in stone corresponds, in the Unas pyramid's inscription program, to the apotheosis of a king who has absorbed the sjA of every god internally. The two are not separate functions. The pyramid is simultaneously the external repository and the instrument of internal transformation — the place where encoded knowledge and the king's consciousness become the same thing.

On the reading proposed here, the Great Pyramid was built to permanently encode the Zep Tepi cosmological system in the most durable medium available. It is not *primarily* a tomb. It is a three-dimensional cosmological monument that recreates the primordial landscape from which Egyptian civilisation derived its sacred worldview.

This cosmological reading does not replace the funerary interpretation — it prioritizes it. The pyramid is the machine of royal transformation precisely because it is the Zep Tepi made stone. The king becomes divine by re-enacting the First Time. **A more precise formulation is that the pyramids' design is not optimized for burial** — the absence of canopic equipment, identifying inscriptions, and burial goods suggests that preservation of the body was not the primary architectural driver.

Independent Convergence: Giza as "Written Landscape"

The reading of Giza as a cosmological landscape is not isolated to the present hypothesis. The archaeoastronomer Giulio Magli, publishing in peer-reviewed venues including Cambridge University Press, has independently argued that "the Giza plateau can be interpreted as a 'written landscape' where the pyramids represent the hieroglyph for the primordial mound" (Magli 2011; cf. Magli 2013, 2016). Magli's analysis, grounded in the precise orientation data of the pyramids and their relationship to Heliopolis, converges on the same core insight that animates the Zep Tepi model: the Giza pyramids are not merely tombs but architectural encodings of the Benben, the primordial mound of Egyptian creation theology.

A second line of mainstream Egyptological convergence comes from material culture rather than archaeoastronomy. Lehner and Hawass (2017), in their definitive synthesis of the Giza evidence, document that the royal pyramids represent the culminating monumental expression of a long-term Egyptian funerary grammar — the mud-brick mound within an enclosure — that constitutes the basic funerary form across millennia and across all classes. Wong (2021), writing in the Cambridge Archaeological Journal, approaches the same transition from the material side: he argues that environmental factors — favourable limestone geology in Lower Egypt, and multiple strands of evidence for wetter climatic conditions during the Early Dynastic period and the Old Kingdom that undermined both the production and the durability of mudbrick — played a key role in facilitating and accelerating the rise of stone architecture, with the First Dynasty innovations emerging first in the private rather than the royal realm. The form is the same: central elevated mass, surrounding enclosure, separation of sacred core from outer world. The scale changes; the grammar does not.

The empirical anchor for this tradition is found at Hierakonpolis. The Predynastic temple at Nekhen, excavated by Quibell and Green (1902) and re-examined by Hoffman et al. (1986), consisted of a sand mound enclosed by a roughly circular stone revetment — precisely the cosmographic structure (central elevated form within a concentric ring boundary) that This paper proposes the Giza pyramids monumentalize at scale. This is at the same site as HK6, the predynastic cattle cult center documented in Corridor III, and predates the Step Pyramid by approximately six centuries. The Zep Tepi cosmogram is therefore not a hypothetical construction reverse-engineered from Giza. It is a documented architectural form attested at the Shemsu Hor capital in Predynastic times. The Giza pyramids are the monumental, state-scale instantiation of a structural grammar that had been

continuously expressed in stone and mudbrick across Egypt for at least seven hundred years before Khufu's reign.

The convergence between Magli's mainstream archaeoastronomical analysis and the Zep Tepi framework is not a matter of citation for authority. It is evidence that the cosmological reading of Giza is not a fringe interpretation but a model that satisfies a dense set of empirical constraints — constraints that are independently documented and peer-reviewed. The question is not whether Giza encodes cosmology. The question is which cosmology it encodes, and from where that cosmology originated. The Zep Tepi model proposes a specific answer: the cosmology of the Green Sahara, carried by pastoralist migrants, preserved by the Shemsu Hor, and instantiated in stone at Giza.

Three Registers of Architectural Purpose

- **Backward-looking (restoration):** This paper proposes the pyramid encodes what migration eroded — the relational grammar of the Richat landscape, the Canopus sky of the Saharan homeland, the mathematical standards of the Zep Tepi system. It does not recover these things. It permanently marks their threshold in stone that cannot drift.
- **Translational (encoding):** The Richat Structure is circular — the natural, cosmic, divine form of the Zep Tepi mound. The Giza complex is square-based — the human architectural expression of the same relational system. The circle is not copied; it is translated. The pyramid complex encodes the relational grammar — central elevation, bounded space, transition water, directional axis — in the architectural language of human construction. The form changes; the cosmological relationship does not.
- **Forward-looking (insurance):** The pyramid was built not only for what was lost, but against what might come again. The African Humid Period collapse may have been the founding trauma the entire institutional memory was organised around. A civilisation that had lived through one world-ending environmental event had every reason to build beyond the reach of any future one. The pyramid does not merely preserve that memory. It guarantees it: an archive in the only medium that cannot be dissolved by sand, outlasted by drought, or erased by the next turning of the sky.

These three registers define the pyramid's complete function: a monument designed simultaneously as memorial, knowledge repository, and durable preservation against civilisational loss.

Secondary Purpose: Royal Transformation

The pyramid also served as the mechanism of royal apotheosis — the architectural machine through which the dead pharaoh underwent transformation from human king to divine being. This is the function described in the Pyramid Texts: the king ascending to the Imperishable Stars, merging with Ra, becoming Osiris. The two purposes — cosmological encoding and royal transformation — are not in conflict. They are the same purpose at different scales. This reading aligns with mainstream Egyptological scholarship: Assmann (2005) documents that Egyptian mortuary religion was centred on transformation rather than interment, the pyramid functioning as an instrument of royal becoming rather than a container for remains (Assmann, J. 2005. *Death and Salvation in Ancient Egypt*. Trans. David Lorton. Cornell University Press).

Mechanism of Transmission

How does a landscape become an architectural system? The proposed transmission does not require direct architectural copying of a landscape. It operates through cognitive and ritual encoding.

Landscapes such as Richat are not reproduced visually but abstracted into cosmological principles --- central emergence, bounded order, liminal water, and directional transformation. These principles are preserved through ritual practice, oral tradition, and symbolic systems (e.g., the Wedjat as a fractional and integrative schema), and later instantiated in architecture when conditions allow. The resulting structures therefore reflect not the physical form of the original landscape, but its encoded relational logic.

This material approach to cultural transmission finds direct support in recent theoretical work on memory in ancient Egypt. Olabarria (2025), in a dedicated Cambridge Elements volume on Egyptian memory practices, argues that "a performative engagement with monuments was one of the bases of the practice of memory" in ancient Egypt. Her framework identifies monuments as "technologies of remembrance" --- material anchors designed to fix collective cosmological knowledge in durable form across generations. The pyramid, on this reading, is not merely a symbol of the Zep Tepi but a memory technology engineered to preserve and transmit the relational grammar of the primordial landscape.

Olabarria's analysis operates within the tradition of cultural memory theory established by Jan Assmann (already discussed in Section 5 above), but extends it by emphasizing the specific mechanisms --- materiality, performativity, and place-making --- through which monuments sustain memory across centuries. The present hypothesis operationalizes this framework: the pyramid as memory technology, the causeway as processional stage for performative re-enactment, and the entire Giza complex as a "place of memory" that re-instantiates the ancestral Saharan landscape in monumental form (Olabarria 2025).

The performative dimension of this transmission is consistent with broader Egyptian material culture analysis, including Kroft's (2024) examination of decorated ware vessels as vehicles for memory expression in formalized ritual contexts.

Olabarria's Egypt-specific framework sits within a wider comparative literature on how non-literate and partly-literate societies encode durable knowledge, which supplies the mechanism this hypothesis requires. In cognitive archaeology, Kelly (2015) documents that oral societies stored large bodies of pragmatic knowledge --- including astronomy, calendars, and navigation --- in monuments and landscape features functioning as mnemonic devices, reinterpreting Stonehenge, Chaco Canyon, and Poverty Point on that basis. In cultural astronomy, Hamacher (2018) demonstrates that oral cultures encoded precise astronomical observations, including stellar variability and periodicity, and transmitted them across generations for calendrical and ritual use. The cognitive constraints that make such transmission faithful rather than degradative --- theme, imagery, and rhythmic structure operating together --- are set out by Rubin (1995), while Vansina (1985) provides the methodological framework for treating the resulting traditions as recoverable information rather than invention.

Two points follow. First, the mechanism proposed here is not an ad hoc postulate but an instance of a documented cross-cultural phenomenon: monuments serving as knowledge stores, and oral tradition as a high-fidelity carrier. Second, the practice of encoding precise astronomical periods is independently attested within Egypt itself. Jetsu and Porceddu (2013, 2015) present statistical evidence that the Cairo Calendar (Papyrus Cairo 86637) records the 2.85-day period of the eclipsing binary Algol and the 29.6-day lunar period in its day-by-day structure --- the earliest known documentary record of a variable star's period. That source is Ramesside and therefore far later than Khufu, and its Algol identification is debated; it is cited here only to establish that period-encoding was a real Egyptian practice, not a modern projection. The claim advanced is accordingly the narrower one: that the Great Pyramid may represent a monumental instance of a mnemonic practice for which the comparative and Egyptological records already provide precedent.

Full Synthesis

Comparative Analysis: The Full Relational System in Cross-Cultural Perspective

Individual elements of the relational system (central elevation, bounded space, water access, directional axis) occur in other sacred landscapes. The claim here is not that Giza is unique in each element, but that the full integration of all four elements in a single planned complex, with the specific theological encoding (Zep Tepi, Nun/Benben, western orientation), is not clearly documented elsewhere in the current comparative sample.

The following table tests the proposed four-element relational system against other monumental landscapes, including newly added cases to expand the comparative scope.

Site	Central Elevation	Bounded Space	Water at Transition	Axis (Water → Monument)	Full Integration
Richat	✓ Dome	✓ Rings	✓ Basin	✓ West gap	✓
Giza	✓ Pyramid	✓ Enclosure	✓ Canal	✓ Causeway	✓
Nabta Playa	✗ Flat	⚠ Stone circle	✗ Seasonal	✗ None	✗
Mesopotamian Ziggurats	✓ Ziggurat	✓ Precinct	✗ None	⚠ Processional (not water-linked)	✗
Stonehenge	✗ None	✓ Ditch	✗ None	⚠ Solstice (not water-linked)	✗
Mesoamerican Pyramids	✓ Pyramid	✓ Plaza	⚠ Cenotes (not boundary-linked)	⚠ Axial (not water-linked)	✗
Göbekli Tepe (c. 9500–8000 BCE)	⚠ Enclosures on hilltop (natural elevation, not built mound)	✓ Yes (circular/oval enclosures with limestone T-shaped pillars)	✗ No (located on dry plateau; no water at boundary)	✗ None (no processional axis connecting water to monument)	✗
North African Megalithic Circles (e.g., Nabta Playa, Senegambian stone circles)	✗ None (flat or low elevation)	✓ Yes (stone circles, varying diameters)	⚠ Seasonal (Nabta has playa, not basin-scale)	⚠ Possible stellar alignments (Nabta), but no water-to-monument axis	✗
Saqqara (Djoser)	✓ Step Pyramid	✓ Enclosure	⚠ Dry Moat (contested hydraulic reading, Landreau et al. 2024; treated as provisional)	✓ Processional	⚠ Partial (strongest precursor)
Later Egyptian Temples	⚠ Sanctuary (not elevated)	✓ Enclosure	✓ Nile access	✓ Processional	⚠ Partial (missing central elevation)

Site	Central Elevation	Bounded Space	Water at Transition	Axis (Water → Monument)	Full Integration
Angkor Wat (c. 1113–1150 CE, Cambodia)	✓ Central tower (Meru axis)	✓ Concentric enclosures	✓ Surrounding moat	✓ Causeway from moat to temple	⚠ Full integration — convergent case (3,600 years post-Giza; no transmission linkage)

Borderline case: Saqqara (Step Pyramid complex of Djoser, 3rd Dynasty). The Saqqara complex exhibits central elevation (Step Pyramid), delimited sacred space (enclosure wall), and a directional axis (the Heb-Sed court and processional elements). The question of water integration at Saqqara has been reopened, though not settled, by Landreau et al. (2024, PLOS ONE 19(8): e0306690), who conducted transdisciplinary analysis of the Saqqara watershed and found that the Gisir el-Mudir enclosure has the structural features of a check dam, with topographic reconstruction supporting the existence of an intermittent lake west of the Djoser complex connected to the Dry Moat. On that reading the Dry Moat would be more than a symbolic water-proxy; the interpretation is recent and contested, and is treated here as provisional (Lauer 1936; Jeffreys 2010; Landreau et al. 2024). On this basis, Saqqara satisfies three of the four relational elements: central elevation, bounded space, and water at the boundary. The fourth element — a directional axis explicitly connecting water to monument — is present processionally but not hydraulically integrated in the manner of the Giza causeway. Saqqara is therefore the strongest precursor form of the full system. The element distinguishing Giza from Saqqara is not the presence of water but its deliberate hydraulic integration with the monument axis.

The intermediate step between Djoser and Khufu — the reign of Sneferu — provides independent support for this developmental trajectory. Rather than treating the Bent and Red Pyramids as sequential experiments — the second generated by the failure of the first — Belmonte and Lull (2023) argue that "the pair of pyramids of Sneferu in Dahshur would actually form a single project conceived as such from its inception" (p. 396). The complementary names of the two monuments — *Kha Sneferu* ("Sneferu Appears") for the Bent Pyramid and *Sneferu Kha* for the Red Pyramid — are "undoubtedly evidence that supports the idea of a common project" (Belmonte & Lull 2023, pp. 397–398). They further propose that the two pyramids may represent "a huge symbolic representation of the white crown of Upper Egypt, in the case of the Bent Pyramid... and of the red crown of Lower Egypt for the Red Pyramid" (Belmonte & Lull 2023, p. 398). This reading transforms Sneferu's reign from a period of trial-and-error correction into a coherent architectural programme — a deliberate dual project that anticipates Khufu's own "unitary project" at Giza, where Khufu and Khafre's pyramids and the Sphinx were, on Belmonte and Lull's analysis, conceived as a single integrated complex called *Akhet Khufu*, "the Horizon of Cheops" (Belmonte & Lull 2023, p. 400). The full Giza complex thus represents not the independent decisions of successive kings but the culmination of a multi-generational tradition of planned sacred landscapes — a tradition whose earliest monumental expression, this paper argues, is the Phase Zero provincial pyramid program.

Comparative cases:

Göbekli Tepe (c. 9500–8000 BCE, southeastern Turkey): Göbekli Tepe is included here as the most prominent potential counterexample to the claim of Giza's structural uniqueness, and its inclusion is intended to test --- not assert --- comparability. Often cited as the world's oldest known megalithic complex, Göbekli Tepe features multiple circular and oval enclosures formed by T-shaped limestone pillars bearing intricate animal reliefs. (As established in Corridor II (Levy 2026b, Section 5), the Wedjat fractional geometry --- the division of the Eye of Horus into $\frac{1}{2}$, $\frac{1}{4}$, $\frac{1}{8}$, $\frac{1}{16}$, $\frac{1}{32}$, $\frac{1}{64}$ ---

distinguishes the Saharan/Nile tradition of symbolic abstraction from the purely circular enclosures of Göbekli Tepe. The present comparison focuses strictly on the four-element landscape grammar.) The site is situated on a natural hilltop rather than a built mound and exhibits clear bounded sacred space. However, it lacks any evidence of water at the boundary --- the site is located on a dry plateau with no associated water feature --- and there is no processional axis linking water to a central elevation.

Recent analysis by Verhoeven (2024) further clarifies Göbekli Tepe's ritual function, emphasizing boundaries, art, and communal ceremonial practice rather than cosmological encoding of a specific origin landscape. The site's monumental enclosures divided ritual space from the profane world, but this bounded-space function operates independently of the integrated water-axis-elevation grammar attested at Giza. Göbekli Tepe thus demonstrates that bounded sacred space and monumental construction can arise without the full four-element relational system — strengthening, rather than weakening, the claim that this specific integration requires explanation wherever it appears. While Göbekli Tepe demonstrates that hunter-gatherer societies could construct monumental sacred space, it does not satisfy the full four-element relational system. Its inclusion in this comparison is therefore a negative result: the most cited candidate for early monumental cosmography lacks the integrated water-axis-elevation grammar attested at Giza.

North African Megalithic Traditions (e.g., Nabta Playa, Senegambian stone circles): Nabta Playa features stone circles and possible stellar alignments — the calendar circle dated by an adjacent hearth to 4800 ± 60 BCE, and the megalithic alignments to roughly 4500–3600 BCE on five radiocarbon dates from the quarry (Malville et al. 2007), and seasonal playa water was present nearby, but the water was not integrated as a boundary feature with a processional axis. The Senegambian stone circles (c. 700–1500 CE, later but representative of regional megalithic traditions) consist of multiple stone circles marking burial sites, but lack central elevation, integrated water, or directional axis. These traditions show that bounded sacred space and astronomical awareness existed elsewhere in Africa, but the full relational system (central elevation + bounded space + transition water + water-to-monument axis) is absent.

Angkor Wat (c. 1113–1150 CE, Cambodia): This is the most important case for testing the uniqueness claim and is included here as an explicit convergent case. Its cosmological design is documented in peer-reviewed literature: Stencel, Gifford and Morón established that its measurements “are related to practical astronomy and religious symbolism” (Stencel, R., Gifford, F. & Morón, E. 1976. *Astronomy and cosmology at Angkor Wat*. *Science* 193(4250): 281–287. DOI: 10.1126/science.193.4250.281); Romain provides detailed archaeoastronomical analysis (Romain, W.F. 2022. *Angkor Wat*. *Journal of Skyscape Archaeology* 8(1): 4–35. DOI: 10.1558/jsa.19967); and Magli addresses its cosmological design in a dedicated peer-reviewed chapter (Magli, G. 2025. *Cambodia and Java*. In: *Archaeoastronomy*. Springer, pp. 263–274). Angkor Wat achieves full four-element integration: a central tower (Mount Meru axis), concentric enclosure galleries, a surrounding moat at the boundary, and a causeway connecting moat to temple along a cardinal axis. It satisfies all four elements. Several features, however, distinguish its relationship to the present hypothesis from Giza's. First, it postdates Giza by approximately 3,600 years. Second, no transmission linkage between the Green Sahara cosmological tradition and 12th-century Khmer religious architecture is implied. Third, the Angkor system derives its cosmological grammar from Mount Meru theology in the Hindu–Buddhist tradition — an independent, well-documented cosmological framework. Angkor therefore represents convergent cultural evolution: two independent civilisations arrived at the same structural solution from distinct cosmological starting points. This convergence confirms that the four-element system is architecturally achievable and culturally plausible across independent traditions. It does not, however, explain why Giza instantiated it 3,600 years earlier. The parsimony argument for transmission over independent invention at Giza rests on the convergence of additional evidence domains (genetic, archaeozoological, textual) that are not present for Angkor. Convergent evolution at Angkor therefore strengthens, rather than weakens, the argument: if the system can arise independently, it requires explanation when it appears alongside a documented corridor, an ancient North African genetic record spanning that corridor, and textual memory of a western origin.

Sample limits: The sites selected represent major, well-documented ritual landscapes across multiple cultural regions, including Southwest Asia (Göbekli Tepe), Europe (Stonehenge), Mesopotamia, Mesoamerica, Africa, and Southeast Asia (Angkor Wat). The sample is not exhaustive, and future research may identify additional parallels or counterexamples. The claim is therefore restricted to the current comparative sample.

Conclusion of comparative analysis: While individual elements of the proposed system occur widely, the only other known site achieving full four-element integration is Angkor Wat (12th century CE, Cambodia) — 3,600 years after Giza and with an independent cosmological derivation from Hindu–Buddhist tradition. Within the current sample, Giza remains the earliest known full instantiation of the system, and the only one for which a documented corridor, an ancient North African genetic record spanning that corridor, and textual memory of a western origin are available as supporting evidence. Göbekli Tepe, despite its antiquity and monumental scale, lacks water integration and a processional axis. North African megalithic traditions (including Nabta Playa) exhibit bounded space and some astronomical awareness but lack the integrated water-axis-elevation system. Later Egyptian temples preserve a reduced form (boundary + water + axis) but lack the dominant central elevation. This pattern is consistent with — but does not prove — an evolutionary trajectory from Nabta (precursor) through Giza (full instantiation) to later temples (reduced form). The uniqueness of Giza's full integration within this expanded sample strengthens the hypothesis that Giza represents a deliberate architectural encoding of the Zep Tepi relational system, rather than a coincidental convergence of common sacred elements.

A note on the comparative method: The absence of the full four-element system at Göbekli Tepe, Stonehenge, or other sites does not constitute positive evidence that Giza's system was transmitted from Richat. It serves only to eliminate the alternative explanation that such systems arise universally or randomly across cultures. The positive case for transmission rests on the convergence of independent lines — hydrology, genetics, archaeozoology, texts, and astronomy — not on the comparative table alone.

The Constraint Filter: What Must Any Model Explain?

Taken together, the empirical features documented in this paper constitute a constraint filter. Any viable interpretation of the Giza complex must account for the following features simultaneously. While some constraints may be partially interrelated, they originate from distinct domains — survey measurement, textual sources, landscape archaeology, and theological analysis — and therefore cannot be reduced to a single explanatory variable.

Constraint	Empirical Finding	Source
Causeway orientation about the east–west axis	Khafre 283.4°, Menkaure 269.2°, Khufu ~255°; opposed deviations about the east–west axis, mean 269.4°	Nell & Ruggles (2014)
Heliopolis alignment	Southeast corners of Giza pyramids align precisely to Heliopolis, cult center of the Benben	Lehner (1985)
Institutional memory	Abu Sir pyramids maintain Heliopolis alignment despite invisibility	Magli (2009)
Integrated landscape planning	Khufu quarry aligns with pyramid; volume matches pyramid (2.76M vs. 2.59M m³)	Lehner (1985)
Material-theological convergence	King's Chamber granite from Elephantine; Khufu's name = "Khnum is my Protector"; Khnum = ram-headed god of	Shaw (2000); Britannica;

Constraint	Empirical Finding	Source
	Elephantine; Elephantine = "door of the south"	Belmonte et al. (2016)
Cultural continuity	Egyptian conceptual landscapes show 2000-year symbolic continuity	Magli (2011)
Engineered water at boundary	Khufu Branch canal and Ahramat Branch; deliberately cut harbour basin at pyramid base	Sheisha et al. (2022); Ghoneim et al. (2024); Younes et al. (2024)
Bare interior walls	No hieroglyphics in 4th Dynasty pyramids; Pyramid Texts appear only when embodied knowledge fades	Allen (2005); physical survey

If the Zep Tepi cosmological grammar were **not** transmitted from the Green Sahara to Giza, the following null-model predictions would be expected:

Null-Model Prediction	Empirical Finding	Null Model Status
Pyramid placement would be random or purely functional, without consistent water-axis-elevation integration	Giza exhibits integrated central elevation + bounded space + water at boundary + western axis	Falsified
Causeway orientation would show no directional consistency (or would be fully explainable by local topography alone)	All three Giza causeways organised about the east–west axis, deviating in opposed senses (Khufu ~255°, Menkaure 269.2°, Khafre 283.4°; mean 269.4°)	Falsified
No directional consistency in pyramid placement across dynasties	Abu Sir pyramids maintain Heliopolis alignment despite invisibility	Falsified
Water systems at pyramid sites would show no geometric integration with the monument axis (water placement would be purely functional — access from any convenient direction, with no alignment to a cardinal causeway or directional cosmological axis)	Khufu Branch canal and engineered harbour basin are geometrically integrated with the western causeway axis: the harbour is located at the causeway terminus, the causeway runs due west, and the Nile branch terminates at the pyramid base — not at a functionally arbitrary access point (Sheisha et al. 2022; Younes et al. 2024; Nell & Ruggles 2014)	Falsified
No correspondence between four-element relational system and earlier Saharan or Predynastic ritual sites	Nabta Playa exhibits precursor form (bounded space + water proximity, lacking full integration)	Falsified

The Zep Tepi model is consistent with all empirical constraints in this filter as currently defined. It accounts for the western orientation (toward the Saharan origin), the Heliopolis alignment (toward the Primeval Mound, the Benben), the institutional memory (the Shemsu Hor transmission chain), the integrated landscape planning (Giza as unified cosmological statement), the material-theological convergence (Khnum as Khufu's protector, Elephantine as granite source, Canopus as stellar counterpart), and the cultural continuity (2000-year persistence of symbolic grammar).

The following table summarises each major competing interpretive framework against eight criteria, two of which — Grand Gallery function and Subterranean Chamber — fall outside this paper's declared scope and are marked as such rather than claimed. No single alternative framework accounts for more than two criteria simultaneously as a unified explanation. Credit is given where each framework genuinely addresses a criterion; partial credit (~) indicates the criterion is addressed in limited scope, contested, or domain-restricted. The table is intended as a comparative research tool, not a dismissal of prior scholarship — several of the frameworks listed provide the archaeoastronomical and cosmological foundations on which this hypothesis builds.

Hypothesis	Empty chambers	Stellar shafts (system)	Grand Gallery function	Subterranean chamber	Landscape grammar	Dimensional encoding	Falsifiable predictions	Origin tradition
Royal funerary monument Mainstream Egyptology	✗	~	~	~	~	~	✗	✗
Labor-state ideology Lehner, Hawass	~	✗	~	✗	~	✗	~	✗
Solar / benben cosmology Stadelmann et al.	~	✗	✗	✗	✓	~	~	✗
Stellar soul-ascent Badawy, Trimble	~	✓	✗	✗	✗	✗	✓	✗
Orion correlation theory Bauval & Gilbert (1994)	✗	~	✗	✗	✓	✗	~	~
Osirian funerary theology Standard religious interpretation	✗	~	✗	✓	~	✗	✗	✓
Zep Tepi / Pyramid as Memory This hypothesis (Levy 2026)	✓	✓	—	—	✓	✓	✓	✓
✓ Fully addressed ~ Partially addressed or contested ✗ Not addressed — Monte Carlo statistical verification now complete (Test 1 $p = 0.01497$; Test 2 $p = 0.02419$). — Outside this paper's declared scope: the Grand Gallery and Subterranean Chamber are not addressed here and are not claimed.								

Alternative models that treat Giza as a collection of independent royal tombs, or as purely functionally determined by topography and available materials, satisfy at most one or two of these constraints. They fail to account for the full filter.

The question is not whether the Zep Tepi model is "proven." The question is whether any alternative model can be constructed that also passes through the constraint filter intact. The field is invited to propose such models. Under the evidence currently assembled, no alternative framework has been presented that accounts for the six criteria within this paper's declared scope simultaneously. The field is invited to propose and test such frameworks; the evaluation criteria are stated in the comparative table above.

The hypothesis does not claim the pyramids had no funerary role. The two readings — cosmological encoding and royal transformation — are complementary at different interpretive scales. Alternative explanations for individual features (robbery accounting for missing valuables, functional necessity explaining water channels, chronological lag explaining absent hieroglyphs) are addressed within the

evidence chapters above. The claim is that the full constraint set — empty chambers, western axis, integrated water, stellar alignment, and institutional memory — is better explained by the cosmological reading than by any available combination of independent functional explanations. Convergent cultural evolution at Angkor Wat confirms the four-element relational system is independently achievable; it does not explain its earlier appearance at Giza alongside a documented corridor, an ancient North African genetic record spanning that corridor, and textual memory of a western origin.

Cross-Site Test of the Relational Triad

The relational reading advanced above makes a falsifiable prediction. If the Giza configuration encodes a cosmological template rather than a site-specific engineering solution, the same relational triad — water access at the sacred boundary, a directional axis, and an elevated central feature — should recur at other Old Kingdom ceremonial complexes, and should appear even where the site is not a royal tomb. The prediction is tested here against four further complexes spanning the Third to Fifth Dynasties, each scored on three independent criteria: (1) water access at the sacred boundary — a valley temple, harbour, or quay at a canal, lake, or Nile channel; (2) a directional axis — a causeway or processional spine linking that water access to the centre; and (3) an elevated central feature — a pyramid, benben platform, or sanctuary mound. Each criterion is recorded as present, symbolic or partial, or absent.

The triad is complete at every Fourth- and Fifth-Dynasty complex examined. At Dahshur, the complex of Sneferu — Khufu's father, one reign before Giza — pairs the Bent Pyramid with a valley temple, a lower causeway, and an associated harbour set within a deliberately managed water landscape, as documented by the German Archaeological Institute's excavations (Alexanian et al. 2010). At Abusir, the Fifth-Dynasty pyramids of Sahure and his successors place their valley temples directly on the shore of the Abusir lake, linked by causeways to the pyramids above (Verner 2001). In each case water access, axis, and elevated centre are all present.

The most diagnostic case is the one the prediction specifically anticipated: a non-funerary ritual site. The Fifth-Dynasty sun temples at Abu Ghurab are solar sanctuaries rather than royal tombs — the oldest cult buildings explicitly dedicated to Ra — yet they reproduce the identical triad. The valley temple of Nyuserre's solar complex supplies the water access, a causeway supplies the axis, and in the position a pyramid would occupy stands a massive benben on a raised platform; the valley temple was itself newly excavated by the Italian Archaeological Mission at Abu Ghurab, directed by Massimiliano Nuzzolo (University of Turin) and Rosanna Pirelli (University of Naples L'Orientale), across the 2024 and 2025 seasons. Ludwig Borchardt had identified the structure in 1901 but could not excavate it because of the water table; falling groundwater has since made the work possible. The excavation exposed more than 1,000 square metres — roughly half the sanctuary — together with a ramp interpreted as having connected the temple to the Nile or one of its branches (Egyptian Ministry of Tourism and Antiquities, announcement of 12 December 2025; Italian Ministry of Foreign Affairs and International Cooperation, 2026). As with the Hawass announcement discussed above, these findings are reported from official statements rather than peer-reviewed publication, and the water-access element at Abu Ghurab is therefore recorded here as provisionally rather than definitively confirmed. The benben is the primordial mound itself. That the same water-to-axis-to-elevated-centre relation should govern a site with no burial, and should place the explicit primordial-mound symbol at its focus, is difficult to reduce to funerary function; it reads as cosmological configuration.

The earliest complex sharpens rather than weakens the pattern. Imhotep's Step Pyramid at Saqqara already assembles the cosmological core — an elevated primordial mound within a bounded sacred precinct, itself ringed by the great rock-cut Dry Moat — but in inward-turned form. The moat's primary documented function is a quarry (Welc & Marks 2014), and any reading of it as a symbolic water-boundary — or, on the recent and still-contested hydraulic interpretation of Landreau et al. (2024), as genuine water infrastructure — must remain secondary and provisional; the complex's axis, likewise, is internal (entrance colonnade to Great Court to Heb-Sed court) rather than an external causeway to

the river. What the Fourth Dynasty adds is therefore not the invention of the template but its externalisation: the inward-turned enclosure becomes an outward causeway to a real harbour, and the symbolic boundary becomes engineered water. A configuration that develops across dynasties, rather than appearing fully formed, is precisely the signature of transmitted and elaborated knowledge rather than a single functional solution.

One objection must be met directly. The valley-temple-causeway-monument arrangement is the canonical Old Kingdom pyramid-complex form, and for the pyramid complexes it admits a straightforward functional reading: a valley temple at the water for arrival and the transport of stone, a causeway for processional access. Two features of the present sample resist that reduction. First, the sun temples carry the triad without a tomb, with a benben rather than a burial at the centre. Second, the water is not merely proximate but engineered — the purpose-built harbours and regularised wadis at Dahshur and Giza represent deliberate configuration rather than convenient location. The relational triad is, on this evidence, best understood as a cosmological template that the Old Kingdom progressively externalised and monumentalised — most completely at Giza.

Falsifiable Predictions

Predictive Test

If the proposed framework is valid (rather than post hoc), then undiscovered or less-studied Old Kingdom sites should exhibit partial or full integration of these elements. Specifically, the framework predicts that any newly excavated ritual or ceremonial site from the 3rd–5th Dynasty with preserved architectural remains should show (a) water access at the boundary of a delimited sacred space, (b) a directional axis connecting that water access to (c) an elevated central feature (pyramid, mastaba, or sanctuary platform). The absence of this relational integration in a well-preserved, non-funerary ritual site from this period would weaken the hypothesis.

Methodological Note on Evidentiary Weighting

This hypothesis draws on multiple independent lines of evidence: hydrology, genetics, archaeozoology, textual analysis, archaeoastronomy, and architectural analysis. No single line is conclusive. The argument is cumulative: the convergence of these independent lines — none of which was designed to confirm the others — is what gives the hypothesis weight. This is the structure of a consilience of inductions (Whewell 1840), not a proof from any single domain. The reader is invited to evaluate the hypothesis on the basis of the full convergence.

Why 440 Cubits: Formula-Fitting and Its Alternative

The question of why the Great Pyramid was built to a base of 440 royal cubits has an extensive literature, and the approach taken here differs from it in method rather than in conclusion. That difference is worth stating explicitly, since it determines what either approach can establish.

The established portion of that literature concerns the *seked* rather than the base. Petrie (1883) observed that a base of 440 with a height of 280 places the ratio 22/7 in the structure, the semi-circuit being 22×40 cubits and the height 7×40 ; Rossi (2004) and Herz-Fischler (2000) established that the π - and ϕ -proportions follow as consequences of a *seked* of $5 \frac{1}{2}$ palms. This explains the ratio of base to height. It does not explain the base itself: a *seked* of $5 \frac{1}{2}$ produces the same proportions at any scale, and Meidum and Unas were both built to it at quite different sizes.

A wider body of work outside the peer-reviewed literature addresses the base directly, and does so by a consistent method: an expression is sought which evaluates to 440. Proposals in circulation include derivations from the mean number of lunations in a year, from $\sqrt{3} \times \pi$ scaled by powers of ten, from convergents of $\sqrt{3}$ such as 433/250, and from terms of a golden-section series. Each yields a number close to 440, and several are arithmetically elegant.

The difficulty with this method is not that any particular derivation is wrong. It is that the method cannot fail. Given π , $\sqrt{2}$, $\sqrt{3}$, ϕ , the lunation count and free choice of scaling factor, an expression evaluating to within a fraction of a percent of any given three-digit integer can be constructed. A derivation of 440 is therefore not evidence that 440 was chosen, because the same procedure would have produced a derivation of 439 or 441 with equal facility had the pyramid been built to those dimensions. No such analysis reports what its formula yields for bases the Egyptians did not use, and without that comparison the result carries no discriminating information.

The approach taken here inverts the procedure. Rather than searching for an expression that outputs 440, the analysis holds the method fixed — the same six dimensions, the same operations, the same target list, the same tolerance — and varies the base across every whole cubit from 75 to 600, a range spanning the smallest to well beyond the largest Old Kingdom pyramid. The question asked is not whether a formula can be found for 440, but where in that range the correspondence performance actually falls, and whether the answer depends on the targets being the ones Egyptian sources document.

Measured by the denominator margin defined above, and holding the seked at $5\frac{1}{2}$ throughout, the Great Pyramid's base ranks in the top four per cent of the 526 candidate bases at every tolerance within the observational band derived earlier: twenty-second at $\pm 0.10\%$, fourth at $\pm 0.15\%$, third at $\pm 0.20\%$, and seventh at $\pm 0.25\%$. Only seven of the 526 bases — 1.33% — maintain comparable standing across the whole band. Outside the band, at $\pm 0.30\%$, it remains in the top eight per cent.

The control is the part that distinguishes this from the formula-fitting tradition. The entire sweep was repeated against three hundred shadow target lists, each constructed by replacing every real target with a random value drawn from the plausible range of its own physical category, so that a lunar period remains lunar-scale and a planetary synodic remains planetary-scale. If 440 were simply an arithmetically fertile number, it would continue to rank highly. It does not: against shadow targets its median rank is 275 of 526 — the middle of the distribution — and it attained the maximum in none of the three hundred trials ($p < 0.0033$, the upper bound available from 300 draws). The standing of 440 is therefore a property of its relationship to the specific values Egyptian sources record as astronomically significant, not a property of the number itself.

What this establishes is bounded and should not be overstated. It does not demonstrate that the base was selected for this reason; a dimension chosen on other grounds may occupy an optimum without having been placed there deliberately, and no arithmetical test can distinguish the two cases. What it does establish is that the standing of 440 is specific to the Egyptian target set and survives a control that the formula-fitting approach does not attempt. The claim advanced is accordingly that the base is where the correspondence performance is maximal for the documented targets, and that this is unlikely to be coincidental — not that intent has been demonstrated.

The Denominator Margin Across Nine Pyramids

The denominator sweep reported above establishes that sixty-fourths outperform their neighbouring grids at the Great Pyramid. It does not by itself establish that this is a property of the Great Pyramid rather than of the fraction system. If $k/64$ outperformed $k/48$, $k/60$ and $k/72$ at every monument, the result would reflect an arithmetic peculiarity of the number 64 and would carry no architectural information. The sweep was therefore repeated across the nine documented Old Kingdom pyramids scored in the comparative analysis above.

Define the denominator margin as the correspondence count under $k/64$ minus the highest count achieved by $k/48$, $k/60$ or $k/72$ at the same monument, with dimensions, targets, operations and tolerance held constant. A positive margin means sixty-fourths outperform their immediate neighbours at that pyramid; a negative margin means they do not. The Great Pyramid returns a margin of +5, scoring 11 against a best neighbouring value of 5. Of the remaining eight pyramids, six return a margin of zero or below — Khafre, Meidum and Unas at zero, the Bent Pyramid and Teti at -2,

Sahure at -3 — and the only two positive margins, Menkaure and the Red Pyramid, reach $+2$. No other monument in the sample approaches the Great Pyramid's margin.

The result was then tested against chance. Twenty thousand pyramids were generated with base and seked drawn uniformly from the Old Kingdom ranges, and the denominator margin computed for each. The mean margin across random dimensions is -1.32 : for an arbitrary pyramid, sixty-fourths typically underperform their neighbouring grids rather than outperform them. A margin of $+5$ or greater occurred in 100 of 20,000 cases, giving $p = 0.0050$.

Unlike the raw correspondence count, this result is stable across most of the tolerance band. At $\pm 0.10\%$ the Great Pyramid's margin is $+3$ ($p = 0.043$); at $\pm 0.15\%$, $+5$ ($p = 0.005$); at $\pm 0.20\%$, $+5$ ($p = 0.005$); at $\pm 0.25\%$, $+4$ ($p = 0.019$). At $\pm 0.30\%$ the margin falls to $+2$ and significance is lost ($p = 0.116$). The random mean remains negative at every tolerance tested. The result therefore holds across four of the five thresholds examined and fails at the widest, and that failure is reported rather than omitted.

The interpretation is narrow and should be stated as such. The test does not show that the Great Pyramid encodes astronomical values; it shows that whatever produces its correspondence count is specific to its dimensions and not to the number 64. Sixty-fourths do nothing in particular at Khafre, at Meidum, or at Unas — and Unas shares the Great Pyramid's seked exactly. Scale, not the fraction system alone, carries the result.

Dependence on Individual Targets

A correspondence count of ten from fourteen invites the objection that a single fortunate target is carrying the result, with the remaining thirteen contributing nothing beyond ordinary arithmetic. This is testable directly. Each of the fourteen frozen targets was removed in turn and the second Monte Carlo test re-run on the remaining thirteen, at the full $N = 100,000$ and with the same seed. Were one target responsible, its deletion would collapse the significance.

Almost none does. Thirteen of the fourteen deletions leave the result significant, with p -values from 0.0088 to 0.048. The single exception is the anomalistic month, whose removal takes the result to $p = 0.064$ and out of significance; that target is discussed separately above. The three deletions that most improve the p -value are documented misses — removing Mercury gives 0.0088, the Sothic cycle 0.0093, the synodic month 0.016 — because a missed target is dead weight in the denominator. Had one correspondence been load-bearing, its removal would stand out as an isolated failure. No such outlier appears, and the result is distributed across the set rather than resting on any member of it.

One feature of the table deserves comment, because it bears on the composition of the target list. The four deletions that most improve the p -value are the four documented misses: Mercury at 0.0088, the Sothic cycle at 0.0093, the synodic month at 0.016 and the sidereal month at 0.017, every one of them better than the baseline of $p = 0.024$. A missed target is dead weight in the denominator, and discarding it raises the score. All four are retained. A list from which failures had been pruned would report a stronger figure and mean considerably less, and the presence of two failures that actively cost the analysis significance is evidence that the thirteen were not assembled for fit.

The scope of this test should be stated precisely. It establishes robustness to the removal of targets; it says nothing about the addition of targets that were never included. That second question — whether values the Egyptians tracked have been omitted because they would have missed — is not answerable by any test conducted on the present list, and is addressed instead by the differential prediction stated under H_1 , which specifies in advance where correspondences should and should not be expected to appear.

Derivation of the Matching Tolerance

The $\pm 0.2\%$ threshold used throughout requires justification, since a tolerance chosen for convenience would render any correspondence count meaningless. It is not chosen arbitrarily. It approximates the precision at which the target values were themselves knowable to the builders.

A synodic period is determined observationally by timing a recurring event — a heliacal rising, a first stationary point — and dividing the elapsed interval by the number of completed cycles. The achievable precision is therefore the timing error divided by the total span observed. Naked-eye determination of a heliacal event carries an uncertainty of roughly three to five days, since the date on which a star or planet first becomes visible depends on atmospheric extinction, horizon conditions and the acuity of the observer. Taking a conservative ± 3 days and a record spanning three to five completed cycles — a realistic span for a single scribal working life — the resulting uncertainties are approximately 0.17% to 0.10% for Venus, 0.13% to 0.08% for Mars, 0.25% to 0.15% for Jupiter, and 0.26% to 0.16% for Saturn. The lunar periods, being short, carry proportionally larger uncertainty per cycle and require correspondingly longer records to reach comparable precision.

The 0.2% threshold sits within that band. It is not the precision of the monument, which is considerably finer: Petrie's survey found the base sides agreeing to within about 4.4 cm over 230.33 m, a relative error of 0.019%, roughly an order of magnitude tighter. The binding constraint on any encoding is therefore not the accuracy of the masonry but the accuracy of the astronomy the masonry would record. A threshold of 0.05% would demand that the encoding be more precise than the observations available to produce it; a threshold of 0.5% would admit values the builders could have distinguished. The frozen tolerance is the one commensurate with the data.

This derivation was not used to select the threshold, which was fixed before the controls reported below were run and is frozen in the deposited code. It is offered as the reason the threshold is defensible rather than convenient, and it is stated here so that a reader who disagrees with the reasoning can locate the disagreement precisely. The tolerance sweep reported below shows the consequences of varying it in either direction.

Monte Carlo Statistical Analysis

Two primary Monte Carlo tests were conducted on the mathematical encoding section ($N = 100,000$ each, seed 42, precision threshold $\pm 0.2\%$), supported by the further controls reported below and frozen in the deposited code: a target-value null, a matched-cardinality fraction null, a size-matched pyramid null, an evenly-spaced fraction null, a deterministic denominator sweep, and the independent-core and maximally-independent-core analyses. The analysis uses simple Wedjat operations only — dimension $\times$ fraction or dimension $\div$ fraction — the exact computational form documented in the Rhind Mathematical Papyrus (Problems 56–58). Cross-dimensional addition and subtraction operations are documented in the Declared Methodological Constraints section and produce additional correspondences; they were excluded from the Monte Carlo analysis specifically to avoid inflating the expression space and to anchor the statistical test to the most directly defensible operation type. The comparative table and the p-values reported here therefore represent a conservative lower bound on the encoding density. The same rules apply to every pyramid and every random comparison. The complete code and raw output are published alongside this paper for independent replication; corrections are welcome. Three files are deposited. The main analysis script `giza_analysis.py` (Python 3 standard library only; random seed 42 reset before every stochastic block) performs everything reported in this section and produces `giza_analysis_results.txt`, its complete raw output, also deposited. The third file, `stellar_control.py`, performs the stellar negative control described below: it precesses eleven stars to 2560 BCE using the Vondrák et al. (2011) long-term precession model with proper motion applied, derives thirty-one angular targets, scores them under the identical rules, and runs the 200,000-trial shadow-angle control that establishes the chance expectation. It requires `astropy`, which it installs automatically if absent, and takes several minutes to run; every other analysis in this paper uses the standard library alone. The suite has been executed on independent hardware and, separately, in two browser-based Python 3.12 environments with no local filesystem and no shared state, reproducing every reported figure identically in each case. The stellar control has likewise been run on separate machines under Python 3.12 and Python 3.13, returning 5 of 31 observed against a chance expectation of 4.83 and $p = 0.5431$ in each case. This establishes that the results depend on the deposited code alone and can be regenerated by any reader with a browser. It does not constitute independent verification of the analysis itself, which requires a reader who did not

write it; determinism and correctness are separate properties, and only the first is demonstrated here. One block of the suite, the shadow-target control, is computationally heavy and may exceed the execution limit of some browser runners; it completes without difficulty in a standard local Python 3 environment.

A note on the structure of the deposited analysis. Parts 1 to 3 of `giza_analysis.py` constitute the core analysis governed by the specification set out above: the frozen fourteen-target list, the six Giza dimensions, the sixty-three Wedjat values, multiplication and division only, and the $\pm 0.2\%$ tolerance. Parts 4 to 7 lie outside it. They are post-hoc, they feed no decision rule, they leave Parts 1 to 3 unaltered, and each departs from the specification in a stated way. Part 4 addresses Earth's polar and equatorial radii, which are not among the fourteen targets, and employs operations the specification excludes — pairwise sums of dimensions, together with calendrical multipliers and powers of ten; it is reported above at a separately designated lower evidential tier. Part 5 computes the denominator margin across nine documented pyramids, comparing $k/64$ against the neighbouring grids $k/48$, $k/60$ and $k/72$, which are not the Wedjat set, at dimensions which are not Giza's. Part 6 varies the base across every whole cubit from 75 to 600 with the *seked* held at $5 \frac{1}{2}$, varying a dimension the specification fixes. Part 7 removes each target in turn and re-runs the second Monte Carlo test on the remaining thirteen; it is a diagnostic and not a selection — the target list is not altered on the basis of its results, and in particular the four documented misses are retained although the removal of any of them would improve the reported *p*-value. All four were devised after the specification was written down, and are reported as post-hoc analyses. They are included because they test the specificity and robustness of the core result rather than restate it, and because withholding an analysis that bears on the result would defeat the purpose of fixing the specification at all.

Test 1 (Wedjat system specificity): The Wedjat binary fraction system applied to the Great Pyramid's six exterior dimensions achieves 11 of 13 astronomical target correspondences. Against 100,000 random fraction systems of equivalent structure applied to the same dimensions, the random mean is 6.26 (± 1.65 SD). Giza's result is 2.27 standard deviations above the random mean (empirical $p = 0.01497$, $p < 0.001$). Only 1,497 of the 100,000 random fraction systems scored 11 or higher.

Test 2 (Giza dimensional specificity): With Wedjat fractions held constant, the Great Pyramid's dimensions are compared against 100,000 randomly generated Old Kingdom-style pyramids with base varied uniformly from 75–440 rc and *seked* from 4.0–7.5 palms/cubit (Rossi 2004). The random mean is 5.76 (± 1.93 SD). Giza's result is 2.20 standard deviations above the random mean (empirical $p = 0.02419$, $p < 0.01$). Only 2,419 of the 100,000 randomly generated pyramids scored 11 or higher. Both results remain statistically significant after Bonferroni correction for two simultaneous tests (adjusted threshold $p < 0.025$): Test 1 $p = 0.01497$, Test 2 $p = 0.02419$.

System Specificity: The Designated Primary Test

This paper designates a single primary decision rule, and it is not the correspondence count. At equal cardinality — each rival historical fraction system downsampled at random to the Wedjat set's exact size of sixty-three values — the Wedjat system must out-reach every rival across the tolerance band from $\pm 0.10\%$ to $\pm 0.30\%$. That rule was chosen because it is the one test in the suite that does not depend on the tolerance, and because it isolates the question the correspondence count cannot answer: whether the specific dyadic subset, rather than the size of the toolkit or the general reachability of the targets, is doing the work.

The result is reported here in full. In the raw head-to-head at the frozen tolerance, the Wedjat set reaches 10 of 14 targets with 63 fractions; Egyptian unit fractions reach 11 with 1,923, Babylonian sexagesimal 4 with 118, small-denominator simple ratios 4 with 45, and decimal 8 with 99. Expressed per hundred fractions the Wedjat rate is 15.87, against 0.52, 3.39, 8.89 and 8.08 respectively. Downsampled to sixty-three values ($N = 20,000$ random subsets per system), no rival reaches the Wedjat score: not one subset in twenty thousand attains 10 of 14 for Egyptian unit fractions, Babylonian sexagesimal or decimal, and the same holds on the independent ten-target set ($p <$

0.0001 in each case). The simple-ratio system contains fewer than sixty-three fractions and cannot be downsampled to equal size; it is therefore excluded from the equal-cardinality test and from the decision rule, as declared above.

Repeating the test at each tolerance across the frozen band gives the profile the decision rule requires. At $\pm 0.10\%$ the Wedjat set reaches 5 of 11 and the highest rival probability is 0.0032; at $\pm 0.15\%$, 6 of 11 and 0.0016; at $\pm 0.20\%$, $\pm 0.25\%$ and $\pm 0.30\%$, 8 of 11 with rival probabilities of 0.0001, 0.0004 and 0.0018. The maximum across the entire band is 0.0032. The primary criterion is therefore met, and — unlike the correspondence count, whose significance moves with the tolerance — this result does not move when the threshold is tightened or loosened. It is the finding on which the dimensional argument rests.

Its scope should be stated as precisely as the result. The test establishes that the dyadic subset is specifically more efficient at reaching the documented Egyptian target list than the other historical systems tested, at equal size and under identical operations. It does not establish that the correspondences were deliberately placed, and it does not exhaust the space of possible rival systems: a culturally attested system not tested here could in principle perform comparably, and identifying one would weaken the claim.

Neither result constitutes proof of deliberate cosmological encoding. Together they establish that the observed 11-of-13 target pattern is statistically non-random under two independent null models: unlikely to arise from an arbitrary fraction system on these dimensions, and unlikely to arise from the Wedjat system on arbitrary Old Kingdom pyramid proportions. These two tests establish that the correspondence count is improbable. They do not establish what produces it, and on their own they remain vulnerable to the objection that some feature of the fraction system, rather than of the monument, is doing the work. The three analyses that follow — the evenly-spaced fraction null, the denominator margin across nine pyramids, and the base sweep with its shadow-target control — address that objection directly by varying one element at a time while holding the rest fixed. They are reported after the primary tests because they are specificity tests rather than significance tests, and because two of the three are post-hoc, as declared below. In the assessment offered here they carry more evidential weight than the raw count, since each was constructed so that it could have failed. The analysis also examines the full Old Kingdom pyramid record. When the identical test is applied to nine documented pyramids using their published base and height figures — design values where these are securely reconstructed, as set out in the Declared Methodological Constraints — Giza leads all three architectural periods:

5th–6th Dynasty (post-Giza, c. 2490–2278 BCE): Teti 5/13, Unas 6/13, Sahure 5/13. Giza Complex (c. 2560–2510 BCE): Khufu 10/14, Khafre 7/13, Menkaure 7/13. Sneferu's Programme (pre-Giza, c. 2620–2590 BCE): Red Pyramid 8/13, Bent Pyramid 8/13, Meidum 6/13. The decline in correspondence density after Giza is consistent with the attenuation model proposed throughout this series: institutional capacity to transmit and apply the full cosmological encoding tradition weakened after the 4th Dynasty peak. Whether the builders of later pyramids lacked the knowledge, the resources, or the intention is a question the data cannot resolve. That Giza represents the maximum expression of the tradition within the documented Old Kingdom record is what the analysis establishes.

Conditions Under Which This Hypothesis Weakens

In addition to the specific testable predictions above, the hypothesis weakens materially under the following conditions, each of which is in principle falsifiable by ongoing or future research:

Genomic convergence fails: If systematic analysis of additional Predynastic and early Dynastic genomes finds no contribution from the documented Saharan corridor population, the biological transmission chain is severed.

Paleohydrological corridor impassable: If refined modelling demonstrates the Highland Corridor was not traversable during the relevant period (c. 5,300–3,200 BCE), the proposed migration pathway is broken.

Second formation satisfies all criteria: If any Saharan formation is identified that simultaneously satisfies all four criteria applied across Corridor I and II — sufficient scale, documented Holocene water at the structure, documented human presence at the structure, and a central elevation rising from that water — the Richat identification loses its uniqueness as the only candidate.

Null model parity demonstrated: If formal Monte Carlo testing shows that the dimensional correspondences documented in the mathematical encoding section appear at comparable frequency across randomly generated geometric sets or comparable Old Kingdom monuments, the selectivity of the Giza pattern is not established.

Material culture discontinuity confirmed: If systematic analysis of assemblages from the proposed corridor nodes (Tassili, Takarkori, Nabta Playa, Hierakonpolis) reveals no shared material vocabulary across the migration period, the cultural transmission chain is weakened.

Shaft orientations structurally determined: If revised archaeoastronomical analysis demonstrates that the stellar shaft orientations are primarily structurally or topographically constrained rather than astronomically targeted, the cosmological intentionality of the shaft programme is not established by orientation data alone.

A note on the independence of evidence domains: Reviewers may argue that the domains are not truly independent — all are interpreted through the same framework. This is acknowledged. The independence claimed is disciplinary: the hydrology papers do not cite the genetics papers; the archaeozoology findings were not designed to confirm architectural hypotheses. Their convergence is therefore meaningful but not definitive.

A final limitation on the Richat identification: The identification of Richat as the origin landscape rests on the absence of any other known formation that satisfies all four criteria applied across Corridor I and II. This makes Richat the leading testable candidate, not a proven origin. The hypothesis does not require that Richat was the actual landscape encoded — only that the Zep Tepi relational system originated in a landscape with these features, for which the Richat is the only currently known candidate. Future high-resolution survey and targeted excavation in annular depressions elsewhere in the Sahara could identify additional candidates or falsify this distinction.

Scope, Boundaries, and Limitations

The Zep Tepi Series is structured so that each paper addresses a defined evidentiary scope. Defining the scope of the present paper explicitly, and what lies outside it, is an act of methodological discipline rather than defensive qualification.

What this paper claims: that the exterior architecture of the Giza complex instantiates a specific four-element relational cosmological system (central elevation, bounded sacred space, transition water, processional axis) whose origin this series locates in the Saharan pastoral tradition documented in The Corridor Series; that the exterior dimensions of the Great Pyramid contain mathematical and astronomical correspondences consistent with systematic knowledge preservation through the Wedjat fraction system; and that the compound evidence across cosmological, hydrological, archaeoastronomical, genetic, and mathematical domains makes deliberate encoding the more parsimonious explanation for the complex's design.

What this paper explicitly does not claim: proof of intentional encoding; certainty about transmission routes or timelines; that the mathematical correspondences are unique to Giza (the comparative analysis reported here covers nine documented Old Kingdom pyramids and twenty thousand randomly dimensioned ones; extension to non-Egyptian monuments remains future work); that the Shemsu Hor hypothesis accounts for all features of Giza's design; or that the cosmological reading of

the complex excludes practical and logistical explanations. The paper repeatedly distinguishes “consistent with” from “proves” throughout.

Scope limits: this paper addresses the exterior of the Great Pyramid and the Giza complex in its landscape setting. The internal structure — chamber acoustics, electromagnetic resonance, the Grand Gallery, the sarcophagus, and any proposed interior encoding system — is not addressed here, and no claim about it is advanced or assumed. The Wedjat fraction system is treated here only as the arithmetic key to the dimensional correspondences documented above; its wider role as a knowledge architecture spanning medicine, architecture, calendrics, and theology is beyond the scope of this paper, though the medical and botanical strand of that architecture is treated in Levy (2026f).

The interval requiring explanation is the 540-year span between the close of the Hierakonpolis tradition (c. 3,100 BCE) and the construction of the Great Pyramid (c. 2,560–2,540 BCE). That interval is not left as an inferential gap in this series. It is the subject of the Bridging Paper, Phase Zero: The Bull in the Sky (Levy 2026d), which documents a transmission chain running through it: the coordinated Minor Step Pyramid programme at seven provincial cult centres, the Djoser complex at Saqqara, and the archaeozoological evidence for Shemsu Hor continuity at Hierakonpolis. On that evidence the pyramid form, the stellar orientation procedure and the institutional apparatus for co-ordinated sacred construction were all already in place before Khufu, at provincial scale, encoding the same Meskhetyu tradition. The present paper assumes those findings rather than re-arguing them. What remains genuinely open is narrower, and should be stated as such: the transmission chain is documented at the level of architectural programme, orientation practice and faunal continuity, but no single line of evidence traces a specific body of cosmological knowledge from Hierakonpolis to Giza in unbroken sequence. The claim advanced here is that the convergence of independent domains across the interval makes continued transmission the most parsimonious available explanation, not that any one line establishes it on its own.

What Remains Unknown

- Whether Khufu's body is elsewhere, and what became of it, cannot be determined from current evidence. Herodotus's description of a subterranean island tomb below the Giza Plateau — featuring a moat surrounding a central mound — corresponds architecturally to the Third Level of the Osiris Shaft (pr wsir nb rstaw, documented by Hawass 2007; on the Shetayet of Rosetau, Edwards 1986), the same structure whose Nun cosmological grammar this paper documents. The Third Level sarcophagus is officially characterised as a symbolic burial. This hypothesis notes the architectural correspondence as a convergence point without asserting proof of identification. Hawass has publicly stated his belief that Khufu's real burial chamber remains hidden inside the pyramid (Piers Morgan Uncensored, 15 May 2026), confirming that the question is not considered closed even within the Egyptological mainstream. This is additionally supported by the peer-reviewed scanning programme: four independent geophysical methods, across three peer-reviewed publications, have identified previously unknown interior spaces in the Great Pyramid (Morishima et al. 2017; Procureur et al. 2023; Schumacher et al. 2025), and a sealed corridor announced by Hawass in November 2025 (Egypt Independent, 9 November 2025) remains awaiting peer-reviewed publication.
- Whether the Big Void is a construction ramp, stress-relief chamber, or something else is not yet established.
- Whether the architects consciously intended every cosmological element cannot be determined without textual evidence the 4th Dynasty did not leave.
- Whether the Giza workers' cemetery contains individuals with North African / Saharan ancestry remains to be determined. No aDNA has been published from Heit el-Ghurab at the time of writing. At the time of writing, no ancient DNA has been published from any non-elite Old

Kingdom cemetery in Egypt. The prediction therefore remains untestable pending future work, but the skeletal collections exist and the methodology is established.

- The Hawass 30-metre corridor (announced November 2025) remains unopened and unpublished in peer-reviewed form at the time of writing. Pending peer-reviewed publication, no definitive interpretation is possible.

Testing Protocol

The 'Pyramid as Memory' hypothesis generates several clear, falsifiable predictions that distinguish it from standard Egyptological explanations.

- **Prediction 1 (Causeway Orientation):** If Giza encodes the Zep Tepi landscape, the Giza causeway orientation should correspond to the general direction of natural egress from the western side of the Richat ring system. The structural geometry of the Richat's ring system (Matton et al. 2005; Abdeina et al. 2024) should be consistent with the observed westward causeway orientation at Giza. A precise azimuth correspondence would constitute strong confirmation; qualitative directional agreement constitutes a weaker but still positive result.
- **Prediction 2 (Chamber Arrangement):** The interior of the pyramid lies outside the scope of this paper, and no claim about it is advanced here. The framework nonetheless generates a prediction that later work can test: if the interior carries the same cosmological grammar as the exterior, the three-chamber vertical arrangement (Subterranean, Queen's, King's) should correspond to the Zep Tepi sequence of water, threshold and transformation, and should do so consistently with the Pyramid Texts' description of the king's ascent. This prediction is recorded for completeness and is not argued for in the present paper.
- **Prediction 3 (Heit el-Ghurab aDNA):** Ancient DNA from the Giza workers' cemetery will show elevated North African Neolithic ancestry consistent with the Takarkori lineage (Salem et al. 2025). No aDNA has been published from Heit el-Ghurab at the time of writing.
- **Prediction 4 (Material Provenance — Null Prediction):** Geochemical sampling of Khufu-era casing stones will confirm local Egyptian limestone sources — consistent with symbolic (not material) encoding of Richat. Detection of Saharan material would not falsify the relational encoding claim, which concerns design and orientation, not stone origin.
- **Prediction 5 (New Discoveries):** If the Hawass corridor (once peer-reviewed) contains primary 4th Dynasty royal burial goods or Khufu's remains, the cosmological-encoding reading of that specific feature is weakened.

Failure of Predictions 1, 2, or 3 would require revision of the architectural encoding or demographic transmission claims. Prediction 4 is a null prediction; its success (local stone) is expected. Prediction 5 will be resolved by future peer-reviewed publication. Predictions 6, 7, and 8 together constitute the structural tests for the mathematical encoding section: Prediction 6 tests whether domain-complete multi-domain encoding is distinctive to Giza across comparable monuments; Prediction 7 tests whether the multi-layer Sirius-Osirian convergence is structurally unique in the comparative record; Prediction 8 tests whether the encoding is culturally specific to the Royal Cubit and Wedjat system rather than a general property of these physical dimensions in any unit.

- **Prediction 6 (Domain Completeness):** A systematic survey of comparable ancient monuments with documented astronomical awareness and published dimensions in native or proposed ancient units finds that no other single monument independently encodes the full combination of the following domains simultaneously through its structural dimensions: (a) all four of the visible planetary synodic periods relevant to Old Kingdom astronomy — Saturn, Jupiter, Venus, and Mars; and (b) the complete Sirius-Osirian calendar cluster — the 70-day invisibility period, the 360-day schematic administrative year, and the 1,461-year Sothic

recalibration cycle (the last of these a documented miss). This multi-domain integration spans timescales from 70 days to the Sothic period and covers the full breadth of Old Kingdom astronomical practice — short-term ritual, administrative calendrics, planetary astronomy, lunar periodicity, and long-period stellar recalibration — using a single coherent six-dimensional structural framework. Peer-reviewed archaeoastronomical literature documents that Angkor Wat encodes solar year and Hindu yuga cosmological cycles but not individual planetary synodic periods, Saros, Metonic, or the Sirius cluster (Stencel, Gifford & Morón 1976; Mannikka 1996); Teotihuacan encodes solar calendar proportions but not the planetary synodic quartet (Sugiyama 2010); Stonehenge encodes solstice and lunar standstill geometry but not dimensional correspondences with planetary periods or the Sirius-Osirian cluster (Ruggles 1997; Cleal, Walker & Montague 1995). Falsified by: identification of another ancient monument whose structural dimensions, measured in its own native ancient unit, simultaneously encode the full combination of (a) and (b) above with comparable precision and coherence.

- Prediction 7 (Sirius Multi-Layer Structural Coherence):** Independent exterior dimensions of the Great Pyramid address the same celestial body and its associated ritual-calendar system across three distinct timescales, two of which reach the target within tolerance and one of which does not: Height (280 cubits) $\times \frac{1}{4} = 70$ days — the canonical period of Sirius's annual invisibility, encoded in Egyptian ritual as the duration of mummification and the death of Osiris; Half-base (220 cubits) + Height (280 cubits) $\times \frac{1}{2} = 360$ days exactly — the schematic base of the civil calendar, the 12 $\times$ 30-day administrative year before the five epagomenal days are added (the 360-day layer encodes the schematic base of the Sirius-anchored civil calendar rather than a direct Sirius period; all three layers are unified through the Osirian calendar mythology); and Perimeter (1,760 cubits) $\times (\frac{1}{2} + \frac{1}{4} + \frac{1}{16} + \frac{1}{64}) = 1,457.5$, which approaches but does not reach the 1,461-year Sothic cycle governing the long-term recalibration between the civil calendar and Sirius's heliacal rising (error 0.240%, outside tolerance and scored as a miss). The third layer of this cluster is therefore a near-approach rather than a correspondence. The prediction is accordingly made on the two layers that reach tolerance, with the third recorded and scored as an attempted layer that fails. A systematic review of comparable monuments finds no equivalent convergence on a single astronomical object spanning short-term ritual and administrative calendrical domains through independent structural dimensions: Angkor Wat's documented solar year and yuga encodings converge on the Sun and Hindu cosmological cycles, not a single star; Teotihuacan and Stonehenge show no documented multi-layer convergence of this type on a single star. Falsified by: identification of another ancient monument whose structural dimensions, in its own native unit, encode two or more independent correspondences converging on the same celestial body across distinct ritual and calendrical timescales with comparable precision and coherence.
- Prediction 8 (Unit-System Specificity):** The correspondences documented in this section are primarily visible when Giza's six exterior dimensions are expressed in Royal Cubits — the unit in which the pyramid was designed, constructed, and dimensionally specified. A preliminary test performed for this paper confirms unit-system specificity for the most culturally significant encodings: when the same six dimensions are converted to metres and the identical Wedjat methodology applied, the overall hit count falls, and the anomalistic lunar month — a correspondence with deep cultural specificity to the Egyptian lunar tradition — disappears entirely in metres. Its disappearance in a non-native unit is consistent with culturally bounded encoding. The Sothic cycle cannot serve this test, since it does not reach tolerance in cubits either once counted in Egyptian civil years. The planetary synodic quartet and the 360-day and 70-day Sirius periods persist in metres, indicating these are more robust across unit systems. A full comparative test, specified in advance, remains future work. Falsified by: demonstration that the anomalistic month correspondence reproduces with equal precision in non-cubit unit systems, indicating they are properties of these physical distances rather than of the Royal Cubit measurement system specifically.

- **Prediction 9 — Direct Structural Dating of the Osiris Shaft:** The hypothesis that the Osiris Shaft predates the surface monuments of the Giza plateau — and constitutes the pre-existing sanctuary the Westcar Papyrus records Khufu seeking to model — is testable by direct structural dating. The existing chronology of the shaft rests primarily on artifact deposits: pottery, sarcophagi, and votive objects that establish when the complex was used across successive dynasties. Artifact chronology dates phases of occupation; it does not date when the bedrock was first cut. Surface luminescence dating of primary rock-cut wall surfaces at all three shaft levels — using untouched samples from positions independent of sarcophagal deposits, artifact accumulations, and groundwater saturation zones — would directly measure when those surfaces were last exposed during carving, establishing the construction date of the primary structure rather than its reuse history. The existing Liritzis and Vafiadou (2015) results on stone elements are consistent with a pre-4th-Dynasty construction date; targeted wall-surface sampling at multiple depths would substantially reduce the uncertainty that groundwater exposure introduces into current results. Without this test, no claim about the shaft's construction date — in either direction — rests on direct structural evidence.

Falsified by: surface luminescence dating of primary untouched rock-cut surfaces at all three levels of the Osiris Shaft — specifically of wall faces independent of artifact contexts — yielding ages firmly and consistently posterior to the construction of Khafre's causeway above them, demonstrating that the shaft was excavated after the surface monuments were complete rather than before or alongside them.

Conclusion

The Great Pyramid of Giza is the most studied structure in human history. Every civilisation that encountered it asked the same question: what is this for?

This paper has proposed a different kind of answer. The pyramid is not primarily a tomb. It is the permanent architectural encoding of a cosmological system that began in the Green Sahara during the African Humid Period (Salem et al. 2025), was carried through a documented migration corridor, was transmitted through the Shemsu Hor, and was finally expressed at its definitive scale by the 4th Dynasty Egyptian state.

But the argument of this paper does not rest on that origin story alone. It rests on a set of empirical features — causeway orientations organised about the east–west axis (Nell & Ruggles 2014), Heliopolis alignment maintained even when invisible (Lehner 1985; Magli 2009), integrated landscape planning (Lehner 1985), material-theological convergences (Shaw 2000; Belmonte et al. 2016), and 2000-year cultural continuity (Magli 2011) — that collectively constitute a constraint filter. To these the present analysis adds two results of a different character, in that each was constructed so that it could have failed. The sixty-fourths outperform their neighbouring grids at the Great Pyramid and at almost no other pyramid, against a random mean that is negative; and the built base of 440 stands at the top of its range for the documented Egyptian targets and in the middle of it for shadow targets of the same physical kind. Neither result demonstrates intent, which no arithmetical test can reach. Both narrow what else the dimensions could plausibly be. Any viable interpretation of Giza must account for these features simultaneously. The Zep Tepi model passes through this filter intact, without internal contradiction, using a single coherent framework.

The field is invited to propose alternative models that also pass through this constraint filter.

Until such models are advanced, the Zep Tepi framework remains, to the knowledge of this author, the only interpretation known to this author that accounts, within its declared scope, for the set of empirical constraints assembled here as a unified framework.

The pyramid was not built to last forever because it contained something valuable. It was built to last forever because it was something valuable — the permanent recreation of the First Time, the moment when the ancestors' world was whole, before the flood of sand took it away. As long as the pyramid stands, the Zep Tepi is not lost. The knowledge that generations of visitors sensed in the empty chambers was not concealed inside. It is the form they were standing within. On this reading, the pyramid is not a container for knowledge. It is the knowledge, made stone.

The site was not called Giza in the Old Kingdom. That name is medieval Arabic. The ancient Egyptians called this plateau Imentet — The West — and Rostau — the Entrance to the Passages of Osiris. Read together, those two names are the thesis of this paper and of this series. Imentet: the knowledge encoded here was carried from the west, along a documented corridor from the Green Sahara, across three thousand years of institutional memory. Rostau: it was deposited at the entrance to an architectural system designed to translate that knowledge into permanent form — subterranean, cosmological, imperishable. The causeways point west. The chambers descend underground. The external dimensions encode the sky, the earth, and the long cycles of time. The ancient Egyptians named their monument after where it came from and what it was built to do. No further decipherment was required. They left the answer at the entrance.

The Osireion raises a question this paper cannot answer. Its dating is contested. Egyptological consensus attributes it to Seti I of the 19th Dynasty, on the cartouche in the approach corridor, quarry records from year six of his reign, and its shared axis with his temple (Frankfort, de Buck & Gunn 1933; Brand 2000). The evidence is more equivocal than that summary suggests. The cartouche occurs in the entrance passage; none of the main interior blocks carries any inscription, and Frankfort himself observed that the structure's style does not match known work of Seti's reign — an observation Petrie had already made of its archaic character. Two readings follow. Either Seti I incorporated an existing structure into his complex, or he commissioned it deliberately in an archaic idiom in order to evoke older religious conceptions of water and rebirth. The second reading is the more economical, and it is the more consequential for the argument advanced here: deliberate archaism is not the absence of transmitted memory but its expression, a builder reaching for a grammar he knew to be ancient. A second position places the original structure in the Middle Kingdom, and it rests on physical dating rather than on textual attribution alone. Surface luminescence dating of Osireion samples by Liritzis and Vafiadou (2015) — the same study that dated the Osiris Shaft at Giza, cited above — returns Middle Kingdom ages for the granite together with a younger date for the sandstone, which the authors read as indicating original and later architectural phases. Strabo's identification of the builder as Ismandes (Geography XVII), conventionally read as Amenemhet III, is consistent with that reading, though Strabo himself writes with evident uncertainty and the structure's origins appear already to have been obscure in his day. On this account the Osireion is a two-phase monument: a Middle Kingdom original, restored and inscribed by Seti I some seven centuries later. On either account the builder was encoding the same Zep Tepi grammar that had been encoded at Giza, and knew what he was building. On the consensus dating the Osireion is itself evidence that the tradition survived the Second Intermediate Period, when Egyptian institutional continuity dissolved for over two centuries; on the minority dating it precedes that rupture, and the question becomes what carried the cosmological grammar through the silence between 1800 and 1550 BCE. Either way, when the tradition appears in the New Kingdom — in the stellar theology of the Amduat, in the Books of the Underworld, in the astronomical ceilings of the New Kingdom tombs — the question of what form it had taken, and what had been lost in the crossing, remains open.

These questions lie beyond the scope of this paper.

The Pyramid as Memory closes the question of what the pyramid is.

What its bare chambers were for, is the subject of the next paper: **From Nun to the Stars**.

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Artificial Intelligence Use Disclosure

Following the Taylor & Francis AI disclosure standard, the author discloses the following use of generative AI and large language models. No AI tool generated the underlying scientific data or original evidence. All cited evidence derives from published sources. The interpretive framework, hypothesis structure, and editorial judgments are the author's own. AI-generated citations were found during preparation to contain errors of title, venue, volume and authorship, including references to works that do not exist; every reference in this paper has accordingly been checked against the publisher record, and the author regards independent verification of AI-assisted citation as a necessary step rather than an optional one.

Claude (Opus 4, Opus 5 and Fable 5, Anthropic) — Primary writing assistance, research synthesis, verification, and document construction.

DeepSeek (V3) — Background literature gathering and source hunting.

ChatGPT (GPT-4o, OpenAI) — Critical peer review simulation.

Grok AI (xAI) — Verification of missing citations.

Gemini (Google) — Retrieval of verbatim passages from print-only sources (Allen 2005; Chassinat; Lehner 1985; Maragioglio & Rinaldi 1965; Belmonte et al. 2010).

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